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6

Processes

A random vector is a finite collection of random variables. Sometimes however we need to consider an infinite collection of random variables, that is, a stochastic process. The classic example is the position of a particle over time. We might study the particle at integer times $t \in \{0, 1, 2, \dots\}$ or continuously over an interval $[0, T]$. Either way, the trajectory requires an infinite number of random variables to describe in its entirety.

In this chapter we look at how to sample from a stochastic process. After defining some terms we consider those processes that can be sampled in a fairly straightforward way. The main processes are discrete space random walks, Gaussian processes, and Poisson processes. We also look at Dirichlet processes and the Poisson field of lines.

We will describe stochastic processes at an elementary level. Our emphasis is on how to effectively simulate them, not on other issues such as their existence. For a clear introduction to the theory of stochastic processes, see Rosenthal (2000).

Some processes are very difficult to sample from. We need to incorporate variance reduction methods of Chapters 8, 9, and **??** into the steps that sample the process. Those methods include sequential Monte Carlo, described in Chapter 15.

This chapter contains some very specialized topics. A first reading should cover §6.1 for basic ideas and §6.2 for some detailed but elementary examples of discrete random walks. Those can be simulated directly from their definitions. Special cases can be handled theoretically, but simple variations often bring the need for Monte Carlo. The later sections cover processes that are more advanced, some of which cannot be simulated directly from their definition. They can be read as the need arises.

6.1 Stochastic process definitions

A stochastic process (or process for short) is a collection of infinitely many random variables. Often these are $X(t)$ for $t = 1, 2, \ldots$, or $X(t)$ for $0 \leqslant t < \infty$, for discrete or continuous time. In general, the process is $\{X(t) \mid t \in \mathcal{T}\}$ and the index set $\mathcal{T}$ varies from problem to problem. In some examples, such as integer t, it is convenient to use X_t in place of $X(t)$. When we need to index the index, then $X(t_j)$ is more readable than X_{t_j}. Similarly, if there are two processes, we might write them as $X_1(t)$ and $X_2(t)$ instead of $X(t, 1)$ and $X(t, 2)$. Usually X_t and $X(t)$ mean the same thing.

When $\mathcal{T} = [0, \infty)$ the index t can be thought of as time and a description of $X(t)$ evolving with increasing t may be useful. In other important cases, $\mathcal{T}$ is not time, but a region in $\mathbb{R}^d$, such as a portion of the Earth's surface where $X(t)$ might denote the temperature at location t. A stochastic process over a subset of $\mathbb{R}^d$ for $d > 1$ is also called a ***random field***.

Any given realization of $X(t)$ for all $t \in \mathcal{T}$ yields a random function $X(\cdot)$ from $\mathcal{T}$ to $\mathbb{R}$. This random function is called a ***sample path*** of the process.

In a simulated realization, only finitely many values of the process will be generated. So we typically generate random vectors, $(X(t_1), \ldots, X(t_m))$. Sampling processes raises new issues that we did not encounter while sampling vectors. Consider sampling the path of a particle generating $X(\cdot)$ at new locations t_j until the particle leaves a particular region. Then m is the sampled value of a random integer M, so the vector we use has a random dimension. Even if $\mathbb{P}(M < \infty) = 1$ we may have no finite a priori upper bound for the dimension m. Furthermore, the points t_j at which we sample can, for some processes, depend on the previously sampled values $X(t_k)$. The challenge in sampling a process is to generate the parts we need in a mutually consistent and efficient way.

We will describe processes primarily through their finite dimensional distributions. For any list of points $t_1, \ldots, t_m \in \mathcal{T}$, the distribution of $(X(t_1), \ldots, X(t_m))$ is a finite dimensional distribution of the process $X(t)$. If a collection of finite dimensional distributions is mutually compatible (no contradictions) they do correspond to some stochastic process, by a theorem of Kolmogorov.

The finite dimensional distributions do not uniquely determine a stochastic process. Two different processes can have the same finite dimensional distributions, as Exercise 6.1 shows. Some properties of a process can only be discerned by considering $X(t)$ at an infinite set of values t, and they are beyond the reach of Monte Carlo methods. For instance, we could never find $\mathbb{P}(X(\cdot)$ is continuous) by Monte Carlo. We use Monte Carlo for properties that can be determined, or sometimes approximated, using finitely many points from a sample path.

Our usual Monte Carlo goal is to estimate an expectation, $\mu = \mathbb{E}(f(X(\cdot)))$. When f can be determined from a finite number of values $f(X(t_j))$ then

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} f(X_i(t_{i1}), \cdots, X_i(t_{iM(i)})) \tag{6.1}$$

where the i'th realization requires $M(i)$ points, and the sampling locations t_{ij}

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may be randomly generated along with $X(t_{ij})$. To reduce the notational burden, we will consider how to generate just one sample path, and hence one value of $f(X(\cdot))$ for each process we consider. Generating and averaging multiple values is straightforward. Sometimes we only require one sample path. For example, Markov chain Monte Carlo sampling (Chapter 11) is often based on a single sample path.

Formula (6.1) includes as a special case, the setting where f depends on $X(t)$ for t in a nonrandom set $\{t_1, \dots, t_m\}$. In this case our problem reduces to sampling the vector $(X(t_1), \cdots, X(t_m))$. In other settings, μ cannot be defined as an expectation using such a simple list of function values. It may instead take the form $\mu = \lim_{m\to\infty} \mu_m$ where $\mu_m = \mathbb{E}\big(f_m(X(t_{m,1}), X(t_{m,2}), \cdots, X(t_{m,m}))\big)$. The set $\{t_{m,1}, \dots, t_{m,m}\}$ could be a grid of m points and the $m+1$ point grid does not necessarily contain the m point grid. Then Monte Carlo sampling for fixed m provides an unbiased estimate $\hat{\mu}$ of μ_m. There remains a bias $\mu_m - \mu$, that must usually be studied by methods other than Monte Carlo.

6.2 Discrete time random walks

The discrete time random walk has

$$\boldsymbol{X}_t = \boldsymbol{X}_{t-1} + \boldsymbol{Z}_t \tag{6.2}$$

for integers $t \geqslant 1$, where $\boldsymbol{Z}_t$ are IID random vectors. The starting point $\boldsymbol{X}_0$ is usually taken to be zero. If we have a method for sampling $\boldsymbol{Z}_t$ then it is easy to sample $\boldsymbol{X}_t$, starting at $t = 0$, directly from (6.2).

When the terms $\boldsymbol{Z}_t$ have a continuous distribution on $\mathbb{R}^d$ then so do the $\boldsymbol{X}_t$ and, for large enough t, any region in $\mathbb{R}^d$ has a chance of being visited by the random walk. When the $\boldsymbol{Z}_t$ are confined to integer coordinates, then so of course are $\boldsymbol{X}_t$ and we have a discrete space random walk.

Figure 6.1 shows some realizations of symmetric random walks in $\mathbb{R}$. One of the walks has increments $Z_t \sim \mathbf{U}\{-1, +1\}$. The other has $Z_t \sim \mathcal{N}(0, 1)$. Figure 6.2 shows some random walks in $\mathbb{R}^2$. The first is a walk on points with integer coordinates given by $\boldsymbol{Z} \sim \mathbf{U}\{(0, 1), (0, -1), (1, 0), (-1, 0)\}$, the uniform distribution on the four points (N,S,E,W) of the compass. The second has $\boldsymbol{Z} \sim \mathcal{N}(0, I_2)$. The third walk is the Rayleigh walk with $\boldsymbol{Z} \sim \mathbf{U}\{\boldsymbol{z} \in \mathbb{R}^2 \mid \boldsymbol{z}^{\mathsf{T}}\boldsymbol{z} = 1\}$, that is, uniformly distributed steps of length one.

The walks illustrated so far all have $\mathbb{E}(\boldsymbol{Z}) = 0$. It is not necessary for random walks to have mean 0. When $\mathbb{E}(\boldsymbol{Z}) = \mu$, then the walk is said to have drift μ. If also $\boldsymbol{Z}$ has finite variance-covariance matrix Σ, then by the central limit theorem $t^{-1/2}(\boldsymbol{X}_t - t\mu)$ has approximately the $\mathcal{N}(0, \Sigma)$ distribution when t is large. In a walk with Cauchy distributed steps, μ does not even exist.

Sequential probability ratio test

The sequential probability ratio test statistic is a random walk. We will illustrate it with an example from online instruction. Suppose that any student who gets

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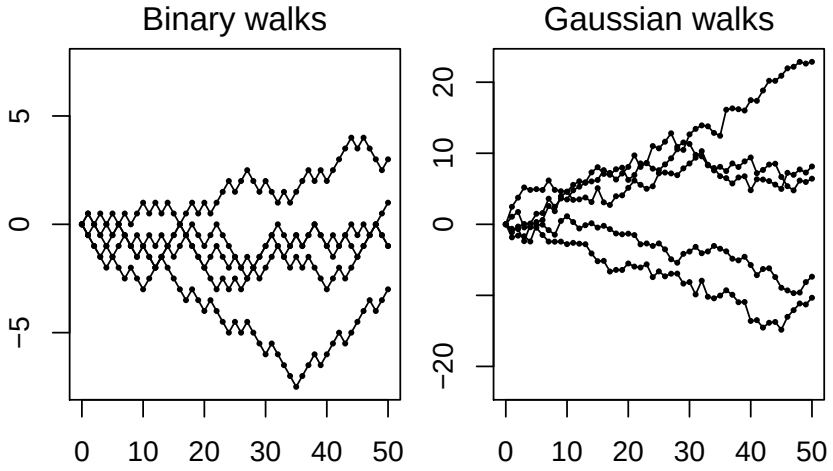


Figure 6.1: The left panel shows five realizations of the binary random walk in $\mathbb{R}$. The walks start at $\boldsymbol{X} = 0$ at time $t = 0$ and continue for 50 steps. Each step is ± 1 according to a fair coin toss. The right panel shows five realizations of a random walk with $\mathcal{N}(0, 1)$ increments.

the right answer 90% of the time or more has mastered a topic, and is ready to begin learning the next topic. Conversely, a student who is correct 75% of the time or less, needs remediation.

We let $Y_i = 1$ for a correct answer to problem i and $Y_i = 0$ for an incorrect answer. Suppose that all the questions are equally hard and that the student has probability θ of being right each time, independently of the other questions. We want to tell apart the cases $\theta = \theta_M = 0.9$ from $\theta = \theta_R = 0.75$. The probability of the observed test scores $\boldsymbol{Y} = (Y_1, \ldots, Y_n)$, if $\theta = \theta_M$, is $\mathbb{P}(\boldsymbol{Y}; \theta_M) = \prod_{i=1}^n \theta_M^{Y_i}(1-\theta_M)^{1-Y_i}$. Sequential analysis uses the ratio

$$L_n(\boldsymbol{Y}) = \frac{\mathbb{P}(\boldsymbol{Y}; \theta_M)}{\mathbb{P}(\boldsymbol{Y}; \theta_R)} = \prod_{i=1}^n \left(\frac{\theta_M}{\theta_R}\right)^{Y_i} \left(\frac{1-\theta_M}{1-\theta_R}\right)^{1-Y_i}. \tag{6.3}$$

A large value of L_n provides evidence of mastery, while a small value is evidence that remediation is required.

Sometimes it is clear for relatively small n whether the student is a master or needs remediation. In those cases continued testing is wasteful. The sequential probability ratio test (SPRT) allows us to stop testing early, once the answer is clear. Under the SPRT, we keep sampling until either $L_n < A$ or $L_n > B$ first occurs, for thresholds $A < 1 < B$. Assume for now that one of these will eventually happen. If $L_n < A$ we decide the student needs remediation while if $L_n > B$ we decide that the student has mastered the topic.

When we can accept a 5% error probability for either decision, then we may use $A = 1/19$ and $B = 19$. These values come from the Wald limits, which treat a likelihood ratio as if it were an odds ratio. The Wald limits are conservative.

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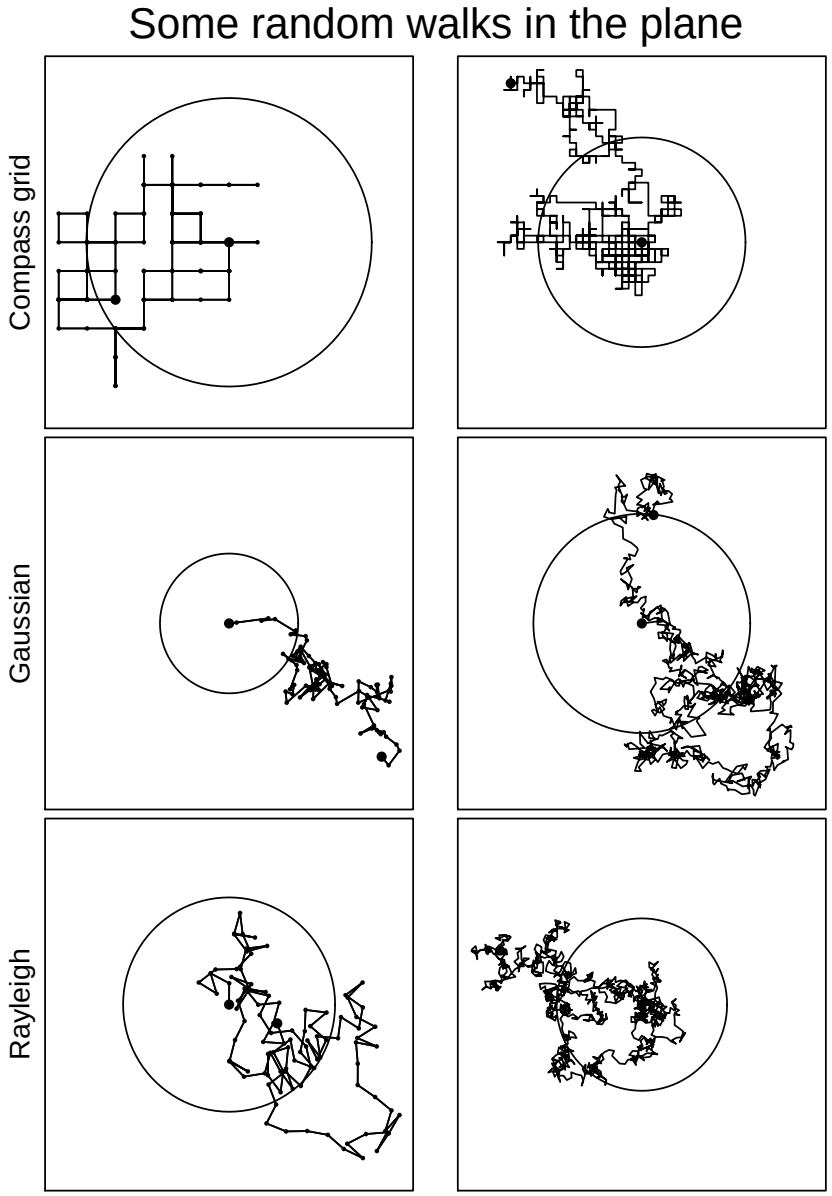


Figure 6.2: This figure shows three random walks in $\mathbb{R}^2$. From top to bottom they are the simple random walk on a square grid, the Gaussian random walk, and the Rayleigh random walk. The left column shows the first 100 steps. The right column shows the first 1000 steps of the same walks. Each panel is centered at $(0, 0)$. There is a reference circle at half the root mean square radius for the final point shown.

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Given that the SPRT has made a decision, the error probabilities are no larger than 5% and are typically slightly smaller. There is a derivation of the Wald limits in Siegmund (1985, Chapter II).

The logarithm of the likelihood ratio (6.3) is a random walk $X_n = \log(L_n) = \sum_{i=1}^n Z_i$ where

$$Z_i = \begin{cases} \log(\theta_M/\theta_R), & \text{with probability } \theta, \\ \log((1-\theta_M)/(1-\theta_R)), & \text{with probability } 1-\theta. \end{cases}$$

If $\theta = \theta_M$, then $\mathbb{E}(Z_i) > 0$ and the walk will tend to drift upwards. If it goes above $\log(B)$ then the student is deemed to have mastered the topic. Conversely when $\theta = \theta_R$, then $\mathbb{E}(Z_i) < 0$ and the walk tends to drift downwards. If it goes below $\log(A)$, the student is offered remedial material.

The log likelihood for students with $\theta > \theta_M$ will drift upwards even faster than for those with $\theta = \theta_M$ and a similarly fast downward drift holds when $\theta < \theta_R$ so it is usual to focus on just the cases $\theta \in \{\theta_R, \theta_M\}$. It is possible that $\theta_R < \theta < \theta_M$ and then testing may go on for a long time. Testing will stop with probability one, as long as Stein's condition $\mathbb{P}(Z_i = 0) < 1$ holds (Siegmund, 1985, page 12). But to avoid long waits, there is often an upper bound $n_{\max}$ beyond which testing will not continue even if all the L_n are between A and B. An SPRT with such a sampling limit is sometimes called a ***truncated SPRT***.

Figure 6.3 illustrates the truncated SPRT assuming $n_{\max} = 75$. The walks take small steps up for correct answers and large steps down for incorrect answers. In 50 samples with $\theta = 0.9$, 44 are correctly scored as masters, 2 are deemed to need remediation, and 4 reached the limit of 75 tests. In 50 samples with $\theta = 0.75$, 44 are correctly scored, 1 is wrongly thought to have mastered the material and 5 reached the testing limit. For undecided cases, the ties are usually broken by treating $\log(L_n)$ as if it had crossed the nearer of boundaries $\log(A)$ and $\log(B)$.

The average number of questions asked in this small example was 33.28 for the students with mastery and 35.64 for those needing remediation. The choice of parameters A, B and $n_{\max}$ involves tradeoffs between the costs of both kinds of errors and the cost of continued testing. For instance, time spent testing is time that could have been spent on the next online lesson instead. One could also choose θ_R and θ_M farther apart which would speed up the testing while creating a larger range (θ_R, θ_M) of abilities that might lead to a student being scored either way.

Self-reinforcing random walks

We can extend the random walk model by letting the distribution of $\boldsymbol{Z}_t$ change at each step. The simplest example is Pólya's urn process. When the process begins, there is an urn containing one black ball and one red ball. At each step, one ball chosen uniformly at random from those in the urn is removed. Then that ball is placed back into the urn, along with one more ball of the same color, to complete the step. Pólya's urn process is a self-reinforcing random walk.

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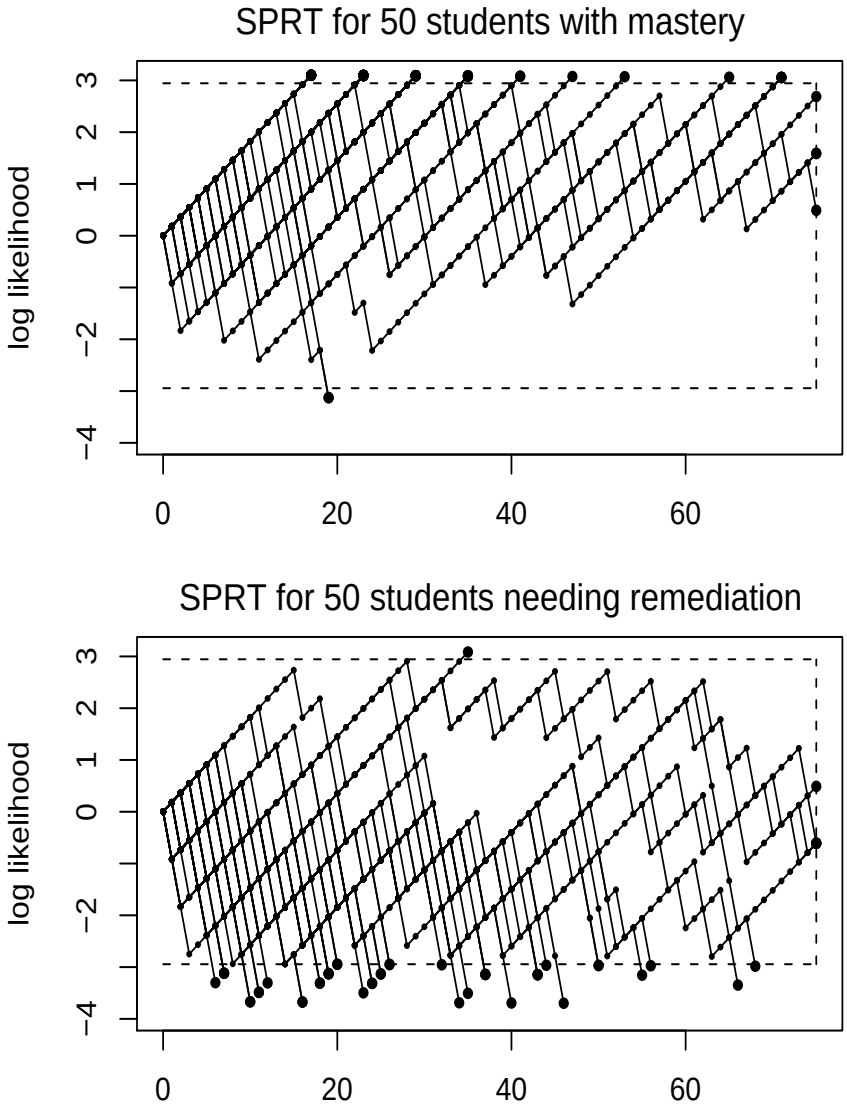


Figure 6.3: This figure shows the progress of the SPRT example from the text. The log likelihood ratio is plotted against the number of questions answered. The dashed lines depict the limits $A = 19$, $B = 1/19$ and $n_{\max} = 75$, The top shows 50 simulated outcomes for students who have mastered the subject. The bottom shows 50 simulated outcomes for students who need remediation.

We can represent the state of the process as $\boldsymbol{X}_t = (R_t, B_t)$ where R_t and B_t are the numbers of red and black balls at time t. The starting point is

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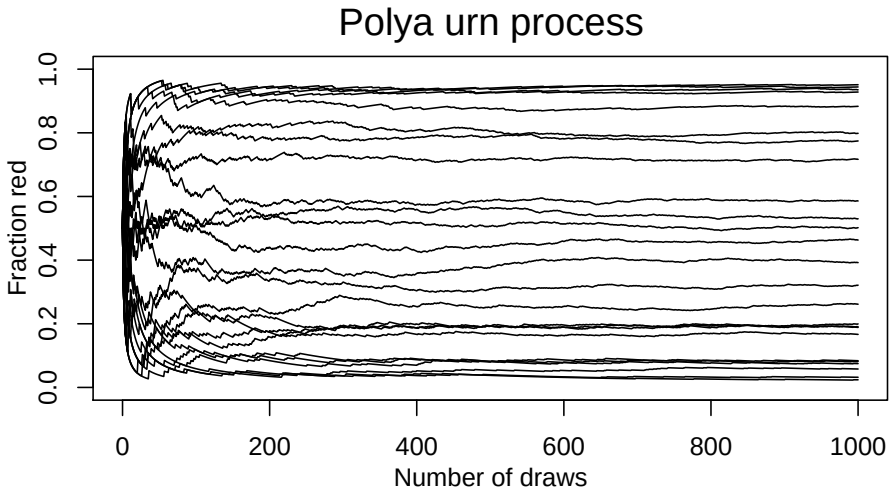


Figure 6.4: This figure shows 25 realizations of the first 1000 draws in Pólya's urn process.

$\boldsymbol{X}_0 = (1,1)$ and $\boldsymbol{X}_{t+1} = \boldsymbol{X}_t + \boldsymbol{Z}_t$ where

$$\boldsymbol{Z}_t = \begin{cases} (1,0), & \text{with probability} \quad R_t/(R_t+B_t), \\ (0,1), & \text{with probability} \quad B_t/(R_t+B_t). \end{cases}$$

The interesting quantity is the proportion $Y_t = R_t/(R_t+B_t)$ of red balls in the urn.

Figure 6.4 shows 25 realizations of the Pólya urn process, taken out to 1000 draws. What we observe is that each run seems to settle down. But they all seem to settle down to different values. Pólya proved that each run converges to a value Y_∞, and that Y_∞ itself is random, from the $\mathbf{U}(0,1)$ distribution. Monte Carlo sampling lets us see how fast this effect takes place, and explore variations of the model.

The Pólya urn process has been used to model the effects of market power in economic competition. Suppose that there are two competing technologies for a newly developed consumer electronics product. Then if new customers tend to buy what their friends have, something like an urn model may hold for the number of customers with each type of product. Under this model the two products are completely identical. Yet they don't end up with equal market shares. Instead, an advantage won early, purely by chance, remains. Naturally, this produces large incentives to be the first mover and get an early advantage, instead of leaving the result to chance.

Slight changes of the urn model can lead to winner-take-all effects. Perhaps

$$\boldsymbol{Z}_t = \begin{cases} (1,0), & \text{with probability} \quad R_t^\alpha/(R_t^\alpha+B_t^\alpha), \\ (0,1), & \text{with probability} \quad B_t^\alpha/(R_t^\alpha+B_t^\alpha), \end{cases}$$

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for some $\alpha > 1$. For example, the product with greater market share might get more business partners or have lower costs and then add customers at a faster than proportional rate. In this case, one firm will end up with all of the market. See Exercise 6.4 for an example where one product is better but the network effects give the lesser product a chance of winning the whole market.

6.3 Gaussian processes

A Gaussian process is one where the finite dimensional distributions are all multivariate normal. Just as a multivariate Gaussian distribution is determined by its mean and variance, a Gaussian process $\{X(t) \mid t \in \mathcal{T}\}$, is defined by two functions, a mean function $\mu(t) = \mathbb{E}(X(t))$ for $t \in \mathcal{T}$ and a covariance function $\Sigma(t,s) = \mathrm{Cov}(X(t), X(s))$ defined for pairs $t, s \in \mathcal{T}$. The finite dimensional distributions of the Gaussian process are

$$\begin{pmatrix} X(t_1) \\ X(t_2) \\ \vdots \\ X(t_m) \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu(t_1) \\ \mu(t_2) \\ \vdots \\ \mu(t_m) \end{pmatrix}, \begin{pmatrix} \Sigma(t_1,t_1) & \Sigma(t_1,t_2) & \cdots & \Sigma(t_1,t_m) \\ \Sigma(t_2,t_1) & \Sigma(t_2,t_2) & \cdots & \Sigma(t_2,t_m) \\ \vdots & \vdots & \ddots & \vdots \\ \Sigma(t_m,t_1) & \Sigma(t_m,t_2) & \cdots & \Sigma(t_m,t_m) \end{pmatrix} \right).$$

We use t instead of $\boldsymbol{t}$ to emphasize the common case, $\mathcal{T} \subset \mathbb{R}$.

While the mean of $X(t)$ can be given by any function $\mu : \mathcal{T} \to \mathbb{R}$, the covariance function Σ has to obey some constraints. It is clear that the covariance matrix of any finite dimensional distribution of $X(t)$ must be positive semi-definite. In fact, that is all we need. Any function Σ for which

$$\sum_{i=1}^{m}\sum_{j=1}^{m} \alpha_i \alpha_j \Sigma(t_i, t_j) \geqslant 0$$

always holds, for $m \geqslant 1$, $t_i \in \mathcal{T}$ and $\alpha_i \in \mathbb{R}$ is a valid covariance function.

The process $X(\cdot)$ is ***stationary*** if $X(\cdot + \Delta)$ has the same distribution for all fixed Δ. For Gaussian processes, stationarity is equivalent to $\mu(t+\Delta) = \mu(t)$ and $\Sigma(t+\Delta, s+\Delta) = \Sigma(t,s)$. Usually $\mathcal{T}$ contains a point 0 and then stationarity means that $\mu(t) = \mu(0)$ and $\Sigma(t,s) = \Sigma(t-s, 0)$ for all $s, t \in \mathcal{T}$.

Standard ***Brownian motion*** is a Gaussian process on $\mathcal{T} = [0, \infty)$. We write it as $B(t)$, or sometimes B_t, and it is defined by the following three properties:

BM-1: $B(0) = 0$.

BM-2: Independent increments: for $0 = t_0 < t_1 < \cdots < t_m$, $B(t_i) - B(t_{i-1}) \sim \mathcal{N}(0, t_i - t_{i-1})$ independently for $i = 1, \ldots, m$.

BM-3: $B(t)$ is continuous on $[0, \infty)$ with probability 1.

We will make considerable use of BM-2, the independent increments property.

Brownian motion is named after the botanist Robert Brown who observed the motion of pollen in water. Standard Brownian motion is also called the

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Wiener process in honor of Norbert Wiener, who proved (Wiener, 1923) that a process does indeed exist with continuous sample paths and the given finite dimensional distributions. While Brownian paths are continuous it is also known that, with probability one, the sample path of Brownian motion is not differentiable anywhere. There are some references on Brownian motion in the end notes.

It is easy to see that $\mu(t) = 0$ and $\Sigma(t,s) = \min(t,s)$ for standard Brownian motion. In particular $B(t) \sim \mathcal{N}(0,t)$, so Brownian motion is not stationary.

We write $B(\cdot) \sim \mathsf{BM}(0,1)$ for a process $B(t)$ that follows standard Brownian motion. When $B(\cdot) \sim \mathsf{BM}(0,1)$ then the process $X(t) = \delta t + \sigma B(t)$ is Brownian motion with drift $\delta \in \mathbb{R}$ and variance $\sigma^2 > 0$, which we denote by $X(\cdot) \sim \mathsf{BM}(\delta, \sigma^2)$. This process has $\mu(t) = \delta t$ and $\Sigma(t,s) = \sigma^2 \min(t,s)$.

It is simple to add a drift and change the variance of Brownian motion. Specifically, to sample $X(\cdot) \sim \mathsf{BM}(\delta, \sigma^2)$ on $[0,T]$ we may use $X(t) = \delta t + \sigma\sqrt{T}B(t/T)$ for $B(\cdot) \sim \mathsf{BM}(0,1)$ on $[0,1]$. As a result we can focus on sampling standard Brownian motion over $[0,1]$.

To sample Brownian motion at any given list of points $t_1 < t_2 < \cdots < t_m$ we can work directly from the definition:

$$\begin{aligned} B(t_1) &= \sqrt{t_1}Z_1, \quad \text{and then} \\ B(t_j) &= B(t_{j-1}) + \sqrt{t_j - t_{j-1}}Z_j, \quad j = 2, \ldots, m, \end{aligned} \tag{6.4}$$

for independent $Z_j \sim \mathcal{N}(0,1)$. In matrix terms we use

$$\begin{pmatrix} B(t_1) \\ B(t_2) \\ \vdots \\ B(t_m) \end{pmatrix} = \begin{pmatrix} \sqrt{t_1} & 0 & \cdots & 0 \\ \sqrt{t_1} & \sqrt{t_2 - t_1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \sqrt{t_1} & \sqrt{t_2 - t_1} & \cdots & \sqrt{t_m - t_{m-1}} \end{pmatrix} \begin{pmatrix} Z_1 \\ Z_2 \\ \vdots \\ Z_m \end{pmatrix}. \tag{6.5}$$

A direct multiplication shows that the matrix in (6.5) is the Cholesky factor of

$$\mathrm{Var}\left(\begin{pmatrix} B(t_1) \\ \vdots \\ B(t_m) \end{pmatrix}\right) = \big(\min(t_j, t_k)\big)_{1 \leqslant j,k \leqslant m} = \begin{pmatrix} t_1 & t_1 & \cdots & t_1 \\ t_1 & t_2 & \cdots & t_2 \\ \vdots & \vdots & \ddots & \vdots \\ t_1 & t_2 & \cdots & t_m \end{pmatrix}. \tag{6.6}$$

The Cholesky connection gives insight, but computationally it is faster to take the cumulative sums of increments in (6.4) than to take the matrix multiplication in equation (6.5) literally.

Standard Brownian motion is perhaps the most important Gaussian process. Section 6.4 is devoted to specialized ways of sampling it, along with related processes: geometric Brownian motion, and the Brownian bridge.

Brownian motion paths are continuous but their non-differentiability is undesirable when we want a model for smooth random functions. Many physical quantities, such as air temperature, CO_2 levels, thickness of a cable, or the

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strain level in a solid, are smoother than Brownian motion. Brownian motion is also nonstationary, i.e., the distribution of $X(t)$ depends on t. Next we consider some alternative Gaussian processes with either smoothness, stationarity, or both.

To get a smooth path, we need $X(t)$ to be close to $X(t+h)$ for small h. A rigorous discussion of differentiability of paths would take us away from Monte Carlo ideas. Instead, we look intuitively at the problem and see that smoothness of $X(\cdot)$ depends on smoothness of $\mu(\cdot)$ and $\Sigma(\cdot,\cdot)$. We suppose that $\mathcal{T}$ is $\mathbb{R}$ or a subinterval of $\mathbb{R}$. We begin by considering, for small $h > 0$, the divided difference

$$\Delta_h(X,t) \equiv \frac{X(t+h) - X(t)}{h}.$$

From properties of the multivariate normal distribution, we find that

$$\Delta_h(X,t) \sim \mathcal{N}\Big(\frac{\mu(t+h)-\mu(t)}{h}, \frac{\Sigma(t+h,t+h) - 2\Sigma(t+h,t) + \Sigma(t,t)}{h^2}\Big).$$

If we informally let $h \to 0$, then we anticipate $X'(t) \sim \mathcal{N}(\mu'(t), \Sigma_{1,1}(t,t))$ where $\Sigma_{1,1} = \partial^2\Sigma(t,s)/\partial t\partial s$. If $\Sigma_{1,1}(t,t)$ and $\mu'(t)$ exist, then $\mathbb{E}((Y - \Delta_h(X,t))^2) \to 0$ as $h \to 0$, for a random variable $Y \sim \mathcal{N}(\mu'(t), \Sigma_{1,1}(t,t))$. For a full discussion of this mean square differentiability see Gikhman and Skorokhod (1996). To get k derivatives in X we require the k'th derivative of μ to exist, along with $\Sigma_{k,k}(t,s)$ (the mixed partial of Σ taken k times with respect to each component) when evaluated with $t = s$.

One application of Gaussian processes is to provide uncertainty estimates for interpolation of smooth functions. Suppose that we obtain $Y_j = f(t_j)$ for $j = 1,\dots,k$. Under the model that $f(t)$ is the realization of a Gaussian process, we can predict $f(t_0)$ by $\hat{f}(t_0) = \mathbb{E}(f(t_0) \mid f(t_1),\dots,f(t_k))$, using the formulas for conditioning in the $k+1$ dimensional Gaussian distribution. By definition $\hat{f}(t_j) = f(t_j)$ for $j = 1,\dots,k$, and so $\hat{f}(\cdot)$ interpolates the known data. The Gaussian model also provides a variance estimate, $\mathrm{Var}(f(t_0) \mid f(t_1),\dots,f(t_k))$.

Modeling a given deterministic function $f(\cdot)$ as a Gaussian process yields a Bayesian numerical analysis. It is usually applied to functions on $[0,1]^d$ for $d > 1$, but we will use the $d = 1$ setup to illustrate Gaussian processes, and then remark briefly on the extension to $d > 1$.

Example 6.1 (Exponential covariance)**.** The Gaussian process $X(t)$ with exponential covariance has expectation $\mu(t) = 0$ and covariance

$$\Sigma(s,t) = \sigma^2 \exp(-\theta|s-t|), \quad \text{where } \sigma > 0 \text{ and } \theta > 0.$$

This process is stationary. The sample paths are continuous, but not smooth. We can get a different mean function $\widetilde{\mu}(\cdot)$ by taking $\widetilde{\mu}(t) + X(t)$.

Example 6.2 (Gaussian covariance)**.** The Gaussian process $X(t)$ with Gaussian covariance (also called the ***squared exponential*** covariance) has expectation $\mu(t) = 0$ and covariance

$$\Sigma(s,t) = \sigma^2 \exp(-\theta(s-t)^2), \quad \text{where } \sigma > 0 \text{ and } \theta > 0.$$

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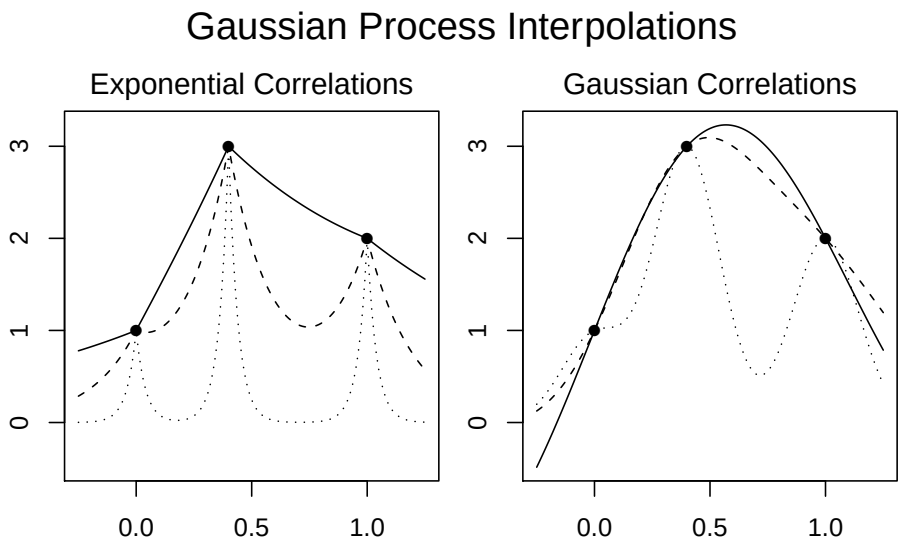


Figure 6.5: This figure shows interpolation at three points using the Gaussian process model, with exponential correlations (left panel) and Gaussian correlations (right panel). Three values of θ are used: 1 (solid), 5 (dashed), and 25 (dotted).

For fixed t and varying s, $\Sigma(s,t)$ is proportional to the density for $s \sim \mathcal{N}(t, 1/(2\theta))$. The process is stationary. The sample paths are very smooth. We can get a different mean function $\widetilde{\mu}(\cdot)$ by taking $\widetilde{\mu}(t) + X(t)$.

Figure 6.5 shows Gaussian process interpolations for both exponential and Gaussian correlation functions. The given data are $f(0) = 1$, $f(0.4) = 3$ and $f(1) = 2$. The interpolations are taken at points separated by 0.01 from -0.25 to 1.25. The interpolations do not depend on σ because σ cancels out of the Gaussian conditional expectation formula. When θ is large, the correlations between points drops off quickly as $|t - t'|$ increases. Absent even tiny correlations with any observed values, the predictions are pulled towards the mean function, in this example $\mu(t) = 0$. For very large θ both predictions come very close to 0 except in small neighborhoods of observed values. When θ is small, the correlation between points t and t' drops off slowly as $|t - t'|$ increases. The predictions for $\theta \ll 1$ (not shown) look very close to those for $\theta = 1$ in the range $[0, 1]$. The exponential model yields nearly piecewise linear interpolation for small θ while the Gaussian model interpolations are much smoother.

The interpolations in Figure 6.5 are made without using any Monte Carlo sampling. The Gaussian process model allows us to do more than just find posterior means and covariances of $f(\cdot)$. Figure 6.6 shows the results of 1000 realizations of Gaussian process $f(\cdot)$ with $\mu(t) = 0$ and $\Sigma(s,t) = \exp(-|s-t|^2)$ conditionally on observing $f(0) = 1$, $f(0.4) = 3$ and $f(1) = 2$. From these simulations we can compute the distribution of the maximizer x^* of $f(\cdot)$. The

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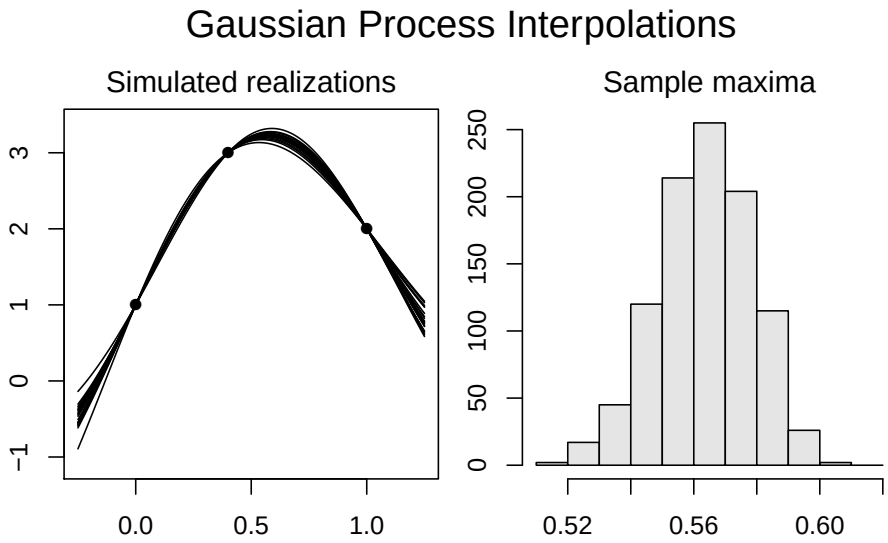


Figure 6.6: The left panel shows 20 realizations of the Gaussian process with the Gaussian covariance, $\theta = 1$ and $\sigma^2 = 1$, conditioned on passing through the three solid points. The right panel shows the locations of the maxima for 1000 such realizations.

second plot in Figure 6.6 shows that the posterior

The Gaussian covariance yields sample paths that may be much smoother than the process we wish to model. The Matérn class provides covariances with smoothness between that of the exponential and the Gaussian.

Example 6.3 (Matérn covariances)**.** The Matérn class of covariances are governed by a smoothness parameter ν. For general $\nu > 0$, the covariance $\Sigma(s,t;\nu)$ is described in terms of a Bessel function, but for $\nu = m + 1/2$ with integer $m \geqslant 0$, the form simplifies. The first 4 of these special cases are

$$\begin{aligned}
\Sigma(s,t;1/2) &= \sigma^2 \exp(-\theta|s-t|),\\
\Sigma(s,t;3/2) &= \sigma^2 \exp(-\theta|s-t|)(1+\theta|s-t|),\\
\Sigma(s,t;5/2) &= \sigma^2 \exp(-\theta|s-t|)\Big(1+\theta|s-t|+\frac{1}{3}\theta^2|s-t|^2\Big), \quad \text{and}\\
\Sigma(s,t;7/2) &= \sigma^2 \exp(-\theta|s-t|)\Big(1+\theta|s-t|+\frac{2}{5}\theta^2|s-t|^2)+\frac{1}{15}\theta^3|s-t|^3\Big),
\end{aligned}$$

where $\sigma > 0$ and $\theta > 0$.

The Matérn covariances include the exponential one, with $\nu = 1/2$ as well as the Gaussian covariance, in the limit as $\nu \to \infty$. Realizations of the process with $\nu = m + 1/2$ have m derivatives. Figure 6.7 shows sample realizations of the Matérn process. Those with higher ν are visibly smoother. Larger θ makes the realizations have greater local oscillations.

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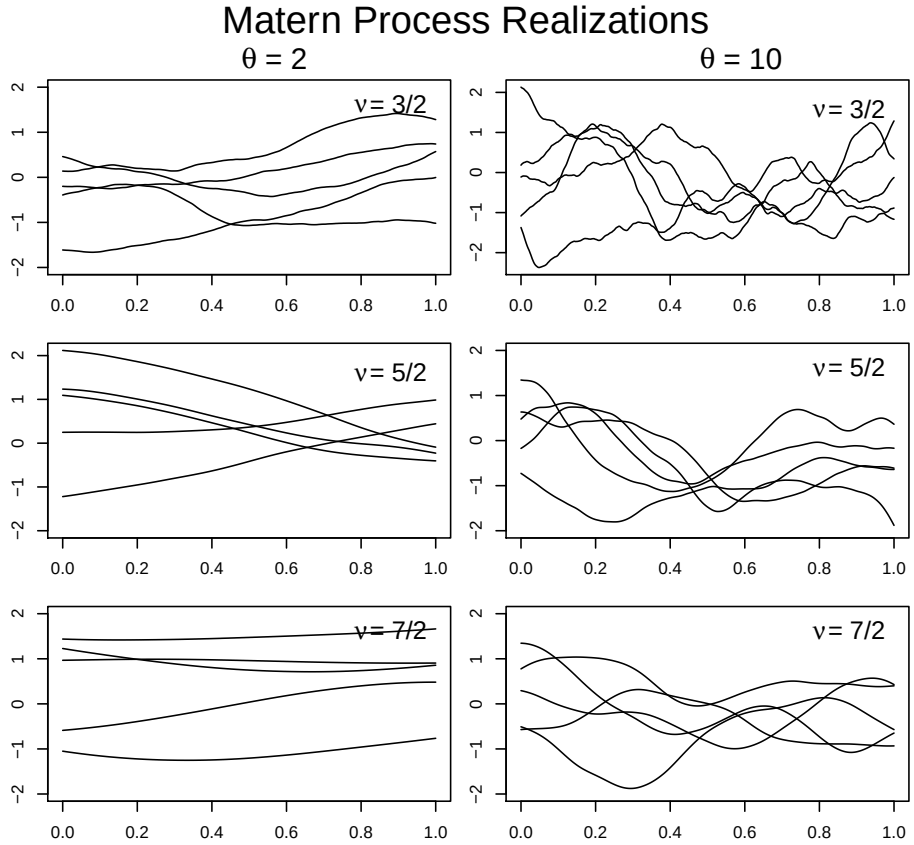


Figure 6.7: This figure shows 5 realizations of the Matérn process on $[0,1]$ for each $\nu \in \{3/2, 5/2, 7/2\}$ and $\theta \in \{2, 5\}$. Every process had $\sigma^2 = 1$.

Example 6.4 (Cubic correlation)**.** The Gaussian process $X(t)$, for $0 \leqslant t \leqslant 1$, with the cubic correlation has expectation $\mu(t) = 0$ and covariance

$$\Sigma(s,t) = \sigma^2\Big(1 - \frac{3(1-\rho)}{2+\gamma}(s-t)^2 + \frac{(1-\rho)(1-\gamma)}{2+\gamma}|s-t|^3\Big)$$

for parameters $\rho, \gamma \in [0,1]$, with $\rho \geqslant (5\gamma^2 + 8\gamma - 1)/(\gamma^2 + 4\gamma + 7)$. This parameters are $\rho = \mathrm{Corr}(X(0), X(1))$ and $\gamma = \mathrm{Corr}(X'(0), X'(1))$. This process was studied by Mitchell et al. (1990). The interpolations from this model are cubic splines. The lower bound on ρ is necessary to ensure a valid covariance.

Prediction and sampling for $d > 1$ work by the same principles as for $d = 1$. It is however more difficult to specify a covariance. In Bayesian numerical

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analysis it is common to take a covariance of product form

$$\Sigma(\boldsymbol{t},\boldsymbol{s}) = \sigma^2 \prod_{j=1}^{d} R_j(t_j, s_j)$$

where $\sigma^2 > 0$ is a variance and each $R_j(\cdot,\cdot)$ is a one dimensional correlation function, stationary or not. In geostatistics it is sometimes preferable to use an ***isotropic covariance***

$$\Sigma(\boldsymbol{t},\boldsymbol{s}) = \sigma^2 R(\|\boldsymbol{t}-\boldsymbol{s}\|)$$

for a correlation function R on $[0,\infty)$. Valid correlation functions include $R(h) = \exp(-\theta h)$ and $R(h) = \exp(-\theta h^2)$. It is also possible to use the Matérn correlations, taking $\Sigma(\boldsymbol{t},\boldsymbol{s}) = \Sigma(0, \|\boldsymbol{t}-\boldsymbol{s}\|; \nu)$.

Unless there is some special structure in Σ, sampling a Gaussian process requires $O(m^3)$ computation to get a matrix square root of Σ. Then it requires $O(nm^2)$ work to generate n sample paths at m points.

The very smooth covariances, like the Gaussian one, often yield matrices Σ that are nearly singular. The singular value decomposition approach to factoring Σ, described in Chapter 5 is then a very good choice. Another technique for near singular Σ is to change the model to incorporate a ***nugget effect***, replacing $\Sigma(t,s)$ by $\Sigma_\epsilon(t,s) = \Sigma(t,s) + \epsilon I_m$ for some small $\epsilon > 0$. If Σ is a valid covariance, then Σ_ϵ is too, with $\Sigma_\epsilon(t_j,t_k) = \mathrm{Cov}(X(t_j)+\varepsilon_j, X(t_k)+\varepsilon_k)$ where the ε's are independent $\mathcal{N}(0,\epsilon)$ random variables which may be thought of as jitter, measurement error or numerical noise. In geostatistics, nuggets might represent very localized fluctuations in the ore content of rock.

6.4 Detailed simulation of Brownian motion

Here we look closely at two alternative strategies for sampling Brownian motion. One strategy uses the principal components factorization, and the other generates points of $B(t)$ one at a time in arbitrary order, using the connection between Brownian motion and the Brownian bridge.

In the principal components method, the matrix on the right side of (6.6) is written in its spectral decomposition as $P\Lambda P^{\mathsf{T}}$ where $\Lambda = \mathrm{diag}(\lambda_1,\ldots,\lambda_m)$ has the eigenvalues in descending order and the columns of P are the corresponding eigenvectors. Then one samples $B(t_1,\ldots,t_m) = P\Lambda^{1/2}\boldsymbol{Z}$ where $\boldsymbol{Z} \sim \mathcal{N}(0, I)$. The variance matrix can be factored numerically. Factoring Σ takes work that is $O(m^3)$ but need only be done once.

In the special case where $t_j = jT/m$ there is a closed form for the eigenvalues and eigenvectors of the covariance matrix due to Akesson and Lehoczky (1998). They show that component i of the j'th eigenvector of the covariance matrix is

$$e_j^{(m)}\left(\frac{iT}{m}\right) = \frac{2}{\sqrt{2m+1}} \sin\Big(\frac{2i-1}{2m+1} j\pi\Big), \quad i = 1,\ldots,m$$

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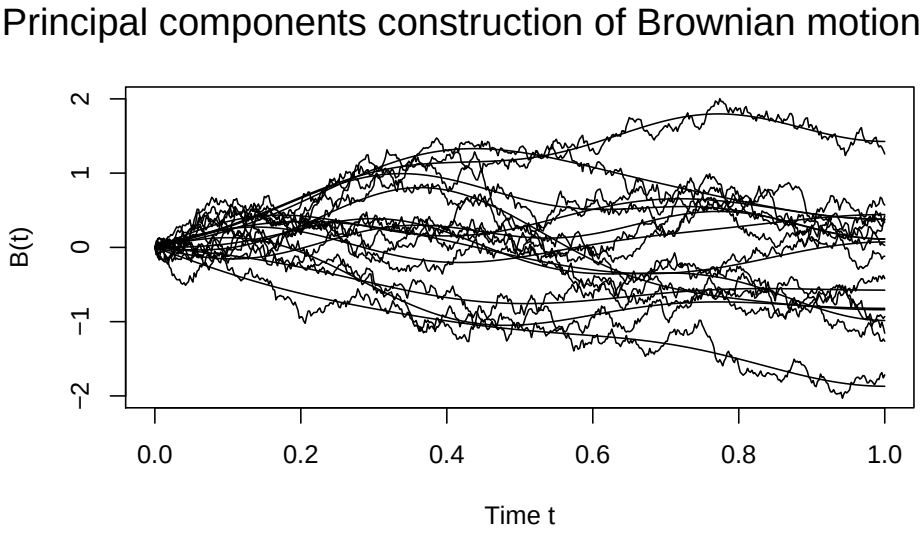


Figure 6.8: This figure shows 12 sample paths of Brownian motion at $m = 500$ equispaced points. Superimposed are the corresponding curves from the first 5 principal components.

and that the j'th eigenvalue is

$$\lambda_j^{(m)} = \frac{T}{4m} \Big/ \sin^2\Big(\frac{2j-1}{2m+1}\frac{\pi}{2}\Big).$$

This leads to the method

$$\begin{aligned} B\left(\frac{iT}{m}\right) &= \sum_{j=1}^{m} Z_j \sqrt{\lambda_j} e_j^{(m)}\left(\frac{iT}{m}\right) \\ &= \sqrt{\frac{T}{2m^2+m}} \sum_{j=1}^{m} Z_j \sin\left(\frac{2i-1}{2m+1} j\pi\right) \Big/ \sin\left(\frac{2j-1}{2m+1}\frac{\pi}{2}\right) \end{aligned} \tag{6.7}$$

for $i = 1, \ldots, m$ using independent $Z_j \sim \mathcal{N}(0,1)$.

The principal components construction offers no advantage for plain Monte Carlo sampling. In fact it is somewhat slower than the direct method. It requires $O(m^2)$ work to generate $B(iT/m)$ even if the square root and all the sines have been precomputed. Direct sampling takes only $O(m)$ work to sum the increments.

The principal components method can offer an advantage when variance reduction techniques, such as importance sampling, stratification or quasi-Monte Carlo are applied to the first few principal components.

Figure 6.8 shows 12 realizations of Brownian motion generated at 500 equispaced points by the principal components method (6.7). The smooth curves

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are generated by truncating (6.7) at $m = 5$. The remaining 495 principal components combine to add the small scale fluctuations around each of the smooth curves. While each curve was sampled from Brownian motion, the 12 curves were not quite independent. Instead the values of the first principal component coefficient Z_1 were generated by stratified sampling as in Exercise 8.7. Stratification is one of the methods alluded to above to increase accuracy. Here it helps to reduce the overlap among the plotted curves.

The principal components construction has a meaningful limit as the sampling rate $m \to \infty$. It is

$$B(t) = \frac{\sqrt{2}}{\pi} \sum_{j=0}^{\infty} Z_j \Big(\frac{2}{2j+1} \Big) \sin\Big(\frac{2j+1}{2} \pi t \Big), \tag{6.8}$$

for $\mathsf{BM}(0,1)$ on $[0,1]$, where $Z_j \sim \mathcal{N}(0,1)$ are independent. Note that the summation starts at $j = 0$. Equation (6.8) allows us to approximate $B(t)$ by truncating the sum at a large value J. The representation (6.8) is known as a Karhunen-Loéve expansion. Adler and Taylor (2007, Chapter 3) give more information on these expansions.

Brownian bridge

We can sample a Gaussian process in any order we like, but we might have to pay an $O(m^3)$ price to get the m'th point. That is expensive, especially if we want to change our mind about the sampling order as the sampling proceeds.

Brownian motion has a Markov property that simplifies conditional sampling. To describe the Markov property, consider sample times $0 = t_0 < t_1 < \cdots < t_m$. The distribution of $B(t_j)$ given $B(t_0), \ldots, B(t_{j-1})$ is the same as that of $B(t_j)$ given just $B(t_{j-1})$, by the independent increments property. Similarly the distribution of $B(t_j)$ given $B(t_{j+1}), \ldots, B(t_m)$ is the same as that of $B(t_j)$ given just $B(t_{j+1})$.

To generate $B(t_j)$ in arbitrary order, we use the following more general result:

$$\mathbb{P}\big(B(t_j) < b \mid B(t_k), k \neq j\big) = \mathbb{P}\big(B(t_j) < b \mid B(t_{j-1}), B(t_{j+1})\big), \tag{6.9}$$

for $0 < j < m$. In other words, the distribution of $B(t_j)$ given some of the past and some of the future depends only on the most recent past and the nearest future.

In this section we will use (6.9) without proof. It applies more generally than for Brownian motion. See Proposition 6.1 in §6.9, which Exercise 6.7 asks you to prove.

Suppose that we want to sample $B(s_1), \ldots, B(s_m)$ for arbitrary and distinct $s_j > 0$. We write s_j instead of t_j because the latter were assumed to be in increasing order and the s_j need not be. We might, for example sample Brownian motion on $[0, T]$ taking $s_1 = T$, $s_2 = T/2$, $s_3 = T/4$, $s_4 = 3T/4$ and so on, putting each new point in the middle of the largest interval left open by the

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previous ones. One such order is to follow s_1 by the nonzero points of the van der Corput sequence (§15.4), all multiplied by T. Or, we might take the points $j2^{-k}T$ for increasing $k \geqslant 0$ and at each k, for $j = 1, 3, \ldots, 2^k - 1$.

Sampling the first point is easy, because $B(s_1) \sim \mathcal{N}(0, s_1)$. After that, if we want to generate a value $B(s_j)$ we need to sample conditionally on the already generated values $B(s_1), \ldots, B(s_{j-1})$. This conditional distribution is Gaussian, and so we can sample it using the methods from §5.2. For an arbitrary Gaussian process, we would have to invert a $j-1$ by $j-1$ matrix in order to sample the j'th value. Equation (6.9) allows a great simplification: we only have to condition on at most two other values of the process.

Suppose first that s_j is neither larger than all of $s_1, \ldots, s_{j-1}$, nor smaller than all of them. Then the neighboring points of s_j are

$$\begin{aligned} \ell_j &= \max\{s_k \mid 1 \leqslant k < j,\ s_k < s_j\}, \quad \text{and} \\ r_j &= \min\{\, s_k \mid 1 \leqslant k < j,\ s_k > s_j\}, \end{aligned}$$

and both are well defined.

Now for $0 < \ell < s < r < \infty$ we find

$$B(s) \mid \big(B(\ell),\ B(r)\big) \sim \mathcal{N}\left(B(\ell) + \frac{s-\ell}{r-\ell}(B(r) - B(\ell)),\ \frac{(s-\ell)(r-s)}{r-\ell}\right), \tag{6.10}$$

(see Exercise 6.5), and so we can take

$$B(s_j) = \frac{(r_j - s_j)B(\ell_j) + (s_j - \ell_j)B(r_j)}{r_j - \ell_j} + Z_j\sqrt{\frac{(s_j - \ell_j)(r_j - s_j)}{r_j - \ell_j}}, \tag{6.11}$$

for independent $Z_j \sim \mathcal{N}(0, 1)$.

We have three more cases to consider, depending on which of ℓ_j and r_j are well defined. For $j = 1$, neither ℓ_j nor r_j is well defined, and we simply take $B(s_1) = \sqrt{s_1}Z_1$ for $Z_1 \sim \mathcal{N}(0, 1)$. If ℓ_j is well defined, but r_j is not because $s_j > \max\{s_1, \ldots, s_{j-1}\}$ then we use the independent increments property and take $B(s_j) = B(\ell_j) + \sqrt{s_j - \ell_j}Z_j$ for $Z_j \sim \mathcal{N}(0, 1)$. Finally, if r_j is well defined, but ℓ_j is not because $s_j < \min\{s_1, \ldots, s_{j-1}\}$ then we take $B(s_j) = B(r_j)s_j/r_j + Z_j\sqrt{s_j(r_j - s_j)/r_j}$. This is simply the first case after adjoining $B(0) = 0$ to the process history.

It is possible to merge all four cases into one by adjoining both $B(0) = 0$ and $B(\infty) = 0$ to the process history prior to s_1. See Exercise 6.6. Any finite value could be used for $B(\infty)$, because that point will always get weight zero. In practice however, all four cases have to be carefully considered in the setup steps for the algorithm, and so merging the cases into one does not bring much simplification.

To sample $\mathsf{BM}(0, 1)$ at points $s_1, \ldots, s_m$ arranged in arbitrary order, based on equation (6.11), we may use Algorithms 6.1 and 6.2. The former algorithm is called once to set up parameter values, and is the more complicated of the

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Algorithm 6.1 Precompute Brownian bridge sampling parameters

BB–parameters(m, s **)**

// $s_1, \ldots, s_m$ are distinct positive values in arbitrary order

for $j = 1$ to m **do**
  $u_j \leftarrow \text{argmax}_k\{\, s_k \mid 1 \leqslant k < j \text{ and } s_k < s_j \,\}$ // $u_j \leftarrow 0$ if set is empty
  $v_j \leftarrow \text{argmin}_k\{\, s_k \mid 1 \leqslant k < j \text{ and } s_k > s_j \,\}$ // $v_j \leftarrow 0$ if set is empty
  if $u_j > 0$ **and** $v_j > 0$ **then**
    $\ell_j \leftarrow s[u_j], \quad r_j \leftarrow s[v_j], \quad w_j \leftarrow \sqrt{(s_j - \ell_j)(r_j - s_j)/(r_j - \ell_j)}$
    $a_j \leftarrow (r_j - s_j)/(r_j - \ell_j), \quad b_j \leftarrow (s_j - \ell_j)/(r_j - \ell_j)$
  else if $u_j > 0$ **then**
    $\ell_j \leftarrow s[u_j], \quad a_j \leftarrow 1, \quad b_j \leftarrow 0, \quad w_j \leftarrow \sqrt{s_j - \ell_j}$
  else if $v_j > 0$ **then**
    $r_j \leftarrow s[v_j], \quad a_j \leftarrow 0, \quad b_j \leftarrow s_j/r_j, \quad w_j \leftarrow \sqrt{s_j(r_j - s_j)/r_j}$
  else
    $a_j \leftarrow 0, \quad b_j \leftarrow 0, \quad w_j \leftarrow \sqrt{s_j}$
return u, v, a, b, w

Notes: $s[u]$ is shorthand for s_u, for readability when u is subscripted.
The algorithm can be coded with ℓ and r in place of ℓ_j and r_j.

two. The latter and simpler algorithm is called n times, once for each Brownian motion sample path that we need. A direct implementation of the setup could cost $O(m^2)$. If $m \ll n$ then the setup cost is negligible compared to the $O(mn)$ cost of generating the points.

Brownian bridge sampling is mildly complicated. A strategy for testing whether an implementation of Algorithms 6.1 and 6.2 is correct is given in Exercise 6.8.

An example of the Brownian bridge approach to sampling $\mathsf{BM}(0,1)$ is shown in Figure 6.9. It samples at times $s = 1, 1/2, 1/4, 3/4, 1/8, 5/8, 3/8, 7/8$ then follows up at the remaining points $s = i/512$ for $i = 1, \ldots, 512$, in this case sequentially but we could as well do them in another order. In the early stages of sampling we have a piecewise linear approximation to the process which, depending on the purpose of the simulation, may capture the most important aspects of the path. Some of the early piece-wise linear approximations are shown.

As with principal components, the main reason to favor the Brownian bridge construction is that it may be exploited by variance reduction methods. The Brownian bridge process offers a further opportunity to improve efficiency. Some finance problems require estimation of $\mu = \mathbb{E}(f(B(T/m), B(2T/m), \ldots, B(T)))$ where the function f has a knockout feature making it 0 if any $B(iT/m)$ fall below a threshold τ. We may sample thousands of paths to evaluate μ. If on a given path we first see that $B(T) < \tau$ then we know $f = 0$ for that path without

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Algorithm 6.2 Brownian bridge sampling of $\mathsf{BM}(0,1)$

BMviaBB(m, s, u, v, a, b, w **)**

// Sample at $s_1, \ldots, s_m$ using u, v, a, b, w precomputed by Algorithm 6.1

for $j = 1$ to m **do**
  if $u_j > 0$ **then**
    $B(s_j) \leftarrow B(s_j) + a_j B(s[u_j])$
  if $v_j > 0$ **then**
    $B(s_j) \leftarrow B(s_j) + b_j B(s[v_j])$
return $B(s_1), \ldots, B(s_m)$

having to sample the rest of it.

The reason that this algorithm is called Brownian bridge sampling, is that the conditional distribution of Brownian motion $B(s)$ on $s \in (\ell, r)$ given $B(\ell)$ and $B(r)$, is called the ***Brownian bridge***. It is also known as tied down Brownian motion. Algorithm 6.2 repeatedly samples one point from Brownian bridge processes on a sequence of intervals.

For the standard Brownian bridge process, $\ell = 0$, $r = 1$, and we condition on $B(0) = B(1) = 0$. Let $\widetilde{B}(t)$ be the standard Brownian $B(t)$ motion on $0 \leqslant t \leqslant 1$ conditionally on $B(1) = 0$. Then $\widetilde{B}$ follows the standard Brownian bridge process, denoted $\widetilde{B} \sim \mathsf{BB}(0,1)$. This is a Gaussian process with $\mathbb{E}(\widetilde{B}(t)) = 0$ and $\mathrm{Cov}(\widetilde{B}(s), \widetilde{B}(t)) = \min(s,t)(1 - \max(s,t))$.

There is a chicken and egg relationship between Brownian motion and the Brownian bridge process. Just as we can sample Brownian motion via the Brownian bridge, we can sample the Brownian bridge by sampling Brownian motion. Specifically, if $B \sim \mathsf{BM}(0,1)$ and $\widetilde{B}(t) = B(t) - tB(1)$, then $\widetilde{B} \sim \mathsf{BB}(0,1)$.

The Brownian bridge process is used to describe Brownian paths between any two points. Suppose that $B(t) \sim \mathsf{BM}(\delta, \sigma^2)$ and we know that $B(a)$ and $B(b)$. We sample this process on $[a,b]$ via

$$B(t) = B(a) + \frac{t-a}{b-a}(B(b) - B(a)) + \sigma\sqrt{b-a}\,\widetilde{B}\Big(\frac{t-a}{b-a}\Big), \quad a \leqslant t \leqslant b,$$

for a process $\widetilde{B} \sim \mathsf{BB}(0,1)$. Notice that the drift δ does not play a role in this distribution, though it does affect the conditional distribution of $B(t)$ for $t < a$ or $t > b$.

Geometric Brownian motion

Brownian motion is used as a model for physical objects being buffeted by particles in their environment. The combined effect of a great many small collisions yields a normal distribution by the central limit theorem. A quite similar pattern is typical of stock prices buffeted by incoming market information. Those

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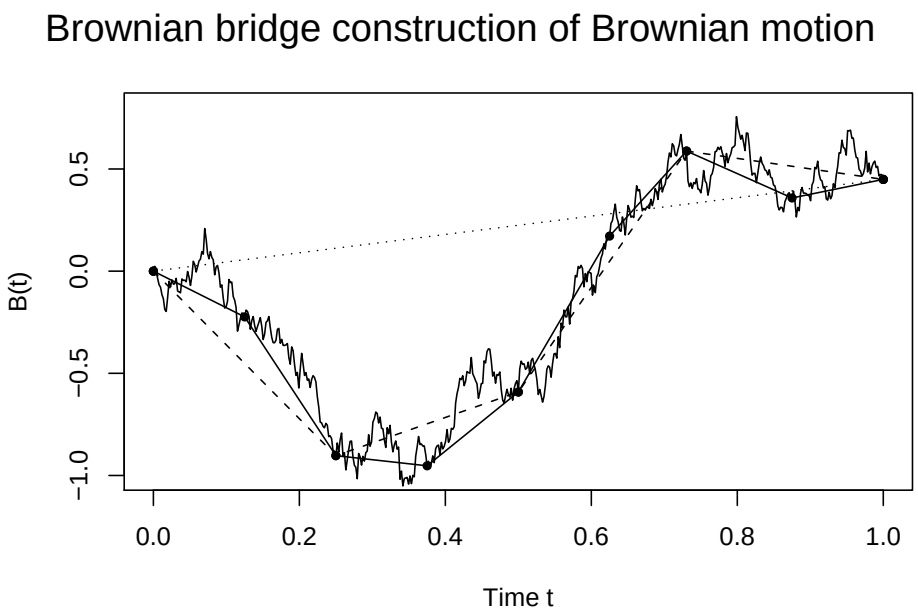


Figure 6.9: This figure shows a sample path of Brownian motion $(s, B(s))$ at $m = 512$ equispaced points. The first 8 points sampled were at $s = 1, 1/2, 1/4, 3/4, 1/8, 5/8, 3/8, 7/8$, in that order. The dotted line connects $(0, 0)$ to $(1, B(1))$. The dashed line connects $(s, B(s))$ for s a multiple of $1/4$. The first 8 points are also connected and shown with a solid circle.

changes are more appropriately modeled as multiplicative, or, additive on the log scale. Those models give lognormal distributions including the geometric Brownian motion model.

Let S_t be the price of a stock or other financial asset at time t. A very basic model for S_t is that

$$\mathrm{d}S_t = \delta S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}B_t \tag{6.12}$$

where $B \sim \mathsf{BM}(0, 1)$ and $S_0 > 0$ is given. Under equation (6.12), the relative change in S_t over an infinitesimal time interval Δ has the $\mathcal{N}(\Delta\delta, \Delta\sigma^2)$ distribution. The process S_t is a geometric Brownian motion, written $\mathsf{GBM}(S_0, \delta, \sigma^2)$. The parameter δ governs the drift, $\sigma > 0$ is the volatility parameter, and $S_0 > 0$ is the starting value.

Equation (6.12) is a stochastic differential equation. It is one of a very few SDEs with a simple closed form solution. We can write

$$S_t = S_0 \exp\bigl((\delta - \sigma^2/2)t + \sigma B_t\bigr). \tag{6.13}$$

If $B_{(\cdot)} \sim \mathsf{BM}(0, 1)$, then $S_{(\cdot)} \sim \mathsf{GBM}(S_0, \delta, \sigma^2)$. Each S_t has a lognormal distribution. Numerical methods for sampling from more general stochastic differential equations are the topic of Section 6.5.

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Given that geometric Brownian motion has small multiplicative fluctuations in value it is not surprising to find that it can be sampled by exponentiating ordinary Brownian motion. What may seem odd at first is that $\sigma^2/2$ has to be subtracted from the drift. Exercise 6.9 asks you to prove it using the Itô formula. Below is a heuristic and more elementary derivation for S_t at one point $t > 0$.

We begin by dividing time $t > 0$ into N time steps of size $\Delta = t/N$ where N is very large. For the first small step, we have S_Δ approximately distributed as $S_0(1+\mathcal{N}(\delta\Delta, \sigma^2\Delta))$. Suppose that Δ is small enough that the multiplicative factor has probability much smaller than $1/N$ of being negative. Then after N such steps, we have probably not multipled by any negative factors and then S_t is roughly

$$\begin{aligned}
&S_0 \prod_{i=1}^{N} \Big(1 + \delta t/N + Z_i\sigma\sqrt{t/N}\Big) \\
&= S_0 \exp\left(\sum_{i=1}^{N} \log(1 + \delta t/N + Z_i\sigma\sqrt{t/N})\right) \\
&\doteq S_0 \exp\left(\sum_{i=1}^{N} \delta t/N + Z_i\sigma\sqrt{t/N} - \frac{1}{2}\Big(\delta t/N + Z_i\sigma\sqrt{t/N}\Big)^2\right) \\
&\doteq S_0 \exp\left(\sum_{i=1}^{N} \delta t/N + Z_i\sigma\sqrt{t/N} - \frac{1}{2}(Z_i^2\sigma^2 t/N)\right)
\end{aligned}$$

for independent $Z_i \sim \mathcal{N}(0,1)$. Now $\sum_{i=1}^N Z_i\sigma\sqrt{t/N} \sim \mathcal{N}(0, t\sigma^2)$ and $\sum_{i=1}^N Z_i^2/N$ is close to 1 by the law of large numbers. As a result S_t is approximately distributed as $S_0 \exp(\mathcal{N}((\delta - \sigma^2/2)t, t\sigma^2)$.

In view of equation (6.13), all the methods for sampling Brownian motion can be applied directly to sampling geometric Brownian motion. We replace the drift δ by $\delta - \sigma^2/2$, generate Brownian motion, and exponentiate the result.

Example 6.5 (Path dependent options)**.** Monte Carlo methods based on geometric Brownian motion are an important technique for valuing path dependent financial options. Here path dependent means that the value of the option depends not only on the asset price at expiration, but also on how it got there.

According to Hull (2008), options are used for three purposes: to hedge against unacceptable risk, to speculate on future prices, and for arbitrage. We will look at an example of the first type.

Consider an airline that is concerned about the price of fuel over the next twelve months. Let the price of fuel at time t be S_t. We pick our units so that $S_0 = 1$, and measure the passage of time in years, taking the present to be time $t = 0$. Suppose that prices $S_t > 1.1$ are problematic for the airline. The airline can hedge this risk by buying an option that pays

$$f(S_{(\cdot)}) = \max\left(0, \frac{1}{12}\sum_{j=1}^{12} S_{j/12} - K\right), \tag{6.14}$$

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where $K = 1.1$, at the end of the year. If the average price goes too high for the airline, then they collect on the option to offset their high costs. If the average price is below the strike price K, then they collect nothing.

This option is called an Asian call option. It is a call option because it is equivalent to having the right, though not the obligation, to buy fuel at an average price of K. By contrast, an Asian put option pays off $\max\bigl(0, K - \frac{1}{12}\sum_{j=1}^{12} S_{j/12}\bigr)$, and it might interest a seller of fuel. The put is equivalent to the right, but not the obligation, to sell at an average price of K. These options are traded globally, not just in Asia. The term Asian refers to their being based on an average price instead of the price at just one time.

The theoretical price for a path dependent option is $e^{-rT}\mathbb{E}(f(S))$, where T is the amount of time until payoff and r is a continuously compounding risk free interest rate. Monte Carlo methods can be used to set a price for the option in (6.14). We repeatedly and independently generate sample paths, compute f on each of them, and average the results. Although this problem arises by considering a geometric Brownian motion, it reduces to a twelve dimensional problem, driven by the distribution of $S_{j/12}$ for $j = 1, \dots, 12$.

6.5 Stochastic differential equations

Brownian motion and geometric Brownian motion are described by stochastic differential equations (SDEs) $\mathrm{d}X_t = \delta\,\mathrm{d}t + \sigma\,\mathrm{d}B_t$ and $\mathrm{d}S_t = S_t\delta\,\mathrm{d}t + S_t\sigma\,\mathrm{d}B_t$ respectively, where $B_t \sim \mathsf{BM}(0,1)$. More general SDEs take the form

$$\mathrm{d}X_t = a(t, X_t)\,\mathrm{d}t + b(t, X_t)\,\mathrm{d}B_t \tag{6.15}$$

for a real-valued ***drift coefficient*** $a(\cdot,\cdot)$ and ***diffusion coefficient*** $b(\cdot,\cdot)$. An interpretation of (6.15) is that $X_{t+\mathrm{d}t} \doteq X_t + a(t, X_t)\mathrm{d}t + b(t, X_t)(B_{t+\mathrm{d}t} - B_t)$ for infinitesimal $\mathrm{d}t$. SDEs arise often in finance and the physical sciences.

In a ***time homogeneous SDE***

$$\mathrm{d}X_t = a(X_t)\,\mathrm{d}t + b(X_t)\,\mathrm{d}B_t. \tag{6.16}$$

Most of our examples are time homogeneous. These processes are also called autonomous: they determine their own drift and diffusion.

The results we discuss for SDEs are based on references listed in the chapter end notes. We give some examples first, before describing how to simulate SDEs.

The ***Ornstein-Uhlenbeck*** process is given by the SDE

$$\mathrm{d}X_t = -\kappa X_t\,\mathrm{d}t + \sigma\,\mathrm{d}B_t, \quad \kappa > 0,\ \sigma > 0. \tag{6.17}$$

The drift term $-\kappa X_t$ causes the process to drift down when $X_t > 0$ and to drift up when $X_t < 0$. That is, it always induces a drift towards zero. This model is used to describe particles in a potential energy well with a minimum at 0.

A generalization of (6.17),

$$\mathrm{d}X_t = \kappa(r - X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t, \quad \kappa > 0,\ \sigma > 0,\ r \in \mathbb{R}$$

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causes the process to revert towards the level r instead of 0. The ***Vasicek model*** for interest rates takes this form, where r is a long term average interest rate. The mean reversion feature is reasonable for interest rates because they tend to remain in a relatively narrow band for long time periods.

A difficulty with the Vasicek model of interest rates is that it allows $X_t < 0$. The ***Cox-Ingersoll-Ross*** (CIR) model has SDE

$$\mathrm{d}X_t = \kappa(r - X_t)\,\mathrm{d}t + \sigma\sqrt{X_t}\,\mathrm{d}B_t, \quad \kappa > 0,\ \sigma > 0,\ r > 0, \tag{6.18}$$

starting at $X_0 > 0$. The $\sqrt{X_t}$ factor in the local standard deviation causes the drift to dominate the diffusion when X_t gets closer to zero, and keep the process positive. The diffusion coefficient is not defined for $X_t < 0$. The process never steps into that inappropriate region, though this is not obvious from the definition. This process is an example of a square-root diffusion, which we return to on page 37.

Not every pair of functions $a(\cdot)$ and $b(\cdot)$ will give rise to a reasonable SDE. A function $c(t, x)$ satisfies the ***Lipschitz condition*** if

$$|c(t, x) - c(t, x')| \leqslant K|x - x'|, \quad \text{for some } K < \infty \tag{6.19}$$

and it satisfies the ***linear growth bound*** if

$$c(t, x)^2 \leqslant K(1 + x^2), \quad \text{for some } K < \infty. \tag{6.20}$$

An SDE satisfies the ***standard conditions*** if the drift and diffusion coefficients each satisfy both the Lipschitz condition and the linear growth bound.

The linear growth bound allows for X_t to grow proportionally to itself, which corresponds to an exponential growth rate. Faster growth raises the possibility of an explosion. For instance, if we have $b = 0$ and $a(x, t) = x^2$, then we violate (6.20) and get $\mathrm{d}X_t/\mathrm{d}t = X_t^2$. One solution of this differential equation is $X_t = 1/(C - X_t)$ which becomes infinite at a finite time, where C is an arbitrary constant.

The Lipschitz condition also rules out some degenerate phenomena. For example, Tanaka's SDE is $\mathrm{d}X_t = \mathbf{sign}(X_t)\,\mathrm{d}B_t$ where $\mathbf{sign}(x) = 1$ for $x \geqslant 0$ and -1 for $x < 0$. The diffusion coefficient of this SDE violates (6.19). In this case a given Brownian path B_t does not determine a unique solution X_t from the starting point X_0. For instance if X_t satisfies Tanaka's SDE then so does $-X_t$.

When the sample path X_t satisfies the SDE (6.15) for a given realization of B_t, then we say that X_t is a strong solution of (6.15). The problem illustrated by Tanaka's SDE is that sometimes there is no unique strong solution. We only want to simulate SDEs where a strong solution exists. Otherwise the Monte Carlo inputs we use to compute B_t and, if necessary X_0, are not sufficient to determine the path X_t.

The standard conditions are enough to ensure that the SDE (6.15) has a unique strong solution. Given an SDE with a strong solution we may try to simulate its paths.

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The ***Euler-Maruyama*** method simulates X_t on $[0,T]$ at a discrete set of times $t_k = kT/N$ for $k = 0, 1, \ldots, N$. We use $\Delta = 1/N$ for the interpoint spacing. Given a starting value $\widehat{X}(0)$, Euler-Maruyama proceeds with

$$\widehat{X}(t_{k+1}) = \widehat{X}(t_k) + a\big(t_k, \widehat{X}(t_k)\big)\Delta + b\big(t_k, \widehat{X}(t_k)\big)\sqrt{\Delta}Z_k \tag{6.21}$$

for independent $Z_k \sim \mathcal{N}(0,1)$. The random variable $\sqrt{\Delta}Z_k$ represents the Brownian increment $B(t_{k+1}) - B(t_k)$. Equation (6.21) can be obtained by approximations $a(t', X) \doteq a(t, \widehat{X}_t)$ and $b(t', X) \doteq b(t, \widehat{X}_t)$ holding over the time window $t' \in [t, t+\Delta)$ and for all X. For nontrivial SDEs, one or both of the functions $a(\cdot,\cdot)$ and $b(\cdot,\cdot)$ will be nonconstant. Then as t and $X(t)$ change, the drift and diffusion functions are altered, and these alterations feed back into the distribution of future process values. The Euler-Maruyama scheme is not exact because it ignores this feedback. The hat on $\widehat{X}(t)$ serves as a reminder that the simulated process is only approximately from the target distribution.

As examples of inexactness, $\widehat{S}_{t+\Delta} \sim \mathcal{N}(\widehat{S}_t(1+\Delta\delta), \sigma^2\widehat{S}_t^2\Delta)$ does not give rise to $\widehat{S} \sim \mathsf{GBM}(\delta, \sigma^2)$ sampled at times $t = k\Delta$. Similarly, an Euler-Maruyama simulation of the CIR model might generate an invalid $\widehat{X}_{t+\Delta} < 0$ that the true process would never give. We look at alternative solutions for the CIR model in the section on square root diffusions below.

The approximation (6.21) is only defined at a finite set of times t_k. The usual way to define $\widehat{X}_t$ at other times is by linear interpolation. Some authors take $\widehat{X}_t$ to be piecewise constant. We have sampled $\widehat{X}$ at equispaced time points, but $\widehat{X}$ can be sampled at unequally spaced times in a straightforward generalization of (6.21).

Theorem 6.1 below shows that the Euler-Maruyama scheme will converge to the right answer, for time homogeneous SDEs (6.16) that satisfy the standard conditions. The more general SDEs (6.15) are included too, but they require an additional condition.

Theorem 6.1. *Let X_t be given by an SDE* (6.15) *that satisfies the standard conditions. If the SDE is not time homogeneous, assume also that*

$$|a(t,x) - a(t',x)| + |b(t,x) - b(t',x)| \leqslant K(1+|x|)|t-t'|^{1/2},$$

for all $0 \leqslant t, t' \leqslant T$, all $x \in \mathbb{R}$, and some $K < \infty$. Let $\widehat{X}_t$ be the Euler-Maruyama approximation (6.21) *for $\Delta > 0$, with starting conditions that satisfy $\mathbb{E}(X_0^2) < \infty$ and $\mathbb{E}((X_0 - \widehat{X}_0)^2)^{1/2} \leqslant K'\Delta^{1/2}$ for some $K' < \infty$. Then there is a constant $C < \infty$ for which*

$$\mathbb{E}(|\widehat{X}_T - X_T|) \leqslant C\Delta^{1/2}. \tag{6.22}$$

Proof. This follows from Kloeden and Platen (1999, Theorem 10.2.2). □

By the Markov inequality, equation (6.22) implies that $\mathbb{P}(|\widehat{X}_T - X_T| \geqslant \epsilon) \leqslant \epsilon C\Delta^{1/2}$. That is, for small Δ, the estimate $\widehat{X}_T$ is close to the unique strong solution X_T with high probability.

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For an approximation $\widehat{X}_t$ on $[0,T]$ at points $t_k = k\Delta$ for $0 \leqslant k \leqslant N$ and $\Delta = T/N$, we say that $\widehat{X}_t$ has ***strong order*** of convergence γ at time T if there exists N_0 and C with

$$\mathbb{E}(|\widehat{X}_T - X_T|) \leqslant CN^{-\gamma}$$

for all $N \geqslant N_0$. The constant C can depend on T. Theorem 6.1 shows that the Euler-Maruyama scheme has strong order of convergence $\gamma = 1/2$.

The order $\gamma = 1/2$ is disappointingly slow. It turns out that the Euler-Maruyama scheme is more accurate than this result suggests. It has better performance by a weaker criterion that we discuss next.

In Monte Carlo sampling, we estimate expectations by averaging over independent realizations of the process. We would be satisfied if the process $\widehat{X}_t$ had the same distribution as X_t even if $\widehat{X}_t$ were not equal to X_t. Such an estimate $\widehat{X}_t$ is called a ***weak solution*** of the SDE. A weak solution would arise, for example, if we could construct $\widehat{X}_t$ as the strong solution of the SDE that defines X_t but using a different Brownian motion $\widetilde{B}_t$ instead of B_t, and starting at $\widetilde{X}_0 = X_0$ or at a point X_0 with the same distribution as $\widetilde{X}_0$. Euler-Maruyama approximates a weak solution much better than it approximates the strong one.

An approximation of $\widehat{X}(t)$ on $[0,T]$ at points $t_k = k/N$ for $0 \leqslant k \leqslant N$ has ***weak order*** of convergence β at time T if, for any polynomial g,

$$|\mathbb{E}(g(\widehat{X}_T)) - \mathbb{E}(g(X_T))| \leqslant CN^{-\beta}$$

holds for all $N \geqslant N_0$ for some $N_0 > 0$ and some $C < \infty$.

Taking the polynomial to be $g(x) = x$ and then $g(x) = x^2$, we find that weak convergence of order β makes the mean and variance of $\widehat{X}_T$ match those of X_T to within $O(N^{-\beta})$. Taking account of higher order polynomials gives an even better match between the distribution of $\widehat{X}_T$ and X_T. Some authors use a different class of test functions g than the polynomials.

The Euler-Maruyama scheme has weak order $\beta = 1$. The sufficient conditions are stronger than the ones in Theorem 6.1: the drift and diffusion coefficients need to satisfy a linear growth bound and be four times continuously differentiable.

It is not necessary to use Gaussian random variables in the Euler-Maruyama scheme. We can replace $Z_k \sim \mathcal{N}(0,1)$ by binary Z_k with $\mathbb{P}(Z_k = 1) = \mathbb{P}(Z_k = -1) = 1/2$. The Euler-Maruyama approximations are cumulative sums which satisfy a central limit theorem and so the distinction between Gaussian and binary increments is minor for large N. Further information on Euler-Maruyama is in Kloeden and Platen (1999, Chapter 13).

Our notions of weak and strong convergence describe the quality of the simulated endpoints X_T. When we seek the expected value of a function $f(X(t_1), X(t_2), \ldots, X(t_k))$ we want a good approximation at more times than just the endpoint. We can reasonably expect convergence at a finite list of points to attain the same rate of convergence that we get at a single point. An intuitive explanation is as follows. For strong convergence, Theorem 6.1 shows $\widehat{X}(t_1)$ is close to $X(t_1)$, and it then serves as a good starting point for $\widehat{X}(t_2)$

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to approximate $X(t_2)$, and so on up to t_k, with Euler-Maruyama computations taking place in each interval $[t_j, t_{j+1}]$. Similarly, for weak convergence the end point error from one segment becomes the starting point error for the next. Close inspection of Theorem 6.1 shows that a mean square accuracy is required of the start point and a mean absolute accuracy is delivered for the end point, and so the intuitive argument above is not a complete theorem.

For some problems, the function f depends on $X(\cdot)$ at infinitely many points. A simple example is the lookback option where

$$f(X(\cdot)) = \exp(-rT)(X_T - \min_{0 \leqslant t \leqslant T} X_t)$$

for fixed $r > 0$. There are weak (Kushner, 1974) and strong (Higham et al., 2002) convergence results relevant for an SDE at infinitely many points but the cited ones come without rates of convergence, and even the sufficient conditions are too complicated to include here. Professor Michael Giles (personal communication) has observed weak convergence with $\beta = 1/2$ empirically for the lookback option for the Euler-Maruyama algorithm.

When simulating an SDE, we have a tradeoff to make. If we simulate n paths using N steps per path and each increment has unit cost, then the simulation has cost $C = Nn$. A larger number N of simulation steps decreases the bias $\mathbb{E}(f(\widehat{X}(\cdot))) - \mathbb{E}(f(X(\cdot)))$ of each simulated outcome. A larger number n of independent simulations reduces the variance of their average.

Suppose that the SDE estimate has a bias that approaches $BN^{-\beta}$ as the number N of steps increases and a variance that approaches σ^2/n as both N and the number n of independent simulations increase. Then the mean squared error approaches

$$\mathrm{MSE} = B^2N^{-2\beta} + \sigma^2 n^{-1} = B^2N^{-2\beta} + \sigma^2 NC^{-1}.$$

This MSE is a convex function of $N > 0$. For our analysis, we'll ignore the constraint that N must be an integer. The minimum MSE takes place at $N^* = KC^{1/(2\beta+1)}$ where $K = (2\beta B^2/\sigma^2)^{1/(2\beta+1)}$. The MSE at $N = N^*$ is $C^{-2\beta/(2\beta+1)}(B^2K^{-2\beta} + \sigma^2 K)$.

Although we don't usually know K (it depends on the unknown B and σ) the analysis above gives us some rates of convergence. The mean squared error decreases proportionally to $C^{-2\beta/(2\beta+1)}$ as the cost C increases. For $\beta = 1/2$ that rate is $C^{-1/2}$, far worse than the rate C^{-1} for mean squared error in unbiased Monte Carlo. Euler-Maruyama's rate $\beta = 1$ corresponds to a mean squared error of order $C^{-2/3}$.

Yet another view of these rates is to see that to achieve a root mean square of $\epsilon > 0$ from Euler-Maruyama, will require simulating a total of $C = O(\epsilon^{-3})$ steps. To get one more digit of accuracy then requires 1000 times the computation, instead of the 100-fold increase usual in Monte Carlo.

The foregoing analysis shows that for the best MSE we should take $n \propto N^{2\beta}$. For Euler-Maruyama then, we would have the number n of replications grow proportionally to N^2 where N is the number of time steps in each simulation.

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It is possible to improve on the Euler-Maruyama discretization. One way is to use algorithms with a higher order of convergence. The other is to use multilevel simulation. We describe these next. The Euler-Maruyama approach is widely used because it is simpler to implement than the alternatives: each time we increase β, we raise the complexity of the algorithm as well as the smoothness required of $a(\cdot,\cdot)$ and $b(\cdot,\cdot)$.

Higher order schemes

There are a great many higher order alternatives to the Euler-Maruyama scheme, perhaps 100 or more. We will only look at two of them. The reasons behind this large number of choices are presented on page 66 of the chapter end notes.

The Euler-Maruyama method is based on a very simple locally constant approximation to $a(\cdot,\cdot)$ and $b(\cdot,\cdot)$. At time t, we have some idea how X will change in the near future, how that will change a and b, and hence how those changes will feed back into X. Higher order approximations use Taylor approximations to forecast these changes over the time interval $[t, t+\Delta]$. In a very small time period, Δ, the function might drift $O(\Delta)$ but the root mean square diffusion will be $O(\sqrt{\Delta})$ which is much larger than the drift. As a result Taylor expansions to k'th order in $\mathrm{d}t$ are combined with expansions taken to order $2k$ in $\mathrm{d}B_t$.

The most important term omitted from the Euler-Maruyama scheme is the linear term for the diffusion coefficient $b(X_t)$. Taking account of this term yields the ***Milstein scheme***

$$\widehat{X}(t_k) = \widehat{X}(t_{k-1}) + a_{k-1}\Delta_k + b_{k-1}\sqrt{\Delta_k}Z_k + \frac{1}{2}b_{k-1}b'_{k-1}(Z_k^2-1)\Delta_k \quad (6.23)$$

where $a_{k-1} = a(\widehat{X}(t_{k-1}))$, $b_{k-1} = b(\widehat{X}(t_{k-1}))$, $b'_{k-1} = b'(\widehat{X}(t_{k-1}))$, $\Delta_k = t_k - t_{k-1}$ and $Z_k \sim \mathcal{N}(0,1)$. This is not Milstein's only scheme for SDEs, but the term 'Milstein scheme' without further qualification refers to equation (6.23).

The Milstein scheme attains strong order $\gamma = 1$, which is better than the Euler-Maruyama order $\gamma = 1/2$. Figure 6.10 shows the improvement in the case of geometric Brownian motion, where we can simulate the exact process. Kloeden and Platen (1999, Chapter 10.3) provide an extensive discussion of the Milstein scheme, giving sufficient conditions for its strong rate, and incorporating nonstationary and vector versions. Exercise 6.13 asks you to investigate the effects of increasing the time period and/or decreasing the number of samples, for the Milstein and Euler schemes.

The Milstein scheme only attains weak order $\beta = 1$, which is the same as for Euler-Maruyama. The Euler-Maruyama scheme is usually preferred over the Milstein one in Monte Carlo applications. Its simplicity outweighs the latter's improved strong order of convergence. Also for an SDE in $d > 1$ dimensions

$$\mathrm{d}\boldsymbol{X}_t = \boldsymbol{a}(t, \boldsymbol{X}_t) + \boldsymbol{b}(t, \boldsymbol{X}_t)\,\mathrm{d}\boldsymbol{B}_t$$

where $\boldsymbol{a} \in \mathbb{R}^d$, $\boldsymbol{b} \in \mathbb{R}^{d\times m}$ and $\boldsymbol{B}$ is a vector of m independent standard Brownian motions, the Euler-Maruyama scheme (6.21) can be readily generalized. The

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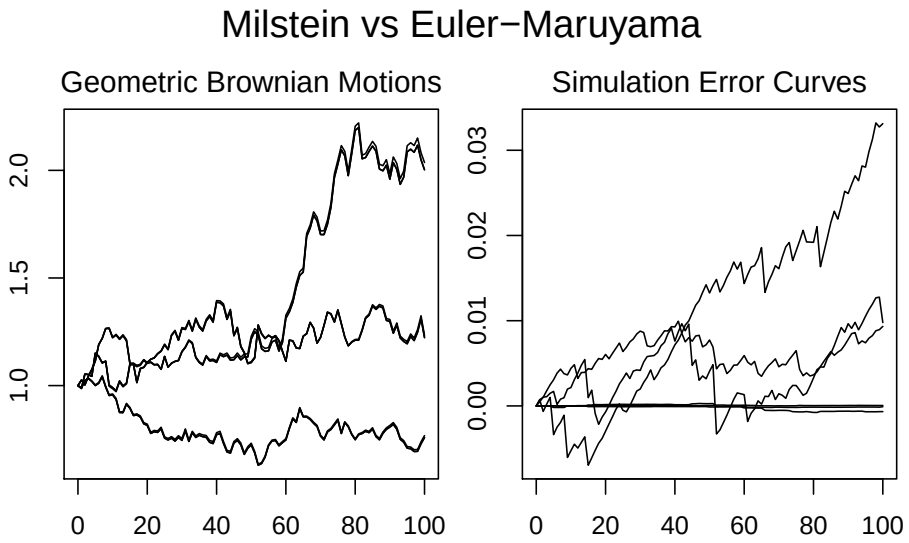


Figure 6.10: The left panel shows three realizations of Geometric Brownian motion ($\delta = 0.05$ and $\sigma = 0.4$) with $N = 100$ steps, on $[0, 1]$. The paths use the exact representation (6.13). Superimposed are the corresponding Euler-Maruyama and Milstein approximations, which closely match the exact paths. The right panel shows the errors of both Euler-Maruyama and Milstein for the three realizations. The Milstein error curves are the three near the horizontal axis.

Milstein scheme in $d > 1$ dimensions requires some additional complicated quantities derived from the Brownian path, variously called Lévy areas or multiple Itô integrals.

Because we are interested in weak convergence, and Euler-Maruyama attains weak order $\beta = 1$, it is worth looking at methods with weak order $\beta = 2$. Such a method would improve the MSE from $O(C^{-2/3})$ to $O(C^{-4/5})$ for computational cost C. The following weak second order scheme

$$\begin{aligned}\widehat{X}(t_k) = \widehat{X}(t_{k-1}) &+ a_{k-1}\Delta_k + b_{k-1}Q_k + \frac{1}{2}b_{k-1}b'_{k-1}(Q_k^2 - \Delta_k)\\ &+ a'_{k-1}b_{k-1}\widetilde{Q}_k + \frac{1}{2}\Big(a_{k-1}a'_{k-1} + \frac{1}{2}a''_{k-1}b_{k-1}^2\Big)\Delta_k^2\\ &+ \Big(a_{k-1}b'_{k-1} + \frac{1}{2}b''_{k-1}b_{k-1}^2\Big)(\Delta_k Q_k - \widetilde{Q}_k)\end{aligned} \tag{6.24}$$

is given by Kloeden and Platen (1999, Chapter 14.2). As before $\Delta_k = t_k - t_{k-1}$. The drift and diffusion coefficients and their indicated derivatives are taken at $\widehat{X}(t_{k-1})$ as before. The random variables Q_k and $\widetilde{Q}_k$ are sampled as

$$Q_k = Z_{k,1}\sqrt{\Delta_k} \quad \text{and} \quad \widetilde{Q}_k = \frac{1}{2}\Delta_k^{3/2}\Big(Z_{k,1} + \frac{1}{\sqrt{3}}Z_{k,2}\Big)$$

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where $Z_{k,j}$ are independent $\mathcal{N}(0,1)$ random variables.

To obtain the rate $O(C^{-4/5})$ with the scheme (6.24), we would take $n \propto N^4$ independent replications. Put another way, we would use a fairly coarse grid of only $n^{1/4}$ times. A second order scheme requires 6 continous derivatives for the drift and diffusion coefficients. It also requires greater smoothness for the function $f(X_T)$ than a first order scheme requires.

There are also well known weak schemes of orders $\beta = 3$ and 4. But such schemes require even greater sampling coarseness and even greater smoothness for the drift and diffusions. Furthermore, the transition from second to third order corresponds to reducing the MSE from $O(C^{-4/5})$ to $O(C^{-6/7})$. For realistic C this may be a very slight gain which could be overshadowed by a less favorable implied constant inside $O(\cdot)$ for the higher order scheme. See Exercise 6.12. That exercise also shows how to adjust for an increased cost per step that usually comes with higher order methods.

Multilevel simulation

Multilevel simulation is a simple and attractive alternative to higher order simulations. Instead of running all n simulations for N time steps, we run simulations of multiple different sizes and combine the results. To see why this might work, consider Figure 6.9, which shows a piecewise linear approximation over eight equal intervals of a Brownian path (on 512 intervals). The full path does not get very far from the approximate one. We might then learn much of what we need about a well behaved function $f(X(\cdot))$ from the first 8 time points that we sample, and only relatively little from the rest of the path. It makes sense to use a large number of coarse paths, and then reduce the resulting bias with a smaller number of fine paths. Multilevel simulation combines paths of many different discretization levels N.

Under favorable circumstances described in Theorem 6.2 below, multilevel schemes achieve a root mean square error below ϵ at a cost which is $O(\epsilon^{-2}\log^2(\epsilon))$ as $\epsilon \to 0$. This is very close to the $O(\epsilon^{-2})$ rate typical for finite dimensional Monte Carlo. By comparison, Euler-Maruyama requires $O(\epsilon^{-3})$ work while schemes that converge at an improved weak order $\beta > 1$ still require $O(\epsilon^{-2-1/\beta})$ work.

Suppose that we seek $\mu = \mathbb{E}(f(X(\cdot)))$ where f is some function of the realization $X(t)$ for $0 \leqslant t \leqslant T$. The strongest theoretical support is for the case where $\mu = \mathbb{E}(f(X(T)))$, a function of just the endpoint. For example, valuing stock options whose payout is determined by the ending share price, known as European options, lead to problems of this type. Multilevel simulations for functions of the entire path, have less theoretical support, but often have good empirical performance.

For integer $\ell \geqslant 0$, the level ℓ simulation produces a sample path $\widehat{X}^\ell(t)$ over $t \in [0,T]$ using $N_\ell = M^\ell$ steps for an integer $M \geqslant 2$. The case $M = 2$ is very convenient, though others are sometimes used.

The level ℓ path is generated at points $t_{k,\ell} = kM^{-\ell}T$ for $k = 1,\dots,N_\ell$ separated by distance $\Delta_{k,\ell} = t_{k,\ell} - t_{k-1,\ell} = M^{-\ell}T$. These grids are equispaced,

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so the level ℓ simulation has spacing $\Delta_\ell = M^{-\ell}T$. For simplicity, we consider a nonrandom starting point $\widehat{X}^\ell(0) = X(0)$.

Let $\widehat{X}^\ell(\cdot)$, the level ℓ simulation, be an Euler-Maruyama scheme (6.21) with spacing Δ_ℓ, and piece-wise linear interpolation between the sampling points. Then define

$$\mu_\ell = \mathbb{E}(f(\widehat{X}^\ell(\cdot))), \quad \ell \geqslant 0, \tag{6.25}$$

and let $\delta_\ell = \mu_\ell - \mu_{\ell-1}$ for $\ell \geqslant 0$, taking $\mu_{-1} = 0$ in order to define δ_0. The multilevel simulation is based on the identity

$$\mu = \mu_0 + \sum_{\ell=1}^{\infty} \delta_\ell = \sum_{\ell=0}^{\infty} \delta_\ell.$$

The multilevel Monte Carlo estimate is

$$\hat{\mu} = \hat{\mu}_0 + \sum_{\ell=1}^{L} \hat{\delta}_\ell = \sum_{\ell=0}^{L} \hat{\delta}_\ell \tag{6.26}$$

for independent Monte Carlo estimates of μ_0 and δ_ℓ. We can also use $\hat{\mu}_K + \sum_{\ell=1}^{L} \hat{\delta}_{K+\ell}$, for $K > 0$, in settings where extremely coarse grids give poor results; we omit the details.

The new ingredient is estimation of δ_ℓ. We estimate δ_ℓ by using the same Brownian path, sampled at both spacings, Δ_ℓ and $\Delta_{\ell-1}$. We will define below the sample path $\widetilde{X}^\ell(\cdot)$ which corresponds to the Brownian motion defining the level ℓ path, as sampled at the coarser level $\ell - 1$. Using that definition, our estimate of δ_ℓ is

$$\hat{\delta}_\ell = \frac{1}{n_\ell} \sum_{i=1}^{n_\ell} f(\widehat{X}_i^\ell(\cdot)) - f(\widetilde{X}_i^\ell(\cdot))$$

where $\widehat{X}_i^\ell(\cdot)$ are n_ℓ independent Euler-Maruyama sample paths and $\widetilde{X}_i^\ell(\cdot)$ are the corresponding coarser versions.

Because $\widehat{X}^\ell$ and $\widetilde{X}^\ell$ are defined from the same Brownian path,

$$|f(\widehat{X}_i^\ell(\cdot)) - f(\widetilde{X}_i^\ell(\cdot))| = O(\Delta_\ell^\gamma)$$

when Euler-Maruyama attains the strong rate γ. The usual strong rate for Euler-Maruyama is $\gamma = 1/2$ and then $\mathrm{Var}(\hat{\delta}_\ell) = O(\Delta_\ell^{2\gamma}/n_\ell) = O(\Delta_\ell/n_\ell)$.

Let us write the variance of $\hat{\mu}$ as $\sum_{\ell=0}^{L} \sigma_\ell^2/n_\ell$ and take the cost to be proportional to $\sum_{\ell=0}^{L} n_\ell/\Delta_\ell$. If we regard n_ℓ as continuous variables and minimize variance for fixed cost, we find that the best n_ℓ are proportional to $\sigma_\ell\sqrt{\Delta_\ell}$. With σ_ℓ itself proportional to $\sqrt{\Delta_\ell}$, based on the strong rate $\gamma = 1/2$, we get $n_\ell \propto \Delta_\ell \propto M^{-\ell}$.

We work with continuous n_ℓ, and to control the bias, we let $L = L_\epsilon$ increase to infinity as ϵ decreases to 0. Taking $n_\ell = cL_\epsilon\Delta_\ell\epsilon^{-2}$ for $c > 0$, the variance is proportional to

$$\sum_{\ell=0}^{L_\epsilon} \frac{\Delta_\ell}{cL_\epsilon\Delta_\ell\epsilon^{-2}} = \frac{L_\epsilon + 1}{cL_\epsilon}\epsilon^{-2} = O(\epsilon^2)$$

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as $\epsilon \to 0$. Having a variance of order ϵ^2, we can attain a root mean square error of order ϵ if we can ensure that the bias $\mu_{L_\epsilon+1} - \mu$ is of order ϵ. The Euler-Maruyama scheme has weak convergence rate 1 and hence the bias is $O(\Delta_{L_\epsilon})$. If we take $L_\epsilon = \log(\epsilon^{-1})/\log(M) + O(1)$, then $\Delta_{L_\epsilon} = M^{-L_\epsilon} = O(\epsilon)$. The total cost is now proportional to

$$\sum_{\ell=0}^{L_\epsilon} \frac{cL_\epsilon \Delta_\ell \epsilon^{-2}}{\Delta_\ell} = cL_\epsilon(L_\epsilon + 1)\epsilon^{-2} = O(\epsilon^2 \log^2(\epsilon)).$$

Theorem 6.2 below gives conditions under which multilevel simulation attains RMSE ϵ at work $O(\epsilon^{-2}\log^2(\epsilon))$. The conditions do not explicitly require the Euler-Maruyama scheme, which corresponds to the case $\beta = 1$.

Theorem 6.2. *Let $\mu = \mathbb{E}(f(X(T)))$ where X has a fixed starting point $X(0)$ and satisfies the SDE $\mathrm{d}X(t) = a(t, X_t)\,\mathrm{d}t + b(t, X_t)\,\mathrm{d}B(t)$ on $[0, T]$, and where f satisfies the uniform Lipschitz bound $|f(x) - f(x')| \leqslant K|x - x'|$. Let $f(\widehat{X}^\ell(T))$ be an approximation to μ based on one sample path using the timestep $\Delta_\ell = M^{-\ell}T$ for integer $M \geqslant 2$, and let $\mu_\ell = \mathbb{E}(f(\widehat{X}^\ell(T)))$.*

Suppose that there exist independent estimators $\hat{\delta}_\ell$ based on n_ℓ Monte Carlo samples, and positive constants $\alpha \geqslant 1/2$, β, c_1, c_2, c_3 such that:

i) $\mathbb{E}(|f(\widehat{X}^\ell(T)) - \mu|) \leqslant c_1 \Delta_\ell^\alpha$,

ii) $\mathbb{E}(\hat{\delta}_\ell) = \delta_\ell = \begin{cases} \mu_0, & \ell = 0 \\ \mu_\ell - \mu_{\ell-1}, & \ell > 0, \end{cases}$

iii) $V(\hat{\delta}_\ell) \leqslant c_2 n_\ell^{-1} \Delta_\ell^\beta$, *and*

iv) the cost to compute $\hat{\delta}_\ell$ is $C_\ell \leqslant c_3 n_\ell \Delta_\ell^{-1}$.

Then there exists a positive constant c_4 such that for any $\epsilon < \exp(-1)$ there are values L and n_ℓ for which the multilevel estimator $\hat{\mu} = \sum_{\ell=0}^L \hat{\delta}_\ell$ satisfies $\mathbb{E}((\hat{\mu} - \mu)^2) < \epsilon^2$ with total computational cost $C = \sum_{\ell=0}^L C_\ell$ satisfying

$$C \leqslant \begin{cases} c_4 \epsilon^{-2}, & \beta > 1 \\ c_4 \epsilon^{-2} \log^2 \epsilon, & \beta = 1 \\ c_4 \epsilon^{-2-(1-\beta)/\alpha}, & \beta < 1. \end{cases}$$

Proof. This is Theorem 3.1 of Giles (2008b). □

The quantity $\alpha \geqslant 1/2$ in Theorem 6.2 is a weak convergence rate, while β can be taken to be at least twice the strong convergence rate. (We have previously used β for the weak rate and γ for the strong one, respectively.) The Euler-Maruyama scheme attains the $(\alpha, \beta) = (1/2, 1)$ rates under the standard conditions on drift and diffusion. The higher strong rate of the Milstein scheme allows one to lower the cost from $O(\epsilon^{-2}\log^2(\epsilon))$ to $O(\epsilon^{-2})$, though the additional effort to use the Milstein scheme might not be worth it.

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A failure for multilevel sampling when the drift and diffusions are not so well behaved is mentioned in the references on page 67.

To implement multilevel sampling, the coarser path $\widetilde{X}^\ell$ has to be coupled with the finer path $\widehat{X}^\ell$. The finer path is sampled at $t_{k,\ell}$ for $0 \leqslant k < M^\ell$, by

$$\widehat{X}^\ell(t_{k+1,\ell}) = \widehat{X}^\ell(t_{k,\ell}) + a_{k,\ell}\Delta_\ell + b_{k,\ell}\sqrt{\Delta_\ell}Z^\ell_{k+1},$$

for independent $Z^\ell_{k+1} \sim \mathcal{N}(0,1)$, where

$$a_{k,\ell} = a\big(t_{k,\ell}, \widehat{X}^\ell(t_{k,\ell})\big), \quad b_{k,\ell} = b\big(t_{k,\ell}, \widehat{X}^\ell(t_{k,\ell})\big).$$

The coarser path has

$$\widetilde{X}^\ell(t_{k+1,\ell-1}) = \widetilde{X}^\ell(t_{k,\ell-1}) + \widetilde{a}_{k,\ell-1}\Delta_{\ell-1} + \widetilde{b}_{k,\ell-1}\sqrt{\Delta_{\ell-1}}\widetilde{Z}_{k+1,\ell-1},$$

for $0 \leqslant k < M^{\ell-1}$, where

$$\widetilde{a}_{k,\ell-1} = a\big(t_{k,\ell-1}, \widetilde{X}^\ell(t_{k,\ell-1})\big), \quad \widetilde{b}_{k,\ell-1} = b\big(t_{k,\ell-1}, \widetilde{X}^\ell(t_{k,\ell-1})\big),$$

and the Brownian increment $\sqrt{\Delta_{\ell-1}}\widetilde{Z}_{k+1,\ell-1}$ is the sum of M Brownian increments that the finer path made on the interval $(t_{k,\ell-1}, t_{k+1,\ell-1}]$. That is

$$\sqrt{\Delta_{\ell-1}}\widetilde{Z}_{k+1,\ell-1} = \sum_{j=1}^{M}\sqrt{\Delta_\ell}Z_{Mk+j,\ell},$$

or

$$\widetilde{Z}_{k+1,\ell-1} = \frac{1}{\sqrt{M}}\sum_{j=1}^{M} Z_{Mk+j,\ell}, \quad 0 \leqslant k < M^{\ell-1}.$$

Square root diffusions

The SDE (6.18) for the CIR process has a diffusion coefficient $b(X_t) = \sigma\sqrt{X_t}$ that does not satisfy the Lipschitz condition (6.19). Though the CIR process fails to satisfy the standard conditions, a unique strong solution does exist.

Before sampling the CIR, we report on its statistical properties. The SDE $\mathrm{d}X_t = \kappa(r - X_t)\,\mathrm{d}t + \sigma\sqrt{X_t}\,\mathrm{d}B_t$ will remain above 0 for all time if $X_0 > 0$ and the Feller condition $\kappa r \geqslant \sigma^2/2$ holds. Otherwise, the CIR process can reach $X_t = 0$ for some $t < \infty$, though it immediately reflects away from that boundary. This SDE has a solution if $\kappa r \geqslant 0$ and $\kappa \in \mathbb{R}$, with starting point $X_0 \geqslant 0$ and $\sigma > 0$. (Moro and Schurz, 2007).

One common approach to simulating the CIR process is to replace $\sqrt{X_t}$ by $\sqrt{|X_t|}$. The simulated process $\widehat{X}_t$ using Euler-Maruyama may then take negative values but will return to positive values shortly thereafter. The estimate converges to the strong solution as $\Delta \to 0$ despite possibly taking negative values. One can also use $\sqrt{\max(0, X_t)}$.

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There is an exact simulation strategy for square root diffusions. The distribution of X_T given the process up to time $t < T$ is

$$\begin{aligned} X_T &= \frac{e^{-\kappa(T-t)}}{n(t,T)}\chi'^2_d(\lambda), \quad \text{where} \\ n(t,T) &= \frac{4\kappa e^{-\kappa(T-t)}}{\sigma^2(1-e^{-\kappa(T-t)})}, \quad d = \frac{4\kappa r}{\sigma^2}, \quad \text{and} \quad \lambda = X_t n(t,T). \end{aligned} \tag{6.27}$$

The noncentral chi-squared random variable $\chi'^2_d(\lambda)$ can be sampled from the definition in Example 4.17 or from a mixture representation in §4.9. Equation (6.27) can be used to sample the square root diffusion exactly at a list of times $0 = t_0 < t_1 < \cdots < t_N$ given the start point X_0. If $r = 0$, then $d = 0$ and the Poisson mixture representation for the noncentral chi-squared distribution gives $\mathbb{P}(X(t_k) = 0 \mid X(t_{k-1})) = \exp(-\lambda/2)$. Then we know that the sampled process reached 0 in the interval $(t_{k-1}, t_k]$.

Example 6.6 (Stochastic volatility)**.** Stochastic volatility models capture the empirically observed fact that the volatility of many traded assets is not constant over time. Heston's (1993) stochastic volatility model takes the form

$$\begin{aligned} \frac{\mathrm{d}S(t)}{S(t)} &= \delta\,\mathrm{d}t + \sqrt{V(t)}\,\mathrm{d}B_1(t) \quad \text{where,} \\ \mathrm{d}V(t) &= \kappa(\theta - V(t))\,\mathrm{d}t + \sigma\sqrt{V(t)}\,\mathrm{d}B_2(t). \end{aligned}$$

It is driven by two Brownian motions $B_1(t)$ and $B_2(t)$, and has positive parameters δ, κ and σ. Given starting values $V(0)$ and $S(0)$, we can sample the volatility process $V(t)$ and then conditionally on $V(t)$ sample the price $S(t)$.

There is one remaining parameter, not visible in the equations above. The two Brownian motions have an instantaneous correlation ρ. This correlation can be any value in $[-1, 1]$. We might take $\rho < 0$ to model stocks with prices that tend to move downwards at the same time that their volatility increases. Or we might take $\rho > 0$ to model commodities that become more volatile as their price increases. Exercise 6.14 has you value a European call option, under this model.

The ***constant elasticity of variance*** (CEV) model generalizes square root diffusions. A CEV process has SDE

$$\mathrm{d}X_t = \delta X_t \mathrm{d}t + \sigma X_t^{\beta+1}\mathrm{d}B_t. \tag{6.28}$$

It includes geometric Brownian motion ($\beta = 0$) and the square root diffusion ($\beta = -1/2$).

The SDE (6.28) has a strong solution if $\beta \geqslant -1/2$. If $\beta < 0$, then the volatility $\sigma X_t^{\beta+1}/X_t = \sigma X_t^{\beta}$ increases as X_t decreases. That property, called the ***leverage effect*** in financial applications, is missing from geometric Brownian motion. For $0 > \beta \geqslant -1/2$, the process can reach 0 in finite time, and it will remain there. Thus the CEV process provides a model which includes the possibility of bankruptcy.

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6.6 Poisson point processes

A point process describes a random list of points $\boldsymbol{T}_i$ in a set $\mathcal{T} \subset \mathbb{R}^d$. The domain $\mathcal{T}$ will often be $[0,\infty)$, or the unit square $[0,1]^2$. Point processes on $[0,\infty)$ are used to model arrival times for phone calls, earthquakes, web traffic and news stories. For multidimensional $\mathcal{T}$, the points $\boldsymbol{T}_i$ could represent the positions of flaws in a silicon lattice, trees in a forest, or galaxies in a cluster.

The number of points in the process is $N(\mathcal{T})$, which may be fixed or random, finite or countably infinite. For $A \subset \mathcal{T}$ let $N(A)$ be the number of points of the process that lie within A. That is

$$N(A) = \sum_{i=1}^{N(\mathcal{T})} 1\{\boldsymbol{T}_i \in A\}. \tag{6.29}$$

We will look at simulating processes that are non-explosive, which means that any set A of finite volume has $\mathbb{P}(N(A) < \infty) = 1$.

The role of finite dimensional distributions is played here by the number of points in a finite list of non-overlapping sets A_j. That is, we specify the distributions of $(N(A_1), \dots, N(A_J))$ for all $J \geqslant 1$ and all disjoint sets $A_j \subset \mathcal{T}$. In this section, we consider Poisson processes which are much simpler than general processes, postponing non-Poisson point processes to §6.7.

The points $\boldsymbol{T}_i \in \mathcal{T}$ are a ***homogeneous Poisson process*** on $\mathcal{T}$ with intensity $\lambda > 0$ if

$$N(A_j) \sim \mathrm{Poi}\big(\lambda \mathbf{vol}(A_j)\big) \tag{6.30}$$

independently, whenever $A_1, \dots, A_J \subset \mathcal{T}$ are disjoint sets with $\mathbf{vol}(A_j) < \infty$. We write $(\boldsymbol{T}_1, \boldsymbol{T}_2, \dots) \sim \mathsf{PP}(\mathcal{T}, \lambda)$.

We often find that real world processes are not homogeneous: earthquakes are more common in some regions than other, fires and hurricanes are more prevalent at certain times of the year, digital and automobile traffic show strong time of day and day of the week patterns. It is a great strength of Monte Carlo methods that we can take account of known non-uniformity patterns in our models.

We incorporate non-uniformity into a Poisson process by replacing the constant intensity λ by a spatially varying intensity function $\lambda(\boldsymbol{t}) \geqslant 0$ for $\boldsymbol{t} \in \mathcal{T}$. We require that the intensity function satisfy $\int_A \lambda(\boldsymbol{t})\,\mathrm{d}\boldsymbol{t} < \infty$ whenever $\mathbf{vol}(A) < \infty$. This does not mean that λ has to be bounded. For example, with $t \in [0,\infty)$ we could have $\lambda(t) = t$. For a ***non-homogeneous Poisson process*** on $\mathcal{T}$ with intensity function $\lambda(t) \geqslant 0$,

$$N(A_j) \sim \mathrm{Poi}\left(\int_{A_j} \lambda(\boldsymbol{t})\,\mathrm{d}\boldsymbol{t}\right) \tag{6.31}$$

independently, whenever $A_1, \dots, A_J \subset \mathcal{T}$ are disjoint sets with $\mathbf{vol}(A_j) < \infty$. We write $(\boldsymbol{T}_1, \boldsymbol{T}_2, \dots) \sim \mathsf{NHPP}(\mathcal{T}, \lambda)$.

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The many techniques for sampling random vectors in Chapter 5 carry over directly to let us sample non-homogeneous Poisson processes on a region of interest. If $\rho(\boldsymbol{t})$ is a density function from which we can sample, then we can sample any Poisson process with $\lambda(\boldsymbol{t}) \propto \rho(\boldsymbol{t})$, as the next theorem shows.

Theorem 6.3. *Let $\boldsymbol{T}_i$ be the points of a Poisson process on $\mathcal{T}$ with intensity function $\lambda(\boldsymbol{t}) \geqslant 0$, where $\Lambda(\mathcal{T}) = \int_{\mathcal{T}} \lambda(\boldsymbol{t})\,\mathrm{d}\boldsymbol{t} < \infty$. Then $\boldsymbol{T}_i$ can be sampled by taking*

$$N(\mathcal{T}) \sim \mathrm{Poi}(\Lambda(\mathcal{T}))$$

and then given that $N(\mathcal{T}) = n \geqslant 1$, taking independent $\boldsymbol{T}_i$ with

$$\mathbb{P}(\boldsymbol{T}_i \in A) = \frac{1}{\Lambda(\mathcal{T})} \int_A \lambda(\boldsymbol{t})\,\mathrm{d}\boldsymbol{t}$$

for $i = 1, \dots, n$.

Proof. For $A \subset \mathcal{T}$, let $\Lambda(A) = \int_A \lambda(\boldsymbol{t})\,\mathrm{d}\boldsymbol{t}$. For $J \geqslant 1$, let $A_1, \dots, A_J$ be disjoint subsets of $\mathcal{T}$ and define $A_0 = \{\boldsymbol{t} \in \mathcal{T} \mid \boldsymbol{t} \notin \cup_{j=1}^J A_j\}$. Pick integers $n_j \geqslant 0$ for $j = 1, \dots, J$, let

$$P_* = \mathbb{P}(N(A_1) = n_1, \dots, N(A_J) = n_J) = \sum_{n_0=0}^{\infty} \mathbb{P}(N(A_0) = n_0, \dots, N(A_J) = n_J)$$

and set $n = n_0 + \cdots + n_J$. Then under the given sampling scheme

$$\begin{aligned} P_* &= \frac{n!}{n_0! n_1! \cdots n_J!} \sum_{n_0=0}^{\infty} \frac{e^{-\Lambda(\mathcal{T})}\Lambda(\mathcal{T})^n}{n!} \prod_{j=0}^{J} \Big(\frac{\Lambda(A_j)}{\Lambda(\mathcal{T})}\Big)^{n_j} \\ &= \sum_{n_0=0}^{\infty} \prod_{j=0}^{J} \frac{e^{-\Lambda(A_j)}\Lambda(A_j)^{n_j}}{n_j!} = \prod_{j=1}^{J} \frac{e^{-\Lambda(A_j)}\Lambda(A_j)^{n_j}}{n_j!} \end{aligned}$$

which matches the joint distribution of $N(A_1), \dots, N(A_J)$ of the Poisson process. □

We cannot use the method of Theorem 6.3 if $\Lambda(\mathcal{T}) = \infty$, because we cannot generate an infinite number of points. In practice we choose $\mathcal{T}$ to be a large region covering the area of most interest. The region $\mathcal{T}$ can have infinite volume, as long as $\Lambda(\mathcal{T}) < \infty$. We could have ruled out $\Lambda(\mathcal{T}) = 0$ because the distribution for $\boldsymbol{T}_i$ is not well defined in that case. But when $\Lambda(\mathcal{T}) = 0$, the sample size is always 0 points and we don't need a well defined distribution for that.

If we can sample points uniformly from $\mathcal{T}$ then we can sample a homogeneous Poisson process on $\mathcal{T}$, by the following Corollary to Theorem 6.3.

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Corollary 6.1. *Let $\boldsymbol{T}_i$ be the points of a homogeneous Poisson process on $\mathcal{T}$ with intensity $\lambda > 0$ where $\mathbf{vol}(\mathcal{T}) < \infty$. Then we may sample the process by taking*

$$N(\mathcal{T}) \sim \mathrm{Poi}(\lambda \mathbf{vol}(\mathcal{T}))$$

and $\boldsymbol{T}_i \sim \mathbf{U}(\mathcal{T})$ independently for $i = 1, \dots, N(\mathcal{T})$.

Proof. We apply Theorem 6.3 with a constant $\lambda(\boldsymbol{t})$. In this case $\mathbb{P}(T_i \in A) = \Lambda(\mathcal{T})^{-1} \int_A \lambda \, \mathrm{d}\boldsymbol{t} = \mathbf{vol}(A)/\mathbf{vol}(\mathcal{T})$, so $\boldsymbol{T}_i \sim \mathbf{U}(\mathcal{T})$. □

Example 6.7 (Poisson process in the disk)**.** Let $\mathcal{T} = \{\boldsymbol{x} \in \mathbb{R}^2 \mid \boldsymbol{x}^{\mathsf{T}}\boldsymbol{x} \leqslant 1\}$. To sample the $\mathsf{PP}(\mathcal{T}, \lambda)$ we take $N \sim \mathrm{Poi}(\pi\lambda)$ and then

$$\boldsymbol{T}_i = \big(\cos(2\pi U_{i1}), \sin(2\pi U_{i1})\big) \times \max(U_{i2}, U_{i3}) \tag{6.32}$$

for independent $\boldsymbol{U}_i \sim \mathbf{U}(0,1)^3$, $i = 1, \dots, N$. Exercise 6.19 asks you to justify equation (6.32).

Poisson processes on $[0, \infty)$

Many applications of Poisson processes describe events happening in the future. Letting time 0 be the present, we use the state space $\mathcal{T} = [0, \infty)$. In this case it is usual to suppose that the points are generated in order, with $T_1 < T_2 < \cdots$. The process can be represented by the counting function

$$N(t) \equiv N\big([0,t]\big) = \sum_{i=1}^{\infty} 1\{T_i \leqslant t\}, \quad 0 \leqslant t < \infty. \tag{6.33}$$

To simulate such a process we usually generate T_1 and then for $i \geqslant 2$ generate the gaps $T_i - T_{i-1}$ conditionally on $T_1, \dots, T_{i-1}$. That is we start at time 0 and work forward.

The (homogeneous) ***Poisson process*** on $\mathcal{T} = [0, \infty)$ is defined by these properties:

PP-1: $N(0) = 0$.

PP-2: For $0 \leqslant s < t$, $N(t) - N(s) \sim \mathrm{Poi}\big(\lambda(t-s)\big)$,

PP-3: Independent increments: for $0 = t_0 < t_1 < \cdots < t_m$, $N(t_i) - N(t_{i-1})$ are independent.

We write $(T_1, T_2, \dots) \sim \mathsf{PP}([0,\infty), \lambda)$, or $\mathsf{PP}(\lambda)$ for short. The parameter $\lambda > 0$ is the rate of the process. The increment $N(t) - N(s)$ is the number of events that happen in the interval $(s, t]$.

The Poisson process has the following well-known characterization:

$$T_i - T_{i-1} \sim \mathrm{Exp}(1)/\lambda, \quad \text{independently,} \tag{6.34}$$

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for $i \geqslant 1$. In using (6.34), we define $T_0 = 0$, though T_0 is not part of the process. Hoel et al. (1971, Chapter 9) give a thorough, yet elementary proof of (6.34). For a simple explanation, notice that under equation (6.34), $\mathbb{P}(T_i - T_{i-1} > x) = \exp(-\lambda x)$. If $T_i - T_{i-1} \geqslant x$, then the interval $(T_{i-1}, T_{i-1} + x)$ got zero events. A nonrandom interval of length x is empty with probability $\mathbb{P}(\mathrm{Poi}(\lambda x) = 0) = \exp(-\lambda x)$ as desired. The full proof is longer because the interval $(T_{i-1}, T_{i-1} + x)$ is a random one, and the process is defined in terms of nonrandom intervals. Briefly: it is a memoryless property of the Poisson process that makes the substitution work for $(T_{i-1}, T_{i-1} + x)$.

Equation (6.34), underlies the ***exponential spacings method*** for sampling a Poisson process:

$$T_1 \sim E_1/\lambda, \quad \text{and} \quad T_i = T_{i-1} + E_i/\lambda, \quad i \geqslant 2, \tag{6.35}$$

for independent $E_i \sim \mathrm{Exp}(1)$. The method (6.35) can be run until either the desired number of points has been sampled or the process exits a prespecified time window $[0, T]$.

Corollary 6.1 supplies another simple way to sample the standard Poisson process on a fixed interval $[0, T]$. We can take

$$\begin{aligned} &N = N(T) \sim \mathrm{Poi}(\lambda T), \\ &S_i \sim \mathbf{U}[0, T], \quad i = 1, \ldots, N, \quad \text{then,} \\ &T_i = S_{(i)}. \end{aligned} \tag{6.36}$$

The last step in (6.36) sorts the points into increasing order, which is necessary because we defined T_i to be ordered but S_i are not necessarily in increasing order. For large enough $\mathbb{E}(N) = \lambda T$ the sorting step will dominate the cost of using (6.36). The value of this representation is primarily in applications where the quantity we are averaging depends strongly on $N(T)$. In such cases we may benefit from a stratified sampling (see §8.4) of N.

It is also possible to sample the Poisson process recursively, in a way that is analogous to the Brownian bridge construction used to recursively sample Brownian motion. That is a specialized topic, which we take up on page 48.

As mentioned previously, non-homogeneous phenomena are very common, and Monte Carlo methods can handle them well. For a process on $[0, \infty)$ we have an intensity function $\lambda(t) \geqslant 0$ and we assume that $\int_s^t \lambda(x)\,\mathrm{d}x < \infty$ for $0 < s < t < \infty$.

As before, the events are $T_1 < T_2 < \cdots$, there are $N(\mathcal{T})$ of them, the number of events in the set A is $N(A)$ and we define the counting function $N(t) = N([0, t])$ for $t \geqslant 0$. The ***non-homogeneous Poisson process*** is defined by these rules:

NHPP-1: $N(0) = 0$,

NHPP-2: For $0 < s < t$, $N(t) - N(s) \sim \mathrm{Poi}\big(\int_s^t \lambda(x)\,\mathrm{d}x\big)$,

NHPP-3: $N(t)$ has independent increments.

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We write $(T_1, T_2, \dots) \sim \mathsf{NHPP}([0,\infty), \lambda)$, or $\mathsf{NHPP}(\lambda)$ for short. For a collection of non-overlapping sets $A_j \subset \mathcal{T}$, we have $N(A_j) \sim \mathrm{Poi}(\int_{A_j} \lambda(t)\,\mathrm{d}t)$ independently.

The ***cumulative rate function*** of $\mathsf{NHPP}(\lambda)$ is $\Lambda(t) = \int_0^t \lambda(s)\,\mathrm{d}s$. We will use it to simulate the points T_i in increasing order. At first, we assume that $\lim_{t\to\infty}\Lambda(t) = \infty$. Then $N(\mathcal{T}) = \infty$. We will also assume that $\lambda(t) > 0$ for all t. Then there is a unique inverse function Λ^{-1} with $\Lambda^{-1}(0) = 0$.

Now we define the variables $Y_i = \Lambda(T_i)$ and the counting function

$$N_y(t) = \sum_{i=1}^{\infty} \mathbb{1}_{Y_i \leqslant t} = \sum_{i=1}^{\infty} \mathbb{1}_{T_i \leqslant \Lambda^{-1}(t)} = N(\Lambda^{-1}(t)).$$

Inspecting this function, we see that $N_y(0) = 0$. Next, the increment

$$\begin{aligned} N_y(t) - N_y(s) = N(\Lambda^{-1}(t)) - N(\Lambda^{-1}(s)) &\sim \mathrm{Poi}\left(\int_{\Lambda^{-1}(s)}^{\Lambda^{-1}(t)} \lambda(x)\,\mathrm{d}x\right) \\ &= \mathrm{Poi}\big(\Lambda(\Lambda^{-1}(t)) - \Lambda(\Lambda^{-1}(s))\big) = \mathrm{Poi}(t-s). \end{aligned}$$

Finally, the increments of $N_y(t)$ are the increments of $N(\Lambda^{-1}(t))$. Since the latter are independent increments, so are the former. We have shown that

$$Y_i = \Lambda(T_i) \sim \mathsf{PP}(1).$$

Therefore we may simulate T_i by taking

$$\begin{aligned} Y_i &= Y_{i-1} + E_i \\ T_i &= \Lambda^{-1}(Y_i) = \Lambda^{-1}(\Lambda(T_{i-1}) + E_i), \end{aligned} \tag{6.37}$$

for $i \geqslant 1$, with independent $E_i \sim \mathrm{Exp}(1)$ and $Y_0 = T_0 = 0$. Equation (6.37) is a ***non-homogeneous exponential spacings*** algorithm.

We assumed that $\lim_{t\to\infty}\Lambda(t) = \infty$. Now suppose instead that $\lim_{t\to\infty}\Lambda(t) = \Lambda_0 < \infty$. Then $\Lambda^{-1}(y)$ does not exist for $y > \Lambda_0$. If $\Lambda(T_i) + E_{i+1} > \Lambda_0$ then there is no point T_{i+1} and the process stops with only i points.

Formula (6.37) is the Poisson process counterpart to inversion of the CDF. It is very convenient, at least when we have closed forms for Λ and Λ^{-1}, or reasonable numerical substitutes. We derived it assuming that Λ was continuous and strictly increasing. Neither of those conditions is necessary, just as they are not necessary when we use inverse CDFs to sample random variables. Exponential spacings can be used for cumulative intensities Λ that take finite jumps or are constant on intervals $[t, s)$. We use

$$\Lambda^{-1}(y) = \inf\{t \geqslant 0 \mid \Lambda(t) \geqslant y\}.$$

Thinning and superposition

Just as we had for random number generation, there are flexible alternatives to inversion for sampling a Poisson process. The analog of acceptance-rejection

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sampling is ***thinning***. Suppose that there is a function $\widetilde{\lambda}(t) \geqslant \lambda(t)$ such that we can sample a Poisson process on $\mathcal{T}$ with intensity $\widetilde{\lambda}$. Then thinning works as follows. First we sample $(\widetilde{\boldsymbol{T}}_1, \widetilde{\boldsymbol{T}}_2, \cdots, \widetilde{\boldsymbol{T}}_{\widetilde{N}}) \sim \mathsf{NHPP}(\mathcal{T}, \widetilde{\lambda})$. Note that $\widetilde{N}$ is random, finite and could be 0. Then if $\widetilde{N} > 0$, we accept each $\widetilde{\boldsymbol{T}}_i$ independently with probability $\rho(\widetilde{\boldsymbol{T}}_i) = \lambda(\boldsymbol{T}_i)/\lambda(\widetilde{\boldsymbol{T}}_i)$. Finally, we deliver the accepted points in the list $(\boldsymbol{T}_1, \ldots, \boldsymbol{T}_N)$. We ordinarily have to relabel the points, as the delivered point $\boldsymbol{T}_i$ may not have originated as $\widetilde{\boldsymbol{T}}_i$ for the same value of i.

To see why thinning works, consider the number of points $\boldsymbol{T}_i$ within the set $A \subset \mathcal{T}$. There are $\widetilde{N}(A)$ points $\widetilde{\boldsymbol{T}}_i \in A$ where $\widetilde{N}(A) \sim \mathrm{Poi}(\int_A \widetilde{\lambda}(\boldsymbol{t})\,\mathrm{d}\boldsymbol{t})$. Let the chance that a point $\widetilde{\boldsymbol{T}}_i \in A$ is accepted be $\rho(A)$. Then

$$\rho(A) = \frac{\int_A \rho(\boldsymbol{t})\widetilde{\lambda}(\boldsymbol{t})\,\mathrm{d}\boldsymbol{t}}{\int_A \widetilde{\lambda}(\boldsymbol{t})\,\mathrm{d}\boldsymbol{t}} = \frac{\int_A \lambda(\boldsymbol{t})\,\mathrm{d}\boldsymbol{t}}{\int_A \widetilde{\lambda}(\boldsymbol{t})\,\mathrm{d}\boldsymbol{t}}.$$

Now given $\widetilde{N}(A)$ we have $N(A) \sim \mathrm{Bin}(\widetilde{N}(A), \rho(A))$ and it then follows by an elementary calculation that $N(A) \sim \mathrm{Poi}(\rho(A)\int_A \widetilde{\lambda}(\boldsymbol{t})\,\mathrm{d}\boldsymbol{t}) = \mathrm{Poi}(\lambda(A))$. If $A_1, \ldots, A_J$ are non-overlapping sets than $\widetilde{N}(A_j)$ are mutually independent and then $N(A_j)$ are also mutually independent. As a result $(\boldsymbol{T}_1, \ldots, \boldsymbol{T}_N) \sim \mathsf{NHPP}(\mathcal{T}, \lambda)$.

There is a geometric description of thinning that echoes the one for acceptance rejection sampling. Let

$$\begin{aligned}\mathcal{S}_1(\widetilde{\lambda}) &= \{(\boldsymbol{t}, z) \mid \boldsymbol{t} \in \mathcal{T}, 0 \leqslant z \leqslant \widetilde{\lambda}(\boldsymbol{t})\}, \quad \text{and}\\ \mathcal{S}_1(\lambda) &= \{(\boldsymbol{t}, z) \mid \boldsymbol{t} \in \mathcal{T}, 0 \leqslant z \leqslant \lambda(\boldsymbol{t})\}.\end{aligned}$$

If we pair the proposed points $\widetilde{\boldsymbol{T}}_i$ with independent $\widetilde{U}_i \sim \mathbf{U}(0,1)$ then the points $(\widetilde{\boldsymbol{T}}_i, \widetilde{U}_i\widetilde{\lambda}(\boldsymbol{T}_i))$ form a uniform Poisson process within $\mathcal{S}_1(\widetilde{\lambda}) \supseteq \mathcal{S}_1(\lambda)$. The subset of these points that lie within $\mathcal{S}_1(\lambda)$ is a uniform Poisson process on $\mathcal{S}_1(\lambda)$. Call them $(\boldsymbol{T}_i, U_i)$ Their components $\boldsymbol{T}_i$ are $\mathsf{NHPP}(\mathcal{T}, \lambda)$.

Example 6.8 (Zipf-Poisson ensemble)**.** Let $X_i \sim \mathrm{Poi}(Ni^{-\alpha})$ for $i = 1, 2, \ldots$ with parameters $\alpha > 1$ and $N > 0$. This is a model for long-tailed count data, such as the number of appearances the i'th most popular word within a set of documents, or the i'th most popular baby's name in a given year. The parameter α governs how long the tail is and $N = \mathbb{E}(X_1)$. The term 'Zipf' refers to the Zipf distribution in which a random positive integer Y is chosen with $\mathbb{P}(Y = y) \propto y^{-\alpha}$.

We can sample the larger values by taking $X_i \sim \mathrm{Poi}(Ni^{-\alpha})$ for $i = 1, \ldots, k$. To sample the finite number of nonzero values among the infinite tail, X_i for $k < i < \infty$, we can use thinning as depicted in Figure 6.11. We sample points x from the Poisson process on $[k+1/2, \infty)$ with $\widetilde{\lambda}(x) = N(x-1/2)^{-\alpha}$. Rounding x to the nearest integer yields $r(x) \equiv \lfloor x + 1/2 \rfloor$. We accept generated point x with probability $\lambda(x)/\widetilde{\lambda}(x)$ where $\lambda(x) = Nr(x)^{-\alpha}$. Then X_i is the number accepted points with $r(x) = i$. Exercise 6.20 asks you how to sample from the $\widetilde{\lambda}$ process.

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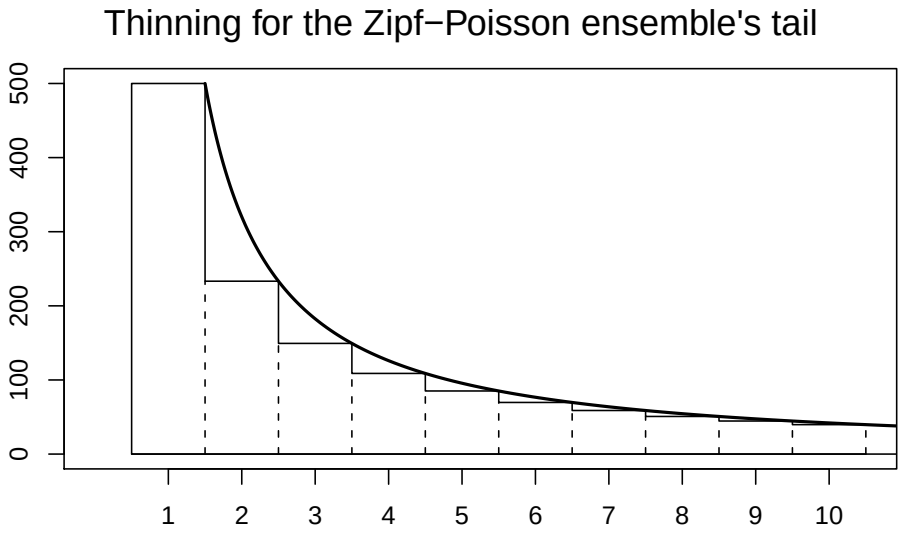


Figure 6.11: This panel illustrates thinning for the Zipf-Poisson ensemble with $N = 500$ and $\alpha = 1.1$. First we take $X_1 \sim \mathrm{Poi}(N)$. Then we generate a Poisson process in the region under the curve $N(x - 1/2)^{-\alpha}$ over $[1.5, \infty)$ shown as the thick curve above the rectangles. For $i \geqslant 2$, X_i is the number of those points within the rectangle $[i - 1/2, i + 1/2) \times [0, Ni^{-\alpha}]$. More generally, we can sample $X_i \sim \mathrm{Poi}(Ni^{-\alpha})$ for $1 \leqslant i \leqslant k$ and then use thinning over the interval $[k + 1/2, \infty)$.

There is also a direct analogue of mixture sampling. Suppose that $\lambda(\boldsymbol{t}) = \sum_{k=1}^{K} \lambda_k(\boldsymbol{t})$ for functions $\lambda_k(\boldsymbol{t}) \geqslant 0$ on $\mathcal{T}$. If we can sample $\boldsymbol{T}_{ik}$ for $i \geqslant 1$ from each $\mathsf{NHPP}(\mathcal{T}, \lambda_k)$ then we can take the union of the generated points as a sample from $\mathsf{NHPP}(\mathcal{T}, \lambda)$. That is $N(A) = \sum_{k=1}^{K} N_k(A)$, where $N_k(A) = \sum_{i \geqslant 1} \mathbb{1}_{T_{ik} \in A}$.

If $\lambda(t)$ is a piecewise constant function for t in the interval $[a, b]$ (i.e., a histogram) then the set under $\lambda(t)$ is a union of rectangles. We may use one mixture component for each of those rectangles.

Example 6.9 (Traffic patterns)**.** Traffic levels, whether on the road or at a web site, are usually far from homogeneous. There is typically a marked cyclical pattern over 24 hours. There is also a day of week pattern, often an annual cycle, and then special exceptional days, some of which are predictable, such as holidays. Figure 6.12 shows traffic levels for one highway segment from Sullivan County, New York, in one direction, for one day. The data was obtained from the New York State Department of Transportation website `http://www.nysdot.gov`.

One way to test a system is to simulate traffic from a model. Another way is to play back actual recorded data. Both yield insights, with recorded data

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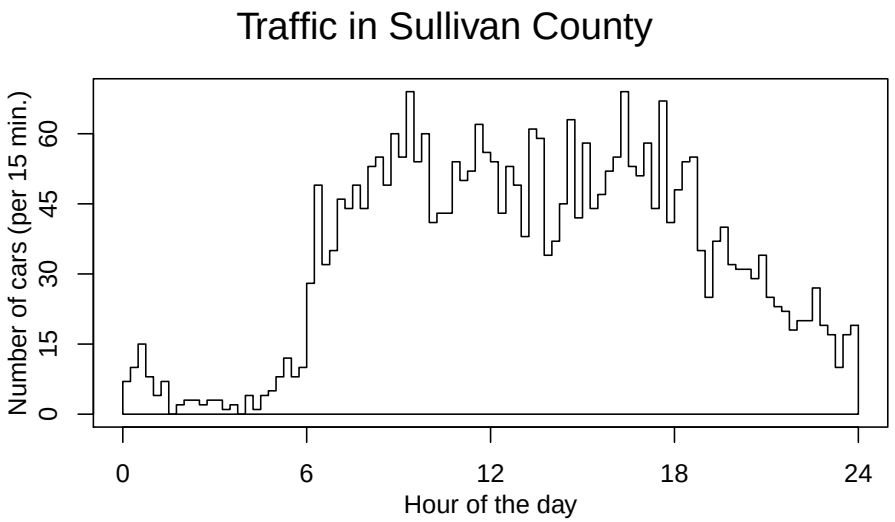


Figure 6.12: This figure shows the number of cars per 15 minute period, for one highway segment over one day, in Sullivan County, New York.

capturing some anomalies that we might not have included in our model. For a stress test, we can replay historical data at an increased intensity level. For example, we could randomly select days from historical records, and resample from them at higher intensity. Letting the data in Figure 6.12 represent a function $\lambda(t)$ on $[0, 24)$ we would then sample a Poisson process with intensity $1.2\lambda(t)$, as one way to model a 20% increase in traffic. See Exercise 6.22. The values from Figure 6.12 are given in Table 6.1.

Poisson field of lines

Sometimes we need to generate random lines in the plane or more generally $\mathbb{R}^d$. For example, chemical engineers use random lines to model the distribution of fibers in a mat and geologists use them to model networks of cracks. A line can be parameterized by its slope and intercept, but then it is tricky to put just the right joint distribution on these because vertical lines have infinite slope.

The default uniform distribution on random lines is the Poisson field. We work in polar coordinates and write the line as

$$L(r, \theta) = \{(x, y) \mid x\cos(\theta) + y\sin(\theta) = r\}$$

where $r \in \mathbb{R}$ is a signed radius and $\theta \in [0, \pi)$ is an angle. The ***Poisson field of lines*** comprises the lines $L(r_i, \theta_i)$ from a Poisson process of intensity λ for $(r, \theta) \in \mathcal{T} = \mathbb{R} \times [0, \pi)$.

The importance of this distribution for lines arises from some invariance properties of the Poisson field. Suppose that we shift the generated lines, re-

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Hour	AM				PM			
1	7	10	15	8	54	43	53	49
2	4	7	0	2	38	61	59	34
3	3	3	2	3	37	45	63	42
4	3	1	2	0	58	44	47	52
5	4	1	4	5	55	69	53	51
6	8	12	8	10	58	44	67	41
7	28	49	32	35	48	54	55	35
8	46	44	49	44	25	37	40	32
9	53	55	49	60	31	31	29	34
10	55	69	54	60	25	23	22	18
11	41	43	43	54	20	20	27	19
12	50	52	62	56	17	10	17	19

Table 6.1: Number of cars, per quarter hour, to pass over a highway segment in Sullivan County, New York. These are the values shown in Figure 6.12.

placing them by $\{(x_0, y_0) + (x, y) \mid x\cos(\theta) + y\sin(\theta) = r\}$ for an arbitrary new origin (x_0, y_0). The distribution of the lines would be unchanged by this shift. The distribution of the lines is also invariant if we rotate our coordinate axes through some fixed angle. The motivation for choosing such an invariant distribution is that cracks or fibers or similar physical objects are not affected by the coordinate system we use. There are no other non-trivial invariant distributions for lines. For more details about the Poisson field, including the uniqueness of the invariant distribution, see the end notes on page 66.

The lines of the Poisson field have infinite extent. When we only want to see their intersection with a bounded region $\mathcal{R} \subset \mathbb{R}^2$ then we only need to consider lines with $r \leqslant r_0 = \sup_{\boldsymbol{x}\in\mathcal{R}} \|\boldsymbol{x}\|$. To get n lines in our region, we sample $R_i \sim \mathbf{U}(0, r_0)$ and $\theta_i \sim \mathbf{U}[0, \pi)$ (all independently) for $i \geqslant 1$, keeping the first n lines $L(r_i, \theta_i)$ that intersect $\mathcal{R}$.

Figure 6.13 shows a sample of Poisson lines generated to intersect a circle of radius $\sqrt{2}$ about the origin. Most such lines intersect the unit square $[-1, 1]^2$ shown the left side of the figure. The right side has lines simulated from a process that prefers lines more nearly parallel with the coordinate axes. In this case the angle was $\theta = \pi(X + Y)/2$ where $X \sim \text{Beta}(1/4, 1/4)$ independently of $Y \sim \mathbf{U}\{0, 1\}$.

If we naively sampled points along the x-axis and then generated lines intersecting it at uniformly distributed angles, we would *not* get lines with an invariant distribution. Theorem 2 of Miles (1964) describes the angles that the Poisson lines make when intersecting some non-random line ℓ, like the x-axis. The angles ϕ made between the random lines and ℓ have density $\sin(\phi)/2$ on $0 \leqslant \phi < \pi$.

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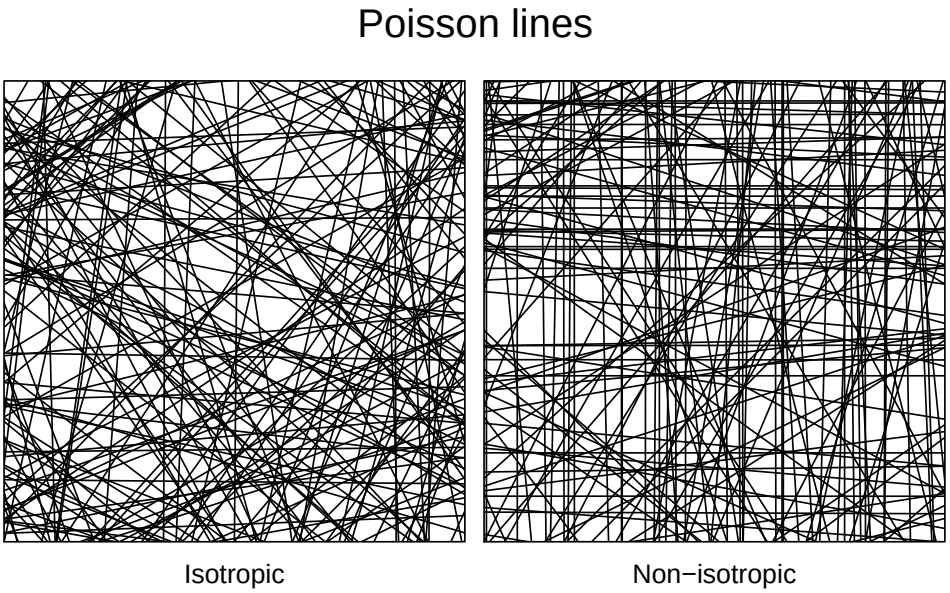


Figure 6.13: One hundred and fifty lines from the Poisson line process were generated to intersect the circle $\{\boldsymbol{x} \in \mathbb{R}^2 \mid \|\boldsymbol{x}\| \leqslant \sqrt{2}\}$. Their intersection with the unit square $[-1,1]^2$ is shown in the left panel. A non-isotropic version favoring lines nearly parallel to the axes, is shown in the right panel.

Recursive sampling for the Poisson process

The Poisson process can also be sampled in a way that parallels the Brownian bridge sampling of Brownian motion from §6.4. The conditional distribution of $N(T/2)$ given $N(T)$ is $\mathrm{Bin}(N(T), 1/2)$. More generally, suppose that $0 < \ell < t < r < T$ where times ℓ, r, and t are either fixed, or are random but independent of the Poisson process we're generating. Then the conditional distribution of $N(t)$ given $N(\ell)$ and $N(r)$ for $\ell < t < r$ is

$$N(t) \mid \big(N(\ell), N(r)\big) \sim \mathrm{Bin}\left(N(r) - N(\ell), \frac{t-\ell}{r-\ell}\right).$$

Because $N(t)$ has independent increments, the conditional distribution in an interval (a,b) given the process over $[0,a]$ and $[b,\infty)$ is the same as the conditional distribution in (a,b) given $N(a)$ and $N(b)$. Under this distribution the $N(b)-N(a)$ points of the process within $[a,b)$ may be sampled independently from the $\mathbf{U}[a,b)$ distribution (and then sorted). We may therefore sample the Poisson process on $[0,T]$ by the following method

$$\begin{aligned} N(T) &\sim \mathrm{Poi}(T\lambda) \\ N(T/2) &\sim \mathrm{Bin}(N(T), 1/2) \end{aligned}$$

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Algorithm 6.3 Recursive sampling of Poisson process on $[0,T]$

PoiRecursive(λ, m, s, u, v, b **)**

// Find $N(s_j)$ for distinct $s_1,\dots,s_m \in [0,T]$
// using u, v, b, precomputed by Algorithm 6.1

for $j = 1$ to m **do**
  if $u_j > 0$ **and** $v_j > 0$ **then**
    $N(s_j) \sim \mathrm{Bin}\big(N(s[v_j]) - N(s[u_j])\ , b_j\big)$
  else if $u_j > 0$ **and** $v_j = 0$ **then**
    $N(s_j) \leftarrow N(s[u_j]) + \mathrm{Poi}(\lambda(s_j - s[u_j]))$
  else if $u_j = 0$ **and** $v_j > 0$ **then**
    $N(s_j) \sim \mathrm{Bin}\big(N(s[v_j])\ , b_j\big)$
  else if $u_j = 0$ **and** $v_j = 0$ **then**
    $N(s_j) \sim \mathrm{Poi}(\lambda s_j)$
return $N(s_1),\dots,N(s_m)$

Note: $s[u]$ is shorthand for s_u.

$$N(T/4) \sim \mathrm{Bin}(N(T/2), 1/2)$$
$$N(3T/4) \sim \mathrm{Bin}(N(T) - N(T/2), 1/2),$$

and so on, producing values $N(s_1),\dots,N(s_m)$. Algorithm 6.3 generates $N(s_j)$ at distinct times $s_j \in [0,T]$ presented in any order, for which $\max_j s_j = T$. When we need the actual event times we may sample them as follows. Let $0 < s_{(1)} < \cdots < s_{(m)} = T$ be the s_j sorted. Then for $j = 1,\dots,m$ we draw $N(s_{(j)}) - N(s_{(j-1)})$ points from $\mathbf{U}(s_{(j-1)}, s_{(j)}]$ interpreting $s_{(0)}$ as 0.

6.7 Non-Poisson point processes

The Poisson assumption is a great simplification. Given the number N of points in the process, their locations $\boldsymbol{T}_1,\dots,\boldsymbol{T}_N$ are IID from a density proportional to λ. Some phenomena are not well modeled by this independence. For example, in forestry, the positions of trees often exhibit some non-Poisson behavior. The existence of a tree at point $\boldsymbol{T}$, may make it more likely that there is another one nearby, if the trees spread their seeds locally. The presence of a tree at $\boldsymbol{T}$ could also reduce the number of nearby trees, due to competition for sunlight or space. Figure 6.14 shows two spatial data sets: some insect cell centers which avoid coming near to each other, and some tree locations that appear to come closer to each other than independent random points would.

First, we consider processes that induce more clustering than homogeneous Poisson processes do. Many of them can be simulated directly from their definitions. A ***Cox process*** is a point process generated as follows: a random function $\lambda(\cdot) \geqslant 0$ is generated on $\mathcal{T}$, and then given $\lambda(\cdot)$, the points $\boldsymbol{T}_i$ are

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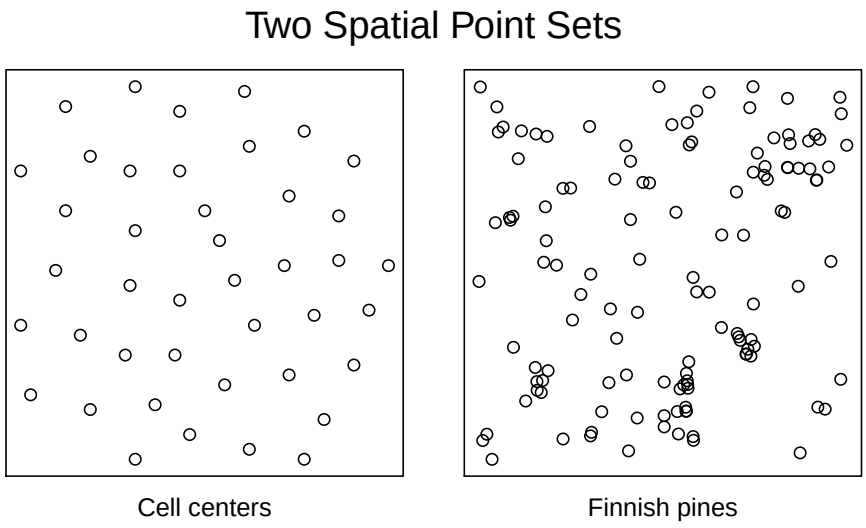


Figure 6.14: The left panel shows centers of some cells in a developing insect (Ripley, 1977). The right panel shows locations of pine trees from a site in Finland (Van Lieshout, 2004). The cell data originated with F. Crick, the tree data with A. Penttinen. Both data sets are in the R package `spatstat`.

sampled as a Poisson process with intensity function λ. In ***Matérn's cluster process***,

$$\lambda(\boldsymbol{t}) = \mu \sum_{i=1}^{\infty} \mathbb{1}_{\|\boldsymbol{t}-\boldsymbol{x}_i\| \leqslant R} \tag{6.38}$$

where $\boldsymbol{x}_i$ are the sampled values of a homogeneous Poisson process with intensity λ_0, and μ and R are positive parameters. The ***Thomas process*** has

$$\lambda(\boldsymbol{t}) = \mu \sum_{i=1}^{\infty} \frac{1}{2\pi\sigma} \exp(-\|\boldsymbol{t} - \boldsymbol{x}_i\|^2/(2\sigma^2)), \tag{6.39}$$

where once again $\boldsymbol{x}_i$ are from a homogeneous Poisson process with parameter λ_0, and μ and σ are positive parameters. The Thomas process has a smooth intensity function. We can generalize (6.38) and (6.39) to incorporate spatial distributions other than uniform in a circle, or Gaussian, and we can use dimension $d > 2$ as well. In the ***log Gaussian Cox process***

$$\lambda(\boldsymbol{t}) = \exp(Z(\boldsymbol{t})) \tag{6.40}$$

for a Gaussian random field $Z(\cdot)$.

Although it is conditionally a Poisson process, the Cox process does introduce dependence. When we observe a point of the process at $\boldsymbol{T}$ then it is more

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likely that $\lambda(\boldsymbol{T})$ is large there. If $\lambda(\cdot)$ is reasonably smooth, then it is likely that $\lambda(\cdot)$ is also large in a neighborhood of $\boldsymbol{T}$. As a result, seeing a point at $\boldsymbol{T}$ makes it more likely that there is another point nearby.

Some of the Cox processes can be simulated directly from their definitions. Given the seeds $\boldsymbol{x}_i$, we can generate Matérn's cluster process by taking $N_i \sim \mathrm{Poi}(\mu\pi R^2)$ points uniformly within the circle $\{\boldsymbol{T} \mid \|\boldsymbol{T} - \boldsymbol{x}_i\| \leqslant R\}$. We sample independently for each seed $\boldsymbol{x}_i$. If we want to sample $\boldsymbol{T}_i$ over the rectangle $\mathcal{R} = [a_1, b_1] \times [a_2, b_2]$ then we can be sure to sample all the relevant seeds by taking $\boldsymbol{x}_i$ to be a point process on $\mathcal{R}_+ = [a_1 - R, b_1 - R] \times [a_2 + R, b_2 + R]$. For the Thomas process, we include $N_i \sim \mathrm{Poi}(\mu)$ points from the $\mathcal{N}(\boldsymbol{x}_i, \sigma^2 I_2)$ distribution. We should widen the sampling region for $\boldsymbol{x}_i$ by a multiple of σ to ensure that most of the relevant seed points are generated.

For more general functions λ, we can get a good approximation, at least for small dimensions d, by breaking $\mathcal{T}$ into smaller regions and working as if λ were constant within those regions. For example, if $\boldsymbol{T} = [0,1]^2$ then we can form an $M_1 \times M_2$ grid of values

$$G = \{\boldsymbol{g}_{j_1 j_2} \mid 1 \leqslant j_1 \leqslant M_1,\ 1 \leqslant j_2 \leqslant M_2\}, \quad \boldsymbol{g}_{j_1 j_2} = \Big(\frac{j_1 - 1/2}{M_1}, \frac{j_2 - 1/2}{M_2} \Big)$$

and sample $\lambda(\boldsymbol{g})$ for all $\boldsymbol{g} \in G$. Then, we can proceed as follows:

$$\begin{aligned}
&\Lambda = \frac{1}{M_1 M_2} \sum_{j_1=1}^{M_1} \sum_{j_2=1}^{M_2} \lambda(\boldsymbol{g}_{j_1 j_2}),\\
&N \sim \mathrm{Poi}(\Lambda), \quad \text{then for } i = 1, \ldots, N,\\
&\boldsymbol{C}_i = \boldsymbol{g}_{j_1 j_2}, \quad \text{with probability proportional to } \lambda(\boldsymbol{g}_{j_1 j_2}),\\
&\boldsymbol{U}_i \sim \mathbf{U}(-1,1)^2, \quad \text{and}\\
&\boldsymbol{T}_i \sim \boldsymbol{C}_i + \Big(\frac{U_{i,1}}{2M_1}, \frac{U_{i,2}}{2M_2} \Big).
\end{aligned} \tag{6.41}$$

The quantity Λ is our estimate of $\int_0^1 \int_0^1 \lambda(\boldsymbol{t})\, \mathrm{d}\boldsymbol{t}$. Then N is the number of points in the process, $\boldsymbol{C}_i$ are their grid centers, and $\boldsymbol{U}_i$ give their offsets within rectangles surrounding the grids points.

Nonsquare rectangles, $M_1 \neq M_2$, are useful when the process $\lambda(\cdot)$ varies more strongly in one direction than another. They also are helpful in debugging. A graphical display of $\lambda(\cdot)$ on a 50×51 grid can expose some errors one might not see in a 50×50 grid.

Cox processes are better at generating clumping than they are at generating points that avoid each other. If $\lambda(\cdot)$ consists of many well separated narrow modes, then $\boldsymbol{T}_i$ will tend to be spaced as far apart as those nodes. But we still have the problem of finding a way to generate such a $\lambda(\cdot)$, and the Poisson sampling that follows could put two or more points into one of those narrow modes. As a result, Cox processes are not a good choice when the points of the process must not come close together.

The simplest model for points that cannot approach each other too closely is the ***hard core model***. In this model $\boldsymbol{T}_i$ are points of a homogeneous Poisson

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process on $[0,1]^2$, subject to the condition that $\min_{1\leqslant i<i'\leqslant N}\|\boldsymbol{T}_i-\boldsymbol{T}_{i'}\|\geqslant\delta>0$. Sometimes a periodic boundary is used, so that, for example, points $(1/2,\delta/2)$ and $(1/2,1-\delta/2)$ are at distance δ and hence overlap. A naive way to sample the hard core model is to generate points $\boldsymbol{T}_1,\dots,\boldsymbol{T}_N$ from a homogeneous Poisson process, and then reject them all if any interpoint distance is less than or equal to δ.

It is more efficient to sample $\boldsymbol{T}_i$ sequentially, rejecting any point that is too close to one of its predecessors. This approach, known as ***dart throwing*** in computer graphics is usually done for a fixed target number N of points. That is, we condition both the number of points and their separation. For large enough N, we could find that there is no legal place for $\boldsymbol{T}_i$ for some $i\leqslant N$. Then we may discard points $\boldsymbol{T}_1,\dots,\boldsymbol{T}_{i-1}$ and start over. At very high densities, dart throwing becomes very expensive. Then Markov chain Monte Carlo methods in Chapter 11 can be used to get a sample with approximately the hard core distribution.

Under the hard core model, when there are N points, the dependence is simply that no pair of points can be too close. It is interesting to consider more general behavior, such as points that attract each other at some distances, or repel each other but incompletely at other distances. Models of this kind are also sampled using Markov chain Monte Carlo (Chapter 11).

Deciding which spatial process to use can be harder than sampling from them. See Cressie (2003, Chapter 8) for methods of fitting spatial process models to observations.

To describe deviations from a Poisson process, we should look first at the distribution of pairs of points. For a process with homogeneous rate $\lambda>0$, we can use ***Ripley's K function*** (Ripley, 1977). If $\boldsymbol{T}$ is an arbitrary point of the process, let $S(\boldsymbol{T},h)$ be the set of points $\boldsymbol{T}_i$ of the process, not including $\boldsymbol{T}$ for which $\|\boldsymbol{T}_i-\boldsymbol{T}\|<h$. Then

$$K(h)=\lambda^{-1}\mathbb{E}\big(|S(\boldsymbol{T},h)|\big), \tag{6.42}$$

with $|S|$ denoting cardinality. For a Poisson process in $\mathbb{R}^2$, $K(h)=\pi h^2$. For a process with local clustering, $K(h)>\pi h^2$ for small $h>0$, while regularly spaced processes, like the one generating the cell centers in Figure 6.14 have $K(h)<\pi h^2$ for small $h>0$. The K-function does not describe everything about the process's pairwise distribution; for example it does not capture a tendency to cluster more in one direction than another.

6.8 Dirichlet processes

The Dirichlet process is used when we need to model a distribution that has itself been randomly selected from a process that generates distributions. We define the process here, then develop the Chinese restaurant process from it, and apply these for the Dirichlet process mixture model in §6.8.

Let the random distribution be F. To be concrete, we assume that F is the distribution of a random vector $\boldsymbol{X}\in\Omega\subseteq\mathbb{R}^d$. Now suppose that we split Ω up

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as follows: $\Omega = A_1 \cup A_2 \cup \cdots \cup A_m$ where $A_i \cap A_j = \varnothing$ for $i \neq j$. This partition of Ω defines a vector

$$(F(A_1), \ldots, F(A_m)) \in \Delta^{m-1} \equiv \Big\{ (p_1, \ldots, p_m) \mid p_j \geqslant 0, \sum_{j=1}^{m} p_j = 1 \Big\}. \quad (6.43)$$

Here we have written $F(A_j)$ as a short form for $\mathbb{P}(\boldsymbol{X} \in A_j \mid F)$.

We encountered the unit simplex Δ^{m-1} in §5.4 on the Dirichlet distribution. If F is random then the vector on the left of (6.43) is a random point in Δ^{m-1}. It is natural to suppose that F is drawn in such a way that $(F(A_1), \ldots, F(A_m))$ has a Dirichlet distribution. The Dirichlet distribution is one of the simplest distributions on the simplex. The complexity in defining the Dirichlet process is to arrange for a Dirichlet distribution to hold simultaneously for any finite partition of Ω.

The Dirichlet process is defined in terms of a scalar $\alpha > 0$ and a distribution G on Ω. In the Dirichlet process

$$(F(A_1), \ldots, F(A_m)) \sim \mathrm{Dir}\big(\alpha G(A_1), \ldots, \alpha G(A_m)\big).$$

The Dirichlet process is written as either $F \sim \mathsf{DP}(\alpha, G)$ or $F \sim \mathsf{DP}(\alpha G)$, whichever is more convenient for a particular purpose. Because the components of a Dirichlet vector have a Beta distribution, we find that $F(A_j) \sim \mathrm{Beta}(\alpha G(A_j), \alpha(1 - G(A_j)))$. Therefore, using moments of the Beta distribution from Example 4.29,

$$\begin{aligned} \mathbb{E}(F(A_j)) &= G(A_j), \quad \text{and} \\ \mathrm{Var}(F(A_j)) &= G(A_j)(1 - G(A_j))/(\alpha + 1). \end{aligned}$$

The random F has a distribution centered on G and α governs the distance between F and G. See Exercise 6.26 for the covariance of $F(A)$ and $F(B)$ for two sets $A, B \subseteq \Omega$.

The Dirichlet process is used as a prior distribution in nonparameteric Bayesian inference. Suppose that $F \sim \mathsf{DP}(\alpha, G)$ and that conditionally on F, the random vectors $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n$ are independent samples from F. For inference on F we want the posterior distribution of F given $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$.

The problem is simplest for $n = 1$. Suppose that $\boldsymbol{X}_1 \in A_j$. Then the posterior distribution of $(F(A_1), \ldots, F(A_m))$ given $\boldsymbol{X}_1$ is

$$\mathrm{Dir}\big(\alpha G(A_1), \ldots, \alpha G(A_{j-1}), \alpha G(A_j) + 1, \alpha G(A_{j+1}), \ldots, \alpha G(A_m)\big). \quad (6.44)$$

Equation (6.44) holds for any partition. It describes a Dirichlet process. Inspecting it we see that $F \mid \boldsymbol{X}_1$ has the $\mathsf{DP}(\alpha G + \delta_{\boldsymbol{X}_1})$ distribution where $\delta_{\boldsymbol{x}}(A_j) = \mathbf{1}_{\{\boldsymbol{x} \in A_j\}}$. Incorporating $\boldsymbol{X}_2$ through $\boldsymbol{X}_n$, we find that

$$F \mid (\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n) \sim \mathsf{DP}\Big(\alpha G + \sum_{i=1}^{n} \delta_{\boldsymbol{X}_i}\Big).$$

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The Chinese restaurant process

Now suppose that we want to sample $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$ from the two stage model: $F \sim \mathsf{DP}(\alpha, G)$ then $(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n) \mid F$ are IID from F. First we sample $\boldsymbol{X}_1$. Let $A \subset \Omega$. By considering the partition $A_1 = A$ and $A_2 = \Omega - A$ we find

$$\mathbb{P}(\boldsymbol{X}_1 \in A) = \mathbb{E}(\mathbb{P}(\boldsymbol{X}_1 \in A \mid F)) = \mathbb{E}(F(A)) = G(A). \tag{6.45}$$

Therefore when $\boldsymbol{X} \sim F$ for $F \sim \mathsf{DP}(\alpha, G)$ the unconditional distribution of $\boldsymbol{X}$ is just $\boldsymbol{X} \sim G$ for any $\alpha > 0$. We can sample $\boldsymbol{X}$ without first sampling F from its Dirichlet process distribution.

The next step is a little surprising, at least at first. For $i \geqslant 2$ we sample $\boldsymbol{X}_i$ conditionally on $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_{i-1}$. There are two steps: first we identify the conditional distribution of F given $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_{i-1}$. Then we sample $\boldsymbol{X}_i$ taking account of the updated distribution of F. The conditional distribution of F given $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_{i-1}$ is $\mathsf{DP}\bigl(\alpha G + \sum_{j=1}^{i-1} \delta_{\boldsymbol{X}_j}\bigr)$, which we may write as

$$F \mid (\boldsymbol{X}_1, \ldots, \boldsymbol{X}_{i-1}) \sim \mathsf{DP}\Bigl(\alpha + i - 1,\ \Bigl(\alpha G + \sum_{j=1}^{i-1} \delta_{\boldsymbol{X}_j}\Bigr)/(\alpha + i - 1)\Bigr).$$

As a result, we sample $\boldsymbol{X}_i$ from the distribution $(\alpha G + \sum_{j=1}^{i-1} \delta_{\boldsymbol{X}_j})/(\alpha + i - 1)$. This distribution is a mixture which samples a value from G with probability $\alpha/(\alpha+i-1)$ and otherwise repeats observation $\boldsymbol{X}_j$ with probability $1/(\alpha+i-1)$ for $j = 1, \ldots, i-1$. That is

$$\boldsymbol{X}_i = \begin{cases} \boldsymbol{Y} \sim G & \text{with probability } \alpha(\alpha + i - 1)^{-1} \\ \boldsymbol{X}_1 & \text{with probability } (\alpha + i - 1)^{-1} \\ \boldsymbol{X}_2 & \text{with probability } (\alpha + i - 1)^{-1} \\ \vdots & \vdots \\ \boldsymbol{X}_{i-1} & \text{with probability } (\alpha + i - 1)^{-1}. \end{cases} \tag{6.46}$$

The update (6.46) is called the Chinese restaurant process, based on the following metaphor. Suppose that customers $i = 1, \ldots, n$ come to a restaurant. Customer 1 picks table $\boldsymbol{X}_1$ by sampling from G. Customer 2 then either goes to a new table $\boldsymbol{Y}$ freshly sampled from G, with probability $\alpha/(1+\alpha)$ or joins the table $\boldsymbol{X}_1$ of customer 1 with probability $1/(1+\alpha)$. Customer i starts a new table with probability $\alpha/(\alpha+i-1)$ or otherwise goes the table of another randomly chosen customer. A table with more customers has greater chance of having customer i join.

Figure 6.15 shows one realization of the state of the restaurant after the first 25 customers have arrived. This example had $\alpha = 4$. Smaller α tends to lead to fewer unique tables. As n increases, the number of unique tables grows logarithmically.

The CRP is a reinforced random walk, like the Pólya urn process we considered in §6.2. That process has a similar agglomeration feature except that the

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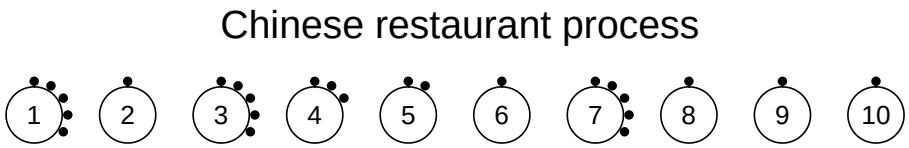


Figure 6.15: This figure shows a realization of the Chinese restaurant process described in the text, for $\alpha = 4$. The first 25 customers have arrived and they occupy 10 distinct tables.

CRP adds new tables from time to time, while the Pólya urn process we saw worked with a fixed set of ball colors.

The expected number of tables in use by the time n customers arrive is

$$\sum_{i=1}^{n} \frac{\alpha}{\alpha+i-1} \leqslant 1 + \int_0^n \frac{\alpha\,\mathrm{d}x}{1+x/\alpha} = 1 + \alpha\log(1+n/\alpha) \sim \alpha\log(n),$$

as $n \to \infty$. For some applications, we want the number of tables to grow more quickly than this. The Pitman-Yor process below allows for faster growth.

The stick-breaking representation

The CRP gives us another way to look at the Dirichlet process. If we were to sample $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$ for a very large n, then the empirical distribution $F_n = (1/n)\sum_{i=1}^n \delta_{\boldsymbol{X}_i}$ should be close to the distribution F that was randomly sampled from $\mathsf{DP}(\alpha, G)$. In the limit

$$F = \sum_{i=1}^{\infty} \pi_i \widetilde{\boldsymbol{X}_i}$$

where $\widetilde{\boldsymbol{X}_i}$ are the unique values (restaurant tables) sampled from G and π_i is the limiting fraction of customers who sit at that table. The values $\widetilde{\boldsymbol{X}_i}$ are IID from G. Suppose that we order the $\widetilde{\boldsymbol{X}_i}$ in decreasing order of their weights π_i. This order is not necessarily the order in which the unique values were observed. For example, it is possible that the second table sampled ends up with the most customers. It can be shown (references on page 68 of the end notes) that these random weights π_i can be generated by the following simple rule. First $\pi_1 = \theta_1$ and for $i \geqslant 2$, $\pi_i = \theta_i \prod_{1\leqslant j<i}(1-\theta_j)$ where θ_i are independent $\mathrm{Beta}(1,\alpha)$ random variables.

The representation

$$F = \sum_{i=1}^{\infty} \Big[\theta_i \prod_{1\leqslant j<i} (1-\theta_j)\Big] \widetilde{\boldsymbol{X}_i}$$

is called the ***stick-breaking*** representation of the Dirichlet process. We start with a stick of unit length, and break off a piece of length $\theta_1 \sim \mathrm{Beta}(1,\alpha)$.

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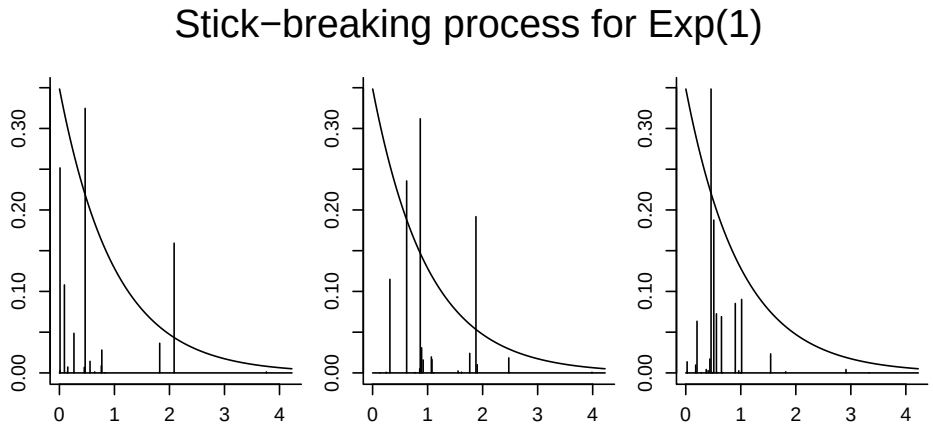


Figure 6.16: This figure shows three realizations of the stick-breaking construction for $\mathsf{DP}(\alpha, G)$ when $\alpha = 2$ and G is Exp(1). Only those components with $\pi_i > 10^{-4}$ are included. The Exp(1) probability density function appears as a reference curve.

Observation $\widetilde{\boldsymbol{X}}_1$ gets weight θ_1 and then all the rest of the observations have to share the remaining weight $1-\theta_1$ which we may think of as a stick of length $1-\theta_1$. Next we break the stick of length $1-\theta_1$ giving a piece of length $\theta_2(1-\theta_1)$ to observation $\widetilde{\boldsymbol{X}}_2$ and sharing the remaining weight $(1-\theta_1)(1-\theta_2)$ among observations $\widetilde{\boldsymbol{X}}_i$ for $i \geqslant 3$. Each new $\widetilde{\boldsymbol{X}}_i$ breaks the remainder of the stick keeping proportion θ_i and passing on the proportion $1-\theta_i$ to the subsequent observations.

Figure 6.16 shows three realizations of the stick-breaking process for $\mathsf{DP}(2, \text{Exp}(1))$. The weights π_i on the selected values $\widetilde{\boldsymbol{X}}_i$ bear no relationship to the height of the Exp(1) probability density function. That density does however influence where the nonzero weights are.

The Dirichlet process mixture model

In the Dirichlet process mixture model, we add one more level of sampling to the Chinese restaurant process. First $F \sim \mathsf{DP}(\alpha, G)$. Then given F, we have $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n \sim F$. Finally, the observations $\boldsymbol{Y}_i$ are conditionally independent from another distribution $H(\boldsymbol{y}; \theta)$ with parameter $\theta = \boldsymbol{X}_i$. Both F and the $\boldsymbol{X}_i$ are unobserved.

As a simple example, suppose that $\boldsymbol{Y}_i \sim \mathcal{N}(\boldsymbol{X}_i, \sigma_1^2 I)$ and that $G = \mathcal{N}(0, \sigma_0^2 I)$. Because the $\boldsymbol{X}_i$ come from a Chinese restaurant process, they will have lots of repeated values among them. Those common values correspond to clusters among the $\boldsymbol{Y}_i$. Figure 6.17 shows three examples with $G \sim \mathsf{DP}(\alpha, \mathcal{N}(0, \sigma_0^2 I))$ and $\boldsymbol{Y}_i \mid \boldsymbol{X}_i \sim \mathcal{N}(\boldsymbol{X}_i, \sigma_1^2 I)$ for $\sigma_0 = 3$ and $\sigma_1 = 0.4$.

We see in Figure 6.17 that the points appear to belong to a small number of

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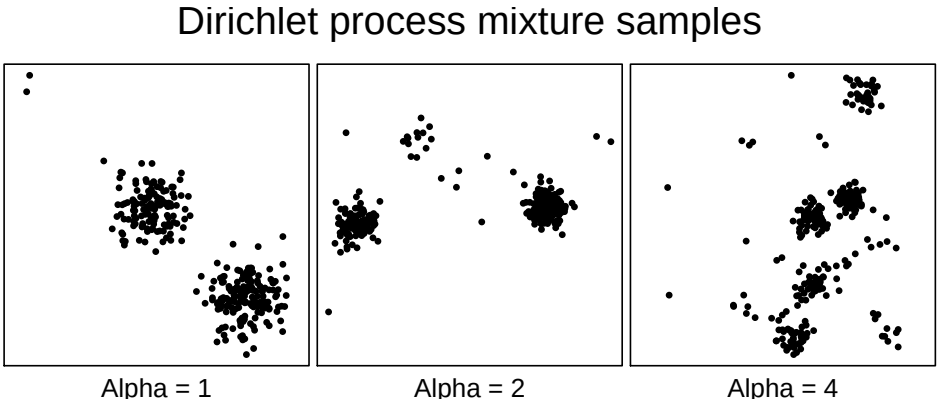


Figure 6.17: This figure shows realizations of the Dirichlet process mixture model described in the text. The parameter α takes values 1, 2, and 4 from left to right. In each case $\boldsymbol{Y}_1, \ldots, \boldsymbol{Y}_{200}$ are shown.

clusters, apart from a few outliers. The clusters typically correspond to points $\boldsymbol{X}_i$ of the CRP that had many repeats, although sometimes a cluster arises from two or more distinct $\boldsymbol{X}_i$ values that are close to each other. Outliers typically arise from values $\boldsymbol{X}_i$ that appeared only a small number of times, perhaps just once, in the CRP.

The large number of tied observations in the CRP is thus very natural when we want to model phenomena that exhibit clusters. The true number of clusters is treated as infinite, but some of the clusters are so rare that they are unlikely to be seen in a reasonably sized sample.

In applications we usually want to reverse this sampling process. For example, we may have data like those shown in Figure 6.17 and then we wish to estimate the cluster locations and their number. Markov chain Monte Carlo methods (beginning at Chapter 11) are well suited to that problem.

Pitman-Yor process

The Pitman-Yor process, $\mathsf{PY}(d, \alpha, G)$ from Pitman and Yor (1997), also has a Chinese restaurant representation. The parameters are a distribution G, and scalars $d \in [0, 1)$ and $\alpha > -d$. We will essentially subtract d (fractional) customers from each table. The case $d = 0$ recovers the Dirichlet process for which the number of tables in use by the first n customers grows proportionally to $\alpha \log(n)$. When $d > 0$, the chance that a customer chooses a new table is increased. The result is a greater number of distinct tables, growing proportionally to αn^d. The description here is based on Teh (2006).

If $F \sim \mathsf{PY}(d, \alpha, G)$ and $\boldsymbol{X}_i \sim F$ independently, then we can sample $\boldsymbol{X}_i$ as follows. First let $\boldsymbol{Y}_1, \boldsymbol{Y}_2, \ldots$ be independent samples from G. We use c_j to count the number of times Y_j has been used (initially $c_j = 0$) and t to represent the number of distinct indices j with $c_j > 0$ (initially $t = 0$) and $c_\bullet = \sum_{j=1}^{t} c_j$

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(initially $c_\bullet = 0$).

In sampling, the next customer to arrive goes to table J, where

$$J = \begin{cases} t+1 & \text{with probability } (\alpha + dt)/(\alpha + c_\bullet) \\ j & \text{with probability } (c_j - d)/(\alpha + c_\bullet) \text{ for } 1 \leqslant j \leqslant t. \end{cases}$$

and set $\boldsymbol{X}_{c_\bullet+1} = \boldsymbol{Y}_J$. The first step always takes $J = 1$ because initially $(\alpha + dt)/(\alpha + c_\bullet) = 1$ and there are no j with $1 \leqslant j \leqslant t = 0$. Therefore $\boldsymbol{X}_1 = \boldsymbol{Y}_1$.

When $J \leqslant t$ then we update the state information by taking $c_J \leftarrow c_J + 1$. When $J = t + 1$ then we put $c_J \leftarrow 1$ and $t \leftarrow t + 1$. In both cases $c_\bullet \leftarrow c_\bullet + 1$.

The CRP above samples a hierarchical model: $\boldsymbol{X}_i \sim F$ where $F \sim \mathsf{PY}(d, \alpha, G)$. Unlike the Dirichlet process, the Pitman-Yor process does not have a convenient form for the distribution of F itself, just the samples from F.

The Indian buffet process

In applications of the Dirichlet process mixture model, every data point belongs to one cluster, and is sampled from a distribution with a parameter appropriate to that cluster. The data point is the customer and the cluster is the table, in the Chinese restaurant process metaphor. The binary variable Z_{ik} takes the value 1 if and only if customer i is at table k, and because each customer is at exactly one table, $\sum_k Z_{ik} = 1$.

In some inference problems a data point may need to belong to more than one group. For example Z_{ik} might be 1 if point i has feature k, for non-exclusive features. For instance if point i is an animal, Z_{i1} might be 1 if i can fly, and Z_{i2} might be 1 if i can swim. It is as if customer i were present at multiple tables.

The Indian buffet process (IBP) accomodates such non-exclusive binary features. In the metaphor, customer i proceeds through a buffet at an Indian restaurant and either samples some food from dish k, setting $Z_{ik} = 1$, or does not, setting $Z_{ik} = 0$.

The process has a parameter $\alpha > 0$. It is sampled as follows. Initially, $Z_{ik} = 0$ for all $i \geqslant 1$ and all $k \geqslant 1$. The first customer draws an integer $D_1 \sim \mathrm{Poi}(\alpha)$ and then samples dishes $1, \ldots, D_1$, setting $Z_{ik} = 1$ for $1 \leqslant k \leqslant D_1$. Here we are choosing to label the first dishes sampled by numbers 1 to D_1, just as the first table used in the CRP is labeled table 1. If $D_1 = 0$, then all Z_{1k} are zero.

By the time the i'th customer enters the buffet, dish k has been sampled $m_k = \sum_{i'=1}^{i-1} Z_{i'k}$ times. If $m_k > 0$, then customer i samples it with probability m_k/i, setting $Z_{ik} = 1$. These dish sampling decisions of customer i are made independently of each other. Customer i then samples $D_i \sim \mathrm{Poi}(\alpha/i)$ new dishes. The number of distinct dishes that have been sampled before customer i arrives is $\overline{D}_{i-1} = \sum_{i'=1}^{i-1} D_{i'}$. To account for the new dishes sampled by customer i, we set $Z_{ik} = 1$ for $k = \overline{D}_{i-1} + 1, \ldots, \overline{D}_{i-1} + D_i \equiv \overline{D}_i$.

Figure 6.18 shows the Z matrix for an IBP with $n = 75$ customers and $\alpha = 20$. The total number of dishes selected was 93. This number is the

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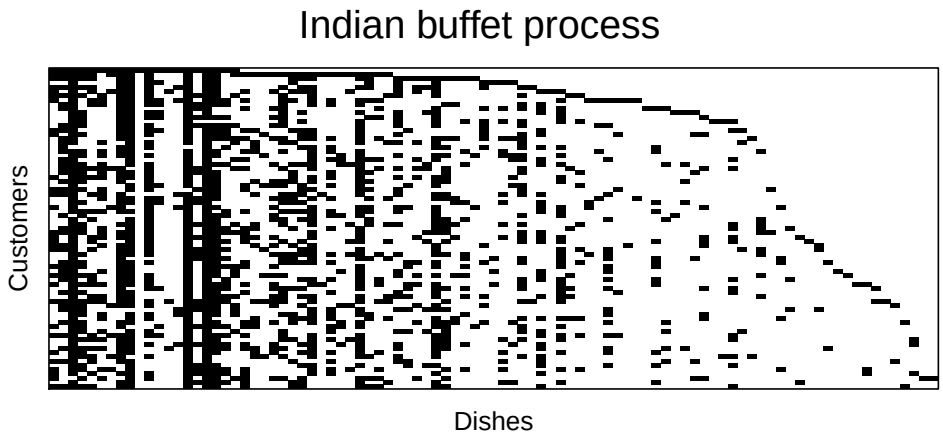


Figure 6.18: This figure shows one realization of the first 75 customers in an Indian buffet process, with $\alpha = 20$. Black squares indicates $Z_{ik} = 1$ for customer i and dish k. Customer 1 is in the top row and dish 1 is in the leftmost column.

realization of Poisson random variable with mean $\sum_{i=1}^{n} \alpha/i \doteq 98.0$. Customer 1 sampled 20 dishes. The third one ultimately became very popular; all but two customers sampled it. Some other dishes were sampled by just one customer.

6.9 Discrete state, continuous time processes

In this section we study processes that evolve in continuous time but take values in a discrete state space. Important examples of this type include chemical reaction processes, biological systems (e.g. predator-prey interactions and epidemics) and industrial processes (e.g. queues and inventory systems).

We will use chemical reaction processes to introduce the ideas. Chemical reaction processes are usually studied by differential equations. But in special circumstances, it is better to simulate them by Monte Carlo. For example, the interior of a cell may have only a small number of copies of a certain protein molecule. Then treating the abundance of that protein as a continuous variable could be misleading.

The main method we consider is ***Gillespie's method*** also called the ***residence-time algorithm***. It samples chemical systems one reaction at a time. In other physical sciences this or very similar methods are known as as ***kinetic Monte Carlo***.

Suppose that a well stirred system contains N different kinds of chemical species (molecules). The system is described by $\boldsymbol{X}(t) \in \{0, 1, \dots\}^N$ where $X_i(t)$ is the number of molecules of type i present at time $t \geqslant 0$. The process $\boldsymbol{X}(t)$ is not constant over time because these molecules participate in reactions of

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various kinds. Some simple examples are:

$$\begin{aligned} S_1 + S_2 &\xrightarrow{c_1} S_3, \\ S_4 + S_4 &\xrightarrow{c_2} S_5, \\ S_3 &\xrightarrow{c_3} S_1 + S_2, \\ \star &\xrightarrow{c_4} S_4, \quad \text{and} \\ S_5 &\xrightarrow{c_5} \star. \end{aligned}$$

The arrows above describe the nature of the change. In the first case one molecule each of S_1 and S_2 combine to form one molecule of S_3. When this happens $\boldsymbol{X}$ changes to $\boldsymbol{X} + \boldsymbol{\nu}_1$ where $\boldsymbol{\nu}_1 = (-1, -1, 1, 0, 0, \dots)$. The quantity c_1 denotes the speed of this reaction, which we discuss further below. The second example (called dimerization) has $\boldsymbol{\nu}_2 = (0, 0, 0, -2, 1, \dots)$. The third example is the reverse of the first one.

Reactions 4 and 5 above indicate spontaneous creation (respectively destruction) of molecules. The creation of S_4 might describe molecules entering the system. Alternatively, the creation of S_4 molecules might consume some other species that are so abundant that the reaction will not meaningfully change their concentrations in a realistic time period. As for the destruction of S_5, it might represent molecules leaving the system or being converted into an output that does not affect any other reactions and that we do not care to count.

The order of a reaction is the number of molecules appearing on the left side of the arrow. The order is strongly related to the speed. A second order reaction can only happen when the two necessary molecules are close together. The chance of "$S_i + S_j \to$ *products*" happening in a small interval $[t, t + \mathrm{d}t)$ is then proportional to $X_i(t)X_j(t)\,\mathrm{d}t$ because there are $X_i(t)X_j(t)$ suitable molecule pairs for this reaction. Whether the reaction happens may depend on the sizes of the molecules, how closely they need to approach each other, or whether the right part of S_i has be adjacent to the right part of S_j. Those variables determine the c's. For example the probability of reaction 1 happening in $[t, t + \mathrm{d}t)$ is modeled as $c_1 X_1(t)X_2(t)\,\mathrm{d}t + o(\mathrm{d}t)$.

The dimerization reaction "$S_i + S_i \to$ *products*" is special. It requires a pair of molecules of the i'th type to interact. There are $X_i(t)(X_i(t)-1)/2$ such pairs and so the reaction takes place at rate of the form $c_j X_i(t)(X_i(t) - 1)/2$.

The model for the probability of a reaction does not depend on the number of reaction products. Reactions $S_1 + S_2 \to S_3$ and $S_1 + S_2 \to S_6 + S_7$ may have different c_i but they both proceed at rate $c_i X_1(t)X_2(t)$.

A first order reaction involves just one kind of molecule. It proceeds at rate $c_i X_j(t)$. A zero'th order reaction, such as $\star \xrightarrow{c_4} S_4$ proceeds at a constant rate c_i.

In all of these cases, we may write the probability of reaction j happening in $[t, t + \mathrm{d}t)$ as $a_j(\boldsymbol{X}(t))\mathrm{d}t + o(\mathrm{d}t)$. The function a_j, called the ***propensity function*** of reaction j, equals the rate constant c_j times the appropriate polynomial in components of $\boldsymbol{X}(t)$. In general, if reaction j consumes $r_i \geqslant 0$ copies

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of molecule S_i then

$$a_j(\boldsymbol{X}) = c_j \prod_{i=1}^{N} \binom{X_i}{r_i}, \tag{6.47}$$

for a constant $c_j > 0$. Equation (6.47) describes what are called 'mass action kinetics'. The most commonly used reactions have at most two contributing molecule types.

We can now simulate the process directly. At time t we assume that reaction j is due to happen after a waiting time of $T_j = E_j/a_j(\boldsymbol{X}(t))$ where $E_1, \dots, E_M$ are independent Exp(1) random variables. The reaction that actually happens is the one with the smallest value of T_j. The step can be likened to setting M alarm clocks and the first one to ring determines the reaction time and type.

Instead of sampling M independent exponential random variables, we can instead sample directly the minimum one because $\min(T_1, \dots, T_M) \sim \mathrm{Exp}(1)/a_0$ where $a_0 \equiv \sum_{j=1}^{M} a_j(\boldsymbol{X}(t))$. To see this, write

$$\mathbb{P}(\min(T_1, \dots, T_M) \geqslant \tau) = \prod_{j=1}^{M} \mathbb{P}(T_j \geqslant \tau) = \exp\Bigl(-\sum_{j=1}^{M} a_j(\boldsymbol{X}(t))\Bigr).$$

The probability that this next reaction is of type j is $a_j(\boldsymbol{X}(t))/a_0$. (Exercise 6.23.) Algorithm 6.4 shows the Gillespie algorithm using this technique. It includes a test for $a_0 = 0$. When that happens, no further reactions are possible and so sampling should stop. It also has a bound S on the number of steps to take because a reaction might take an unreasonably large number of steps in time T.

Sampling all the M times and finding their minimum is sometimes called the ***first reaction method*** while the strategy in Algorithm 6.4 is called the ***direct method***. The direct method seems faster because it uses only two random numbers per time step. The first reaction method can be modified to a version that does not have to update all of the alarm clock times. In the ***next reaction*** method we keep track of critical times (alarm clocks) for each reaction type. Careful updating schemes and bookkeeping require us to only generate one exponential random variable per time step. Keeping the clocks in a priority queue speeds up the task of identifying the next reaction type. This technique is advantageous when M is very large. See page 69 of the chapter end notes for details.

Example 6.10. A discrete version of the Lotka-Volterra predator-prey model uses these reactions:

$$\begin{aligned} R &\xrightarrow{c_1} 2R, \\ R + L &\xrightarrow{c_2} 2L, \\ L &\xrightarrow{c_3} \star. \end{aligned}$$

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Algorithm 6.4 Gillespie's algorithm on $[0, T]$

Gillespie($\boldsymbol{x}_0$, $\boldsymbol{a}$, $\boldsymbol{\nu}$, T, S**)**

// Sample chemical reactions $\boldsymbol{a}$, $\boldsymbol{\nu}$ by Gillespie's algorithm starting at $\boldsymbol{x}_0$.
// Sample until sooner of time T or S steps.

$s \leftarrow 0, \quad t_0 \leftarrow 0, \quad \boldsymbol{X}(t_0) \leftarrow \boldsymbol{x}_0$
while $s < S$ **and** $t_s < T$ **do**
  $a_0 \leftarrow \sum_{j=1}^{M} a_j(\boldsymbol{X}(t_s))$
  if $a_0 > 0$ **then**
    $\tau \sim \mathrm{Exp}(1)/a_0$ // time to next reaction
    $J \leftarrow j$ with prob. $a_j(\boldsymbol{X}(t_s))/a_0$ // type of next reaction
    $\Delta \leftarrow \boldsymbol{\nu}_J$
  else
    $\tau \leftarrow \infty$ // $a_0 = 0 \Rightarrow$ no more reactions
    $\Delta \leftarrow (0, \ldots, 0)$
  $s \leftarrow s + 1$
  $t_s \leftarrow t_{s-1} + \tau$
  $\boldsymbol{X}(t_s) \leftarrow \boldsymbol{X}(t_{s-1}) + \Delta$
return $s, (t_0, \boldsymbol{X}(t_0)), \ldots, (t_s, \boldsymbol{X}(t_s))$

Notes: the simulation stopped early if $t_s < T$. Otherwise $\boldsymbol{X}(T) = \boldsymbol{X}(t_{s-1})$. If $\tau \leftarrow \infty$ is problematic, use some other very large value when $a_0 = 0$. The reaction type J can be sampled by the binary search method in §4.4.

Here R represents a prey species, such as rabbits, while L is a predator, such as lynx. Rabbits reproduce themselves at rate c_1, predation at rate c_2 decreases the rabbit count while increasing the number of lynx, and lynx die of natural causes at rate c_3.

The first reaction is sometimes written $\bar{F} + R \xrightarrow{c_1} 2R$ where $\bar{F}$ represents a food source F for the rabbits that is so abundant that it will not be materially depleted in a reasonable time. An alternative notation for such a reaction is "$F + R \xrightarrow{c_1} F + 2R$". The third reaction is sometimes also written $L \xrightarrow{c_3} Z$. The product, Z, denotes dead predators which we decide here not to keep track of.

This model is very simplistic, and more elaborate reaction sets are used in population models. As simple as it is, the Lotka-Volterra reaction set does demonstrate how predator-prey ecosystems might oscillate instead of converging to a fixed equilibrium point.

Letting X_1 be the number of rabbits and X_2 be the number of lynx, we have

$$\boldsymbol{\nu}_1 = (1, 0), \quad \boldsymbol{\nu}_2 = (-1, 1), \quad \text{and} \quad \boldsymbol{\nu}_3 = (0, -1).$$

The propensity functions are

$$a_1(\boldsymbol{X}) = c_1 X_1, \quad a_2(\boldsymbol{X}) = c_2 X_1 X_2, \quad \text{and} \quad a_3(\boldsymbol{X}) = c_3 X_2.$$

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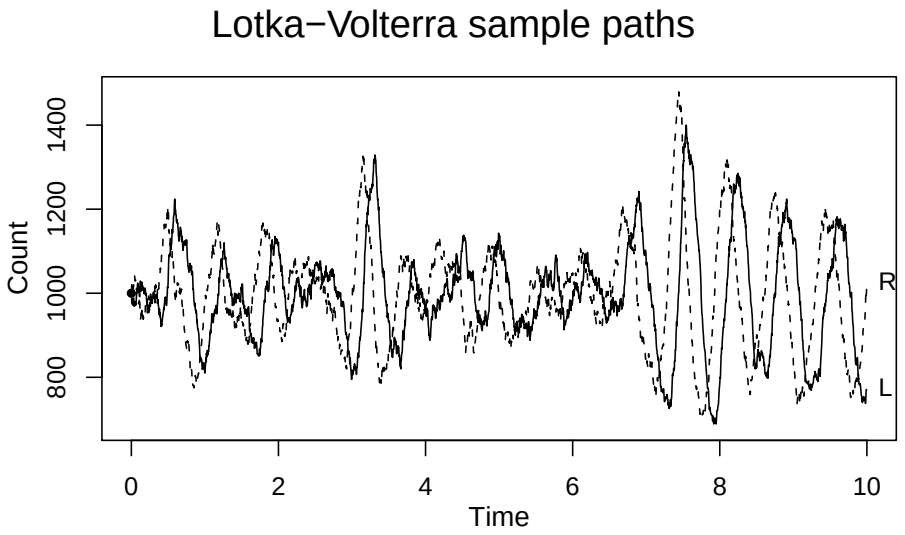


Figure 6.19: This figure shows a realization of predator (solid) and prey (dashed) counts versus time for a Lotka-Volterra model over the time interval $[0, 10]$. Every 100'th time point, out of just over 290,000 is shown. The model starts at $\boldsymbol{X}(0) = (1000, 1000)$ and goes through a series of oscillations of random amplitude.

If $X_1 = 0$ then X_2 will decrease to zero and the system will end up at $\boldsymbol{X} = (0, 0)$. If $X_2 = 0$ then X_1 will increase without bound and the system will approach $\boldsymbol{X} = (\infty, 0)$. In a more realistic model, another reaction would set in to prevent $X_1 \to \infty$.

The typical behavior of the system is to oscillate, at least before $\min(X_1, X_2) = 0$ happens. Figure 6.19 shows sample paths versus time for one realization using rates $c_1 = 10$, $c_2 = 0.01$ and $c_3 = 10$ from Gillespie (1977). When the prey population rises, the predator population grows soon afterwards. The rising predator population reduces the prey population which causes a fall in the predator population. The populations oscillate unevenly and out of phase. Figure 6.20 shows a trajectory giving predator versus prey counts for that same realization. The points $\boldsymbol{X}(t)$ tend to rotate clockwise as t increases.

Faster simulations

When the number of molecules is large, the Gillespie simulation can proceed slowly. If we are willing to make an approximation like the Euler-Maruyama one of §6.5, then it is possible to get a faster algorithm. If a reaction rate stays constant at level a for time τ, then the number of such reactions has the $\mathrm{Poi}(a\tau)$

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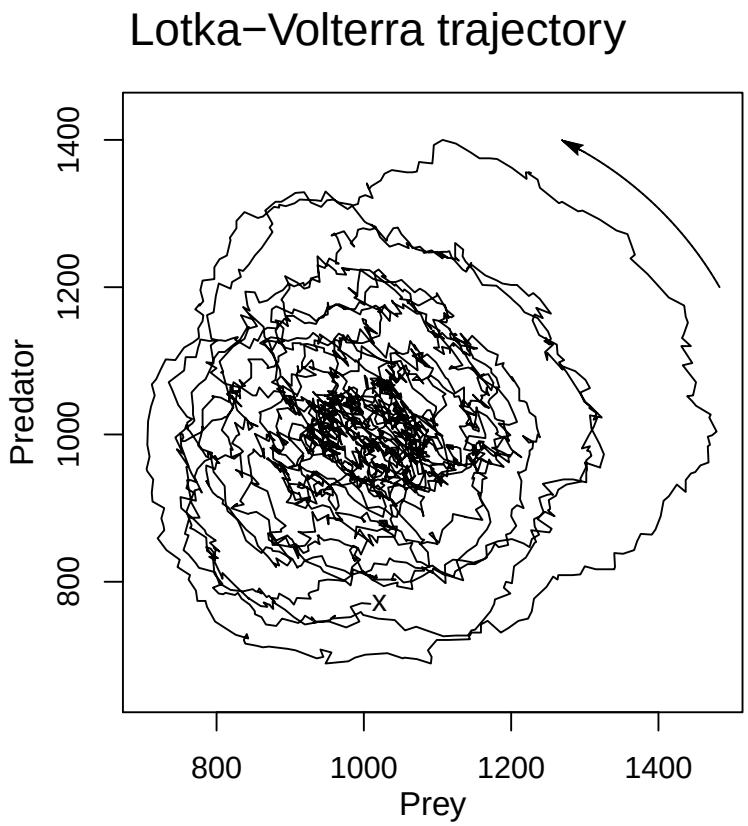


Figure 6.20: This figure plots predator versus prey counts for the sample paths from Figure 6.19. The trajectory started at $(1000, 1000)$ and ended at the X near $(1000, 800)$. It tends to rotate counter clockwise as shown by the reference arrow.

distribution. In a speed-up called ***τ-leaping***, we make the update

$$\begin{aligned} \boldsymbol{X}(t+\tau) &= \boldsymbol{X}(t) + \sum_{j=1}^{M} \boldsymbol{\nu}_j Y_j(t) \quad \text{where,} \\ Y_j(t) &\sim \mathrm{Poi}(\tau a_j(\boldsymbol{X}(t))) \quad \text{(independently).} \end{aligned} \tag{6.48}$$

The approximation ignores the fact that reactions taking place in the time interval $[t, t+\tau)$ cause changes to $\boldsymbol{X}$ which in turn cause changes to the rates. This feedback effect is small if τ is small enough. Small means that each $a_j(\boldsymbol{X}(s))$ is unlikely to undergo a large relative change over $t \leqslant s \leqslant t+\tau$.

One difficulty in τ-leaping is that we might leap to a state in which some component of $\boldsymbol{X}(t+\tau)$ is negative. There are numerous proposals for finding a large but reasonably safe value of τ to use. One method and some supporting

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literature are given on page 68 of the end notes. In some systems we may have a small number of components of $\boldsymbol{X}$ fluctuating near zero causing the system to choose small steps τ and hence proceed slowly.

The multilevel algorithms for stochastic differential equations discussed in §6.5 extend very well to continuous-time Markov chains. A multilevel algorithm makes it easier to handle negative values for chemical species. We may replace the j'th molecule update $X_j(t_s) \leftarrow X_j(t_{s-1}) + \Delta_j$ by $X_j(t_s) \leftarrow \max(X_j(t_{s-1}) + \Delta_j, 0)$. The resulting simulation is biased. But because multi-level simulation is based on a telescoping sum, the bias is largely canceled by each finer level of time resolution that is used. By coupling the finest level simulation to an exact simulation it is possible to remove bias completely using a small number of exact simulations. See page 69 of the end notes.

In certain circumstances τ is large enough that the Poisson distributed $Y_j(t)$ are approximately normally distributed, while also being small enough that feedback effects are negligible. In that case we could make the update

$$\boldsymbol{X}(t+\tau) = \boldsymbol{X}(t) + \left(\sum_{j=1}^{M} \boldsymbol{\nu}_j a_j(\boldsymbol{X}(t))\right)\tau + \sum_{j=1}^{M} \boldsymbol{\nu}_j \sqrt{\tau a_j(\boldsymbol{X}(t))} Z_j \tag{6.49}$$

where Z_j are independent $\mathcal{N}(0,1)$ random variables. Equation (6.49) is an Euler-Maruyama algorithm for the SDE

$$\mathrm{d}\boldsymbol{X} = \left(\sum_{j=1}^{M} \boldsymbol{\nu}_j a_j(\boldsymbol{X}(t))\right)\mathrm{d}t + \sum_{j=1}^{M} \boldsymbol{\nu}_j \sqrt{a_j(\boldsymbol{X}(t))}\,\mathrm{d}B_j(t), \tag{6.50}$$

known as the ***chemical Langevin*** equation. In very large systems, the drift term in (6.50) completely dominates the random diffusion term. In such cases, Monte Carlo is not needed. The system is rewritten using chemical concentrations instead of molecule counts and then solved using differential equations.

Chapter end notes

For background on stochastic processes in general, there is the book by Rosenthal (2000). A key result is Kolmogorov's extension theorem which ensures that if there are no contradictions among a set of finite dimensional distributions then there exists a process with those finite dimensional distributions.

For sequential methods in statistics, see the book by Siegmund (1985). Some applications to educational testing are presented by Finkelman (2008).

The urn process was given in the second part of Pólya (1931). Pemantle (2007) gives a survey of reinforced random walks and their many applications. Applications to economics were made by Brian Arthur, including a well known article, Arthur (1990), in Scientific American.

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Gaussian processes and fields

For background on Brownian motion see Borodin and Salminen (2002). The principal components construction for Brownian motion of Akesson and Lehoczky (1998) was used by Acworth et al. (1997) for some financial simulations. The value of Brownian bridge sampling for Brownian motion was recognized by Caflisch and Moskowitz (1995) who applied it to some financial problems.

For background on Gaussian random fields see Cressie (2003). The Matérn covariance has been strongly advocated by Stein (1999) as a better choice than the Gaussian covariance model. The parameterization of the Matérn covariance in Example 6.3 is in the form used by Ruppert et al. (2003). A different version is sometimes used in machine learning (Rasmussen and Williams, 2006).

Poisson processes

The exponential spacings method for non-homogeneous Poisson processes on $[0,\infty)$ was used by Lewis and Shedler (1976) to simulate $\mathsf{NHPP}(\exp(\alpha_0+\alpha_1 t))$. The time transformation $\Lambda^{-1}(t)$ used to transform a homogeneous process into a non-homogeneous one is in Çinlar (1975). Thinning is due to Lewis and Shedler (1979).

The Zipf-Poisson ensemble of Example 6.8 is from Dyer and Owen (2012). They use it to show how slowly the correct underlying ranking of items emerges in a sample. Their analysis extends to other long-tailed distributions.

The Poisson line field is described here in the way that Solomon (1978) presents it. For a proof that the Poisson field is the unique invariant distribution for lines in the plane, see Kendall and Moran (1963). For further discussions, including random planes and random rotations, see Mathai (1999) or Moran (2006). Abdelghani and Davies (1985) use random line models in chemical engineering and Gray et al. (1976) use them to model networks of cracks.

Stochastic differential equations

Stochastic differential equations are described in the texts by Karatzas and Shreve (1991), Oksendal (2003) and Protter (2004). Kloeden and Platen (1999) provide a comprehensive treatment of methods for their solution. The notation $\mathrm{d}X_t = a(t,X_t)\,\mathrm{d}t + b(t,X_t)\,\mathrm{d}B_t$ is a short form for the integral equation

$$X_t = X_0 + \int_0^t a(s,X_s)\mathrm{d}s + \int_0^t b(s,X_s)\mathrm{d}B_s$$

defined in those texts.

Platen and Heath (2006, Chapter 7) describe the standard conditions under which an SDE has a unique strong solution. In keeping with the level of this book we have left out measurability conditions. Tanaka's SDE example is in Tanaka (1963).

The exact solution for geometric Brownian motion is a straightforward consequence of Itô's well known lemma (Itô, 1951). See Exercise 6.9.

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There are so many SDE sampling schemes because there are several changeable aspects of an SDE algorithm and multiple choices for each aspect that we might change. The first changeable aspect is whether we seek better strong convergence so that $\gamma > 1/2$ versus improved weak convergence, $\beta > 1$. For either strong or weak convergence there are several attainable rates worth considering. The higher the rate we want, the higher are the derivatives of the drift and diffusion coefficients that we must consider. When those derivatives are computationally unpleasant there are schemes which replace some or all of them by judiciously constructed divided differences. Just as the Euler-Maruyama scheme is a natural generalization of Euler's method for solving the deterministic differential equation $\mathrm{d}x/\mathrm{d}t = a(t,x)$ subject to $x(0) = x_0$, other numerical approaches to solving differential equations (e.g. Runge-Kutta and implicit methods) have stochastic counterparts. There are SDE schemes for both stationary and nonstationary processes, versions for vector valued processes, and versions that replace the normal random variables Z_k used at each step by discrete random variables whose first few moments match those of $\mathcal{N}(0,1)$. Combining all of these factors leads to an explosion of choices.

Kloeden and Platen (1999) provide a comprehensive discussion of schemes for SDEs and their properties. The summary appearing just before their Chapter 1 is an excellent entry point for the reader seeking more detailed information.

Multilevel Monte Carlo

Heinrich (1998, 2001) used multilevel Monte Carlo methods to approximate entire families of integrals of the form $\int f(\boldsymbol{x};\theta)\,\mathrm{d}\boldsymbol{x}$ for $\theta \in \Theta \subset \mathbb{R}^d$. Giles (2008b) developed a multilevel method for sampling SDEs and showed how a multilevel Euler-Maruyama scheme can get an MSE within a logarithmic factor of $O(C^{-1})$ where C is the total number of simulated steps. Giles (2008a) gives empirical results on Milstein versions of multilevel Monte Carlo. The cost rate $O(\epsilon^{-2})$ appears to hold for some path-dependent exotic options (Asian, lookback, barrier and digital options). Big improvements for those options come from finding a better strategy for f than simply applying it to a piecewise linear approximation, but then it becomes challenging to couple the coarse and fine paths.

Hutzenthaler et al. (2011) show that the multilevel scheme can diverge if the SDE has drift and diffusion that are not both globally Lipshitz.

Taking $M = 2$ in multilevel Monte Carlo is convenient, but not necessarily optimal. The discussion in Giles (2008b) suggests that $M = 7$ might give the greatest accuracy, reducing mean squared error by about a factor of 2 compared to $M = 2$. In practice, $M = 2$ allows many more levels to be used in the simulation, and having more levels makes it easier to compare observed to expected rates of convergence for bias and variance. For problems where the convergence rate is not yet known, having more levels is advised.

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Square root diffusions

The CIR model is due to Cox et al. (1985). Andersen et al. (2010) survey simulation of square root diffusions. Higham and Mao (2005) provide a justification for replacing $\sqrt{X_t}\mathrm{d}B_t$ by $\sqrt{|X_t|}\mathrm{d}B_t$ in numerical schemes for square root diffusions. Lord et al. (2010) find that taking $\widehat{X}_{t+\Delta} = \alpha(r - \max(0, \widehat{X}_t)) + \sqrt{\max(0, X_t)}\mathrm{d}B_t$ works well, particularly when the square root diffusion is used in the Heston model. They change the drift term too, not just the diffusion term. Moro and Schurz (2007) present strategies for preventing simulated square root diffusions from ever taking negative values. They also describe the conditions under which the CIR process avoids 0. The material on constant elasticity of variance models is based on Linetsky and Mendoza (2010).

Spatial processes

For more information on spatial point processes, including methods to estimate their parameters, see Cressie (1991) and Diggle (2003). Baddeley (2010) discusses software for fitting, graphing and sampling spatial point processes. Fiume and McCool (1992) present a hierarchical dart throwing algorithm for use in computer graphics. Ripley (1977) describes several different, but closely related, hard core models.

Dirichlet processes

The Dirichlet process was introduced by Ferguson (1973). The stick-breaking construction is presented in Sethuraman (1994). A survey of Dirichlet and related models for machine learning problems appears in Jordan (2005), which was presented at NIPS 2005. Teh (2006) uses the Pitman-Yor process for language modeling.

Discrete states and continuous time

The Gillespie method was introduced in a pair of papers Gillespie (1976, 1977) that proved the theoretical and practical utility of the method. While such algorithms had been used earlier, for example by Kendall (1950) and Bartlett (1953), their relevance to chemical reactions was new and somewhat controversial. The algorithm is also known as the ***residence-time algorithm*** of Cox and Miller (1965). The article by Higham (2008) gives a good description of various chemical reaction simulations.

The τ-leaping algorithm is due to Gillespie (2001). Gillespie (2007) gives a survey of methods for selecting τ in τ-leaping. He favors the approach of Cao et al. (2006) which works as follows. The mean and standard deviation of the change $X_j(t+\tau) - X_j(t)$ to component X_j over time period τ can be determined from the reaction equations that X_j participates in. For each j they find the largest τ that keeps both the mean and standard and standard deviation below $\max(\epsilon X_j(t), 1)$. Call that τ_j. They then take $\tau = \min_{1 \leqslant j \leqslant d} \tau_j$. The algorithm

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is governed by the user's choice of $\epsilon \in (0,1)$. Notice that each chemical species is always allowed to have a change of size 1. This choice of τ can still yield negative components. In that case the step is ignored and a replacement is drawn.

The next reaction method is due to Gibson and Bruck (2000). At time $t = 0$, we initialize the system with $\boldsymbol{X}(0) = \boldsymbol{x}(0)$ and clocks set to $T_j \sim \mathrm{Exp}(1)/a_j(\boldsymbol{x}(0))$ for $J = 1, \ldots, M$. We then begin simulating time steps and reactions as follows. At time $t \geqslant 0$, the next reaction will take place at time $t' = \min(T_1, \ldots, T_M)$ and it is of type $j' = \arg\min_j T_j$. We then set $\boldsymbol{X}(t') = \boldsymbol{x}(t) + \boldsymbol{\nu}_{j'}(\boldsymbol{X}(t))$. Before sampling the next reaction we have to maintain proper times $T_1, \ldots, T_M$ for each reaction type. For the reaction type j' that actually took place we set $T_{j'} = t' + \mathrm{Exp}(1)/a_{j'}(\boldsymbol{X}(t'))$. For any reaction type j where $a_j(\boldsymbol{X}(t')) = a_j(\boldsymbol{X}(t))$ we leave T_j unchanged. Such a shortcut is justified by the memoryless property of the exponential distribution and it may apply to a great many of the reactions. For a reaction type j where $a_j(\boldsymbol{X}(t')) \neq a_j(\boldsymbol{X}(t))$ we set $T_j = t' + (T_j - t')a_j(\boldsymbol{X}(t))/a_j(\boldsymbol{X}(t'))$. This update amounts to speeding up or slowing down the j'th clock. Then set $t = t'$ and take another step, unless convergence criteria have been met.

The next reaction method requires some care because it is possible that $a_j(\boldsymbol{X}(t')) = 0$ if a reaction type momentarily becomes impossible perhaps because an input went to 0. Then $T_j = \infty$ which is locally reasonable. Future updates might make reaction type j possible again but the multiplicative update for T_j will then take the ill-defined form $\infty \times 0$. The way to handle this case is to leave $T_j = \infty$ until a time t' arises where $a_j(\boldsymbol{X}(t')) > 0$. Then set $T_j = t' + (t' - \widetilde{t}_j)\widetilde{a}_j/a_j(\boldsymbol{X}(t'))$ where $\widetilde{t}_j$ is the time at which a_j became 0 and $\widetilde{a}_j$ is the value a_j had just prior to becoming 0.

Anderson and Higham (2012) develop multilevel Monte Carlo methods for continuous time Markov chains. Example 8.1 in §8.6 describes the idea behind their strategy for coupling simulations using two different time step sizes. Multilevel simulations can attain a root mean squared error of size $\varepsilon > 0$ with computational cost $O(\varepsilon^{-2}\log(\varepsilon))$. That rate is proved under a Lipschitz condition on the propensity functions which does not hold for certain mass action kinetics models such as $a_j = c_j x_i(x_i - 1)/2$. If the system keeps $\boldsymbol{X}(t)$ inside a bounded region then mass action kinetics (6.47) are Lipschitz.

A Markov property

Markov processes have a convenient property: the present depends on the past and future only through the most recent past and the most immediate future. We used this property in simulating Brownian motion at points taken in an arbitrary order.

Let $Y_1, Y_2, \ldots$ be a real valued Markov chain. That is $\mathbb{P}(Y_j \leqslant y_j \mid Y_1, \ldots, Y_{j-1}) = \mathbb{P}(Y_j \leqslant y_j \mid Y_{j-1})$, for $j \geqslant 1$ and $y_j \in \mathbb{R}$. This chain need not be homogeneous, that is $\mathbb{P}(Y_j \leqslant y \mid Y_{j-1})$ might depend on j.

A well-known consequence of the Markov property is that the past is condi-

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tionally independent of the future, given the present:

$$\begin{aligned}&\mathbb{P}\big((Y_1, Y_2, \dots, Y_{j-1}) \in \mathcal{A},\ (Y_{j+1}, Y_{j+2}, \dots) \in \mathcal{B} \mid Y_j\big)\\ =&\mathbb{P}\big((Y_1, Y_2, \dots, Y_{j-1}) \in \mathcal{A} \mid Y_j\big)\mathbb{P}\big((Y_{j+1}, Y_{j+2}, \dots) \in \mathcal{B} \mid Y_j\big).\end{aligned} \tag{6.51}$$

We need to turn this result inside out to get our statement on the present given the past and future.

Proposition 6.1. *Let $Y_0, Y_1, \dots, Y_m \in \mathbb{R}$ be consecutive points of a Markov chain. Let ℓ and r be integers with $0 \leqslant \ell < r \leqslant m$ and $r - \ell \geqslant 2$. Then for $\mathcal{A} \in \mathbb{R}^{r-\ell-1}$,*

$$\begin{aligned}&\mathbb{P}((Y_{\ell+1}, \dots, Y_{r-1}) \in \mathcal{A} \mid Y_0, \dots, Y_\ell, Y_r, \dots, Y_m)\\ =&\mathbb{P}((Y_{\ell+1}, \dots, Y_{r-1}) \in \mathcal{A} \mid Y_\ell, Y_r).\end{aligned} \tag{6.52}$$

Proof. See Exercise 6.7. □

Exercises

6.1. Let $\boldsymbol{X}_t$ be Brownian motion on $\mathcal{T} = [0, 1]$. Let $T \sim \mathbf{U}(0, 1)$ and define the process $\boldsymbol{Y}_t$ on $\mathcal{T}$ via $\boldsymbol{Y}_t = \boldsymbol{X}_t$ for $t \neq T$ and $\boldsymbol{Y}_T = \boldsymbol{X}_T + 1$. Show that $\boldsymbol{X}_t$ and $\boldsymbol{Y}_t$ have the same finite dimensional distributions.

6.2. For the online education example of §6.2, suppose that testing can only go to $n_{\max} = 25$. The threshold for remediation is reduced to $\theta_R = 0.65$ to compensate. The threshold for mastery is still $\theta_M = 0.9$. The limits remain at $A = 1/19$ and $B = 19$.

a) Use a Monte Carlo simulation to estimate the fraction of students with mastery who end up with $n = 25$ and $\boldsymbol{X}_{25} < 0$ and hence are wrongly deemed to need remediation.

b) Use another simulation to estimate the fraction of students at the new remediation threshold who will end up with $n = 25$ and $\boldsymbol{X}_{25} > 0$ and hence be wrongly deemed to have mastered the material.

c) Present the histogram of the values N at which each of these SPRTs terminated.

6.3. The truncated SPRT used in the online education example suffers a well known inefficiency. We suppose that when the test runs to $n = n_{\max}$ steps that we decide in favor of mastery for $L_{n_{\max}} \geqslant 1$ and remediation otherwise.

Now it is possible to arrive at a log likelihood ratio L_n with $1 < L_n < B$ so that even if $\boldsymbol{Y}_{n+1} = \boldsymbol{Y}_{n+2} = \cdots = \boldsymbol{Y}_{n_{\max}} = 0$ we will still have $L_{n_{\max}} \geqslant 1$. In other words those last $n_{\max} - n$ questions cannot possibly change the decision and so we might as well not ask them. Similarly, we could arrive at a point where even if the student gets all the remaining questions right it will not be enough to show that the topic has been mastered. In a truncated SPRT with curtailment, we stop early if continued sampling is futile in this way.

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For the parameters in the previous exercise estimate the expected number of these futile questions when $\theta = \theta_R$. Repeat for $\theta = \theta_M$. Give 99% confidence intervals. State in clear mathematical notation how you compute the number of futile questions for a given trial. Hand in your source code, and be sure that it has clear internal documentation

6.4. Here we revisit the Pólya urn model of §6.2. As before $\boldsymbol{X}_t = (R_t, B_t)$ with $\boldsymbol{X}_0 = (1, 1)$, but now

$$\boldsymbol{Z}_t = \begin{cases} (1,0), & \text{with probability} \quad 2R_t^\alpha/(2R_t^\alpha + B_t^\alpha), \\ (0,1), & \text{with probability} \quad B_t^\alpha/(2R_t^\alpha + B_t^\alpha), \end{cases}$$

for $\alpha = 3/2$. In the economic context, the red product has an advantage: its customers are much better at convincing their friends to buy their product. So we expect the red company to have at least a 50% chance of winning the lion's share of the market, possibly more, but maybe not 100%. Suppose that the market follows these simplified rules. If $R_T > 20B_T$ for $T = 10{,}000$ then the red side wins. If instead $B_T > 20R_T$ then the black side wins. If neither happens, then the market is still up for grabs. Estimate the probability that the red side wins and give a 99% confidence interval. Repeat this for the probability that the black side wins, and for the probability that the market is not yet decided by time T.

6.5. Equation (6.10) describes how we can sample the Brownian bridge process. Prove equation (6.10) from the formula for the conditional distribution of some subcomponents of a multivariate Gaussian random vector.

6.6. Rewrite equation (6.10) in a form that allows us to assign $\ell_j = 0$ and $B(\ell_j) = 0$ for those j with $s_j < \min\{s_k, 1 \leqslant k < j\}$ and $r_j = \infty$ and $B(r_j) = 0$ for those j with $s_j > \max\{s_k, 1 \leqslant k < j\}$. The formula should work whether $\ell_j = 0$ or not and whether $r_j = \infty$ or not. The formula should not require computing ∞/∞ explicitly. Hint: work with some ratios of the sampling times.

6.7. Prove Proposition 6.1, showing that the middle of a Markov chain depends on the past and future only through the most recent past and the nearest future points.

6.8. Here we look at writing and testing an implementation of Algorithms 6.1 and 6.2. The idea is to ensure that they are equivalent to multiplying a Gaussian vector by a square root of the proper variance matrix. Do the following:

a) Write versions of Algorithm 6.1 and Algorithm 6.2, but modify Algorithm 6.2 so that instead of generating Z_i within the algorithm, the values of Z_i are passed in as a vector $\boldsymbol{Z}$ of length m. In ordinary use $\boldsymbol{Z} \sim \mathcal{N}(0, I_m)$.

b) Write a third function that generates times $s_1, \ldots, s_m$ as independent $\mathbf{U}(0,1)$ random variables, calls the setup Algorithm 6.1 with these times and saves the state vectors u, v, a, b and w. Then this function returns

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a matrix $C \in \mathbb{R}^{m\times m}$ whose i'th column is the output of Algorithm 6.2 when given $\boldsymbol{Z} = e_i$, where $e_i \equiv (0,\dots,0,1,0,\dots,0)$ is the vector with all components 0 except the i'th which is 1.

c) Modify the third function so that it computes $\Sigma = CC^{\mathsf{T}}$ and returns

$$E = \sum_{i=1}^{m}\sum_{j=1}^{m} |\Sigma_{ij} - \min(s_i, s_j)|.$$

Hand in your final source code including comments. Report the largest value of E in 100 trials with $m = 40$. Report the largest value of E in 100 trials with $m = 1$. It would be wrong to use $C^{\mathsf{T}}C$ instead of CC^{T}. Try it anyway and report the largest values of E that you get for $m = 40$ and for $m = 1$. (Make sure nobody uses the wrong version later.)

6.9. If $\mathrm{d}X_t = a(X_t)\,\mathrm{d}t + b(X_t)\,\mathrm{d}B_t$ for standard Brownian motion B_t and f is twice continously differentiable then ***Itô's formula*** is that

$$\mathrm{d}f(X_t) = \bigl(f'(X_t)a(X_t) + \frac{1}{2}f''(X_t)b^2(X_t)\bigr)\,\mathrm{d}t + f'(X_t)b(X_t)\,\mathrm{d}B_t.$$

a) Use Itô's formula to find the SDE satisfied by $S_t = \exp(X_t)$ for $X_t = \alpha t + \sigma B_t$.

b) Use your result from part **a** to find the SDE X_t for which $S_t = \exp(X_t)$ has SDE $\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}B_t$.

6.10 (Exotic options)**.** Here we look at valuing some exotic options. The options are based on an underlying stock price $S(t)$ for $0 \leqslant t \leqslant 1$. We suppose that $S(t)$ follows a geometric Brownian motion with parameters: $S(0) = 1$, $\delta = 0.035$, and $\sigma = 0.25$.

Let $t_j = j/256$ for $j = 0,\dots,256$. The quantity z_+ is $\max(0, z)$, the positive part of z. Estimate the expected value of the following functions of $S(\cdot)$, and give a 99% confidence interval.

a) Asian call, at the money:

$$\left(\frac{1}{256}\sum_{j=1}^{256} S_{j/256} - 1\right)_+.$$

b) Asian call, starting out of the money:

$$\left(\frac{1}{256}\sum_{j=1}^{256} S_{j/256} - 1.2\right)_+.$$

c) Asian put, at the money:

$$\left(1 - \frac{1}{256}\sum_{j=1}^{256} S_{j/256}\right)_+.$$

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d) European call, with down and out barrier:

$$\bigl(S_1 - 1\bigr)_+ \times \prod_{j=1}^{256} \mathbb{1}_{S(j/256) > 0.9}.$$

e) Asian put, with up and in barrier:

$$\left(1 - \frac{1}{256}\sum_{j=1}^{256} S_{j/256}\right)_+ \times \left(1 - \prod_{j=1}^{256} \mathbb{1}_{S(j/256) < 1.1}\right).$$

f) Lookback call (fixed strike):

$$\max_{1 \leqslant j \leqslant 256} (S_{j/256} - 1.1)_+.$$

g) Lookback put (floating strike):

$$\max_{1 \leqslant j \leqslant 256} S_{j/256} - S(1).$$

Notes: The value of the options includes a discount factor $\exp(-\delta)$ that we omit here for simplicity. Many of these options are ordinarily considered in continuous time, not just at equispaced time points. Even for finitely spaced time points, sometimes the spacing is not uniform. For example there could be different numbers of trading days between the listed times.

6.11. Brownian motion is a good model for many processes that evolve in time. When the process is indexed by a spatial line segment instead, we may prefer to treat the left and right ends of the interval symmetrically. For $\mathcal{T} = [0, 1]$, define the process $C(t) = B^{(1)}(t) + B^{(2)}(1 - t)$ where $B^{(1)}$ and $B^{(2)}$ are independent Brownian motions.

a) Find $\mathbb{E}(C(t))$ and $\mathrm{Var}(C(t))$ and $\mathrm{Cov}(C(s), C(t))$ for $0 \leqslant s \leqslant t \leqslant 1$.

b) Find $\mathrm{Corr}(C(0), C(1))$.

c) Show that $\mathbb{E}((C(s) - C(t))^2) = 2|s - t|$, for $s, t \in [0, 1]$.

d) Does the process $C(t)$ have independent increments? Since C has Gaussian finite dimensional distributions, the question reduces to whether

$$\mathrm{Cov}\bigl(C(t_4) - C(t_3), C(t_2) - C(t_1)\bigr) = 0$$

whenever $0 \leqslant t_1 < t_2 \leqslant t_3 < t_4 \leqslant 1$.

e) Generate and plot 10 independent sample paths of C at points $t_i = i/m$ for $i = 0, \ldots, m = 300$. Hand in your code.

f) Suppose that $C(0) = 1$ and $C(1) = -1$. Generate sample paths $C(t)$ for $t_i = i/m$ conditionally on these observed values for $C(0)$ and $C(1)$. Here $i = 1, \ldots, m - 1$ and $m = 300$ as before. Turn in your code along with a mathematical explanation of your strategy.

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6.12. Suppose that one SDE scheme has a mean squared error of $A_1 C_1^{-\beta_1}$ while another has MSE $A_2 C_2^{-\beta_2}$ where $\beta_2 > \beta_1 > 0$, $A_j > 0$ and $C_j > 0$ is the number of steps used by method j. In this exercise, you find the computational cost at which the asymptotically better method attains a meaningfully large speedup. Specifically, we want to achieve a factor $R > 1$ of reduction in cost while holding the MSE fixed.

a) Find C_* so that if $C_1 \geqslant C_*$ and $\mathrm{MSE}_1 = \mathrm{MSE}_2$, then $C_1/C_2 \geqslant R$. Write C_* in terms of A_1, A_2, R, β_1 and β_2.

b) Suppose that $\beta_2 = 6/7$, $\beta_1 = 4/5$, $A_2 = A_1$ and $R = 2$. What then is the cost C_* at which the desired reduction will appear?

c) Repeat the previous part for the more modest goal of a 10% improvement, $R = 1.1$.

d) Repeat the previous two parts for $\beta_1 = 2/3$ and $\beta_2 = 4/5$.

e) Until now, we have measured cost in terms of the number of steps taken. Here we take account of the fact that higher order methods usually have more expensive steps. Suppose that method 2 has twice the cost per simulation step as does method 1. Then the critical values R for the previous analyses are 4 and 2.2. Find the critical costs C_* under this condition.

6.13. Repeat the simulation shown in Figure 6.10 100 times.

a) Plot the mean absolute error for the Milstein scheme and for the Euler scheme versus the input $[0, 1]$.

b) Replace the input interval $[0, 1]$ by $[0, 10]$ and repeat the previous part.

c) Replace the number $N = 100$ of time steps in the simulation by $N = 10$ and repeat the previous two parts.

6.14. Consider the stochastic volatility model in Example 6.6. Let the parameter values be $\delta = .03$, $\kappa = 0.8$, $\theta = 0.2$, $\sigma = .25$, $\rho = -0.4$, $V(0) = .25$ and $S(0) = 1$. Evaluate $\mathbb{E}(e^{-\delta} \max(0, S(1) - 1.1))$ by Monte Carlo.

6.15. We studied multilevel simulation for the Euler-Maruyama scheme and found a strategy that attains a root mean square error of $O(\epsilon)$ at a cost of $O(\epsilon^{-2} \log^2(\epsilon))$. That scheme used levels $\ell = 0$ through L_ϵ and $L_\epsilon \to \infty$ as $\epsilon \to 0$. It is of interest to know how many samples are taken at the finest level that the algorithm uses. To answer this:

a) Find an expression for n_{L_ϵ} in terms of c, M and ϵ. The expression need not take integer values.

b) Does $n_{L_\epsilon} \to \infty$ as $\epsilon \to 0$, does it approach 0, or does it approach a fixed nonzero value?

6.16. Let T_i be the points of a non-homogeneous Poisson process on $[0, \infty)$, with intensity function $\lambda(t) = At^{B-1}$ for constants $A > 0$ and $B > 0$. Describe how to simulate the first N points T_i, $1 \leqslant i \leqslant N$ of this process.

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6.17. Consider the homogeneous Poisson process on $\mathbb{R}$ with rate $\lambda > 0$. Devise a method of sampling the N points of this process that are closest to the origin. Hint: it is not correct to sample $N/2$ points on $[0, \infty)$ and $N/2$ points on $(-\infty, 0]$.

6.18. Consider the non-homogeneous Poisson process on $\mathbb{R}$ with intensity $\lambda(t) = \exp(t)$.

a) Devise a method to sample the 100 points of this process that come closest to the origin.

b) Of interest is the number $Y = \sum_{i=1}^{100} \mathbb{1}_{T_i<0}$ of negative points in the sample. Use Monte Carlo sampling to generate a histogram showing the distribution of Y.

6.19. Prove that the formula in (6.32) gives points uniformly distributed in $\mathcal{T} = \{\boldsymbol{x} \in \mathbb{R}^2 \mid \boldsymbol{x}^{\mathsf{T}}\boldsymbol{x} = 1\}$.

6.20. To sample the tail of the Zipf-Poisson ensemble using thinning we need to generate points (x, z) uniformly in the set

$$\mathcal{S} = \{(x, z) \mid k + 1/2 \leqslant x < \infty, 0 \leqslant z \leqslant Nx^{-\alpha}\}$$

for a threshold integer $k \geqslant 0$ and parameters $\alpha > 1$ and $N > 0$ The number of points to generate has the Poi(μ) distribution where $\mu = \int_{k+1/2}^{\infty} Nx^{-\alpha}\,\mathrm{d}x$.

a) Give a closed form expression for μ.

b) Show how to sample $(x, z) \sim \mathbf{U}(\mathcal{S})$ by a transformation of $\boldsymbol{U} = (U_1, U_2) \sim \mathbf{U}(0, 1)^2$.

6.21. The Zipf-Poisson ensemble was sampled by taking $X_1, \ldots, X_k$ directly from their Poisson distributions and then sampling the tail by thinning the Poisson process with intensity $\widetilde{\lambda}(x) = N(x - 1/2)^{-\alpha}$ on $[k + 1/2, \infty)$. Setting $k = 0$ would amount to sampling the entire ensemble by thinning the $\widetilde{\lambda}$ process and taking none of the X_i directly. Is that possible?

6.22. For the Sullivan County traffic data in Table 6.1, define $\widetilde{\lambda}(t)$ on $0 \leqslant t < 24$ as the average number of cars observed per minute. The function should be piecewise constant with 96 segments. Simulate a Poisson process on $[0, 24)$ with intensity $\lambda(t) = 1.2\widetilde{\lambda}(t)$. Repeat that simulation 10,000 times and report the items described below.

a) The average number of cars seen in 24 hours.

b) The average arrival time, as a number between 0 and 24, of the 1000'th car of the day. Record 24 if the simulated day had fewer than 1000 cars.

c) The average number of cars in the time window from 1.5 to 1.75, not including the endpoints.

d) A histogram of the busiest one hour period from those 1000 simulated days. Consider all periods of the form $[t, t+1)$ where $t \in \{0, 0.25, 0.5, 0.75, \ldots, 23.0\}$. For a day where $[9.25, 10.25)$ is the busiest hour, report $t = 9.25$.

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e) A histogram of the number of cars in that busiest hour.

f) A histogram showing the smallest interarrival time between two cars in a given day.

Be sure to describe the approach you took to simulating this non-homogeneous Poisson process and show your code. It is possible to derive the distribution of the number of cars seen in a random day. This may help you to check your simulation.

6.23. Let $T_j \sim \mathrm{Exp}(1)/a_j$ be independent where $0 < a_j < \infty$ for $j = 1, \dots, M$. Prove that

$$\mathbb{P}\Big(T_j \leqslant \min_{1 \leqslant k \leqslant M, k \neq j} T_k\Big) = \frac{a_j}{a_0}$$

where $a_0 = \sum_{j=1}^{M} a_j$.

6.24. Consider a Cox process with

$$\lambda(\boldsymbol{t}) = \frac{\mu}{2\pi \det(\Sigma)} \sum_{i=1}^{\infty} \exp(-(\boldsymbol{t} - \boldsymbol{x}_i)^{\mathsf{T}} \Sigma^{-1} (\boldsymbol{t} - \boldsymbol{x}_i)/2)$$

where $\Sigma = \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)$, where $\mu > 0$ and $0 \leqslant \rho < 1$ are given constants and $\boldsymbol{x}_i$ are the points of a homogeneous Poisson process with parameter $\lambda_0 > 0$.

a) Describe how to simulate the process $\boldsymbol{T}_i$ in the window $[-1, 1]^2$.

b) Using simulations, investigate numerically the $K(\cdot)$ function for this process. Plot the results for $0 < h < 1/2$ and each $\rho \in \{0, 1/2, 3/4, 9/10, 99/100\}$. Choose a value of μ to use.

c) The i'th contribution to $\lambda(\cdot)$, namely $\exp(-(\boldsymbol{t} - \boldsymbol{x}_i)^{\mathsf{T}} \Sigma^{-1} (\boldsymbol{t} - \boldsymbol{x}_i)/2)/(2\pi)$ has contours

6.25. Devise a process like the Poisson field of lines but with lines whose polar coordinates cluster. There are many ways to do this. Select one and illustrate four realizations of it, each with 30 lines intersecting the region $[-1, 1]^2$. Repeat with lines whose polar coordinates avoid each other.

6.26. Suppose that $F \sim \mathsf{DP}(G, \alpha)$ for a distribution G on $\Omega \subset \mathbb{R}^d$ and $\alpha > 0$. Let A and B be disjoint subsets of Ω. Find an expression for $\mathrm{Cov}(F(A), F(B))$. Now find a more general formula for the case that A and B may intersect. Now specialize your formula to the case $A \subset B$.

6.27. Simulate the first 1000 customers for the Chinese restaurant process with $\alpha = 3$. Keep track of which table each customer goes to. We would like to know the following:

a) The average number of tables occupied.

b) The expected index of the most crowded table, breaking ties in favor of smaller table numbers.

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c) The number of tables with exactly one customer.

d) The average number of customers at the most popular table.

Give an estimate and a confidence interval for each of those numbers. Use $n = 10{,}000$ simulations of the CRP.

6.28. The Pitman-Yor process yields a number of distinct tables that grows proportionally to αn^d where n is the number of customers. For $\alpha = 1$ and $d = 1/2$ simulate the Pitman-Yor process until $n = 10{,}000$ and keep track of the number of distinct tables you obtain. Produce a histogram from 1000 independent simulations of the Pitman-Yor process.

6.29. Simulate the predator-prey curve in Figure 6.20 but with a few changes:

a) Keep the starting points the same and the reaction rates the same but change the $R + L \stackrel{c_2}{\to} 2L$ reaction to $R + L \stackrel{c_2}{\to} 3L$. With more Lynx produced per rabbit consumed, we expect a different cycle behavior, and possibly an early extinction. Simulate the process twice, up to time 10, and plot every 100th point. Plot both realizations. Describe what you see.

b) Repeat the previous, using $R + L \stackrel{2c_2/3}{\to} 3L$. The reaction now produces more offspring but takes place less frequently. What happens this time?

You may use algorithm 6.4 (Gillespie's).

6.30. A rock-paper-scissors dynamic has been observed among certain bacteria. Suppose that species A produces a poison that helps it displace species B. Species B reproduces faster than C and will displace it. Species C is resistant to the poison that A produces. The cost to C of being resistant explains why B reproduces faster than C. The cost of being resistant is lower than the cost of producing the poison, and so C displaces A. The system is described by the reactions:

$$\begin{aligned} A + B &\stackrel{c_a}{\to} A + A, \\ B + C &\stackrel{c_b}{\to} B + B, \quad \text{and,} \\ C + A &\stackrel{c_c}{\to} C + C. \end{aligned}$$

If one species goes extinct, then another will quickly follow.

a) Simulate this model using starting counts $\boldsymbol{X}(0) = (100, 100, 100)$ of each bacteria type where $\boldsymbol{X} = (X_A, X_B, X_C)$ and rates $c_a = c_b = c_c = 1$. Plot the trajectories of the three species counts over the first 30,000 steps or until the time at which two species have gone extinct, whichever happens first. Make 4 such trajectories and describe qualitatively what you see. It helps to use different colors for the three components of $\boldsymbol{X}$. It is also reasonable to thin out the data, plotting every k'th point of the simulation for $k = 10$ or more.

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b) At what value of t did your simulations of $\boldsymbol{X}(t)$ stop in each of the 4 trajectories?

c) Change the simulation so that $\boldsymbol{X}(0) = (1000, 1000, 1000)$ and run for up to 100,000 steps. Now give A an advantage, setting $c_a = 1.1$ while leaving $c_b = c_c = 1$. This should be a disadvantage for B. The case of C is indirect and more interesting. If A is eating faster then it is consuming B which preys on C, so that is good for C. It also increases the supply of A, which is the food for C, which is also good for C. So perhaps C benefits the most. But A has a very direct advantage and C's advantage is indirect. Run this simulation 40 times and report which species comes out best. (Perhaps it is B for some surprising counter-intuitive reason.) There will not be enough extinctions to settle the issue. To keep score, in each simulation record the percent of time steps that left A (respectively B or C) with the largest population. Also record the average population size of each species, averaged over the time steps. It is not worthwhile to use the simulated elapsed time between steps in the weighting. From the gathered data decide which species has the advantage over the first 100,000 time steps.

The biological example of rock-paper-scissors is from Kerr et al. (2002). Their context is a bit different from the experiment here. The bacteria are arranged spatially and interact only with their near neighbors and not as a mixture. The result is much more stable.

Lotka-Volterra wildlife management

These exercises form a small project to investigate the effects of interventions in the Lotka-Volterra model.

6.31. Here we revisit the Lotka-Volterra model of Example 6.10 to investigate the effect of wildlife management schemes. We begin with the same reactions, starting values and rate constants of that example, as given on page 63. Now, when the predator population gets too high we introduce hunting. The hunting reaction removes one predator but leaves the number of prey unchanged. It has a propensity proportional to the number of predators present if that number is above a threshold. Otherwise it has propensity 0.

a) Write a formula for the propensity function $a_4(\boldsymbol{X})$ and the update $\boldsymbol{\nu}_4$ of the hunting reaction. It is not of mass action kinetics form but it can still be simulated by Gillespie's algorithm.

b) Investigate several choices for the threshold at which hunting is allowed and for the rate constant in the hunting reaction. Look at both extreme choices and moderate ones. Which, if any, of the parameter choices you tried stabilize the populations? Which, if any, make it less stable? Which brought prompt extinction of the predator population? Describe how you decided to quantify 'stability', say how many steps you let the simulations run, and how many replicates you made of each. Hand in plots for some of

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the extreme cases you considered. Hand in plots for two more parameter settings that you find interesting.

6.32. Change the hunting model from Exercise 6.31. Now the authorities issue a variable number of licenses for each time step. If the number of predators goes above a threshold then the number of hunting licenses issued is proportional to that excess prey count. The hunting reaction takes place at a rate proportional to the number of licenses times the number of predators.

a) Write the formulas for $a_4(\boldsymbol{X})$ and $\boldsymbol{\nu}_4$.

b) Repeat part b of Exercise 6.31 for this new hunting model.

6.33. Explain why $\bar{F} + R \to 2R$ is a better model for the rabbit population than $\bar{F} + 2R \to 3R$ even though the latter accounts for rabbit offspring having two parents.

There are many elaborations of the Lotka-Volterra model to make it more realistic. The model could incorporate a limit on the carrying capacity of the environment for prey. Then the growth rate of the prey population slows as it approaches that upper bound. Another elaboration is to remove the assumption of perfect population mixing. We simulate some number K of different populations each with Lotka-Volterra reactions taking place within them and specify migration rates between adjacent populations for both predators and prey.

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